

**GSM BSS
GBSS26.1**

Handover Feature Parameter Description

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1 About This Document

1.1 Scope

This document describes the Handover feature, including its technical principles, related features, network impact, and engineering guidelines.

This document covers the following features:

- GBFD-110601 HUAWEI I Handover
- GBFD-115504 AMR FR/HR Dynamic Adjustment
- GBFD-110801 Processing of Measurement Report
- GBFD-510102 Fast Move Handover
- GBFD-510501 HUAWEI II Handover
- GBFD-510103 Chain Cell Handover
- GBFD-110802 Pre-processing of Measurement Report
- GBFD-117101 BTS power lift for handover
- GBFD-110607 Direct Retry
- GBFD-110608 SDCCH Handover

1.2 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements of the operating environment that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

The product supports only the functions mentioned in this document, and the functions can be enabled only in the specified scenarios. If a function not mentioned in this document is enabled or a function is enabled in an unspecified scenario, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

1.3 Change History

This section provides information about the changes in different document versions. There are two types of changes:

- Feature change
Changes in features of a specific product version
- Editorial change
Changes in wording or addition of information that was not described in the earlier version

04 (2025-05-28)

This issue includes the following changes.

Change Type	Change Description	Parameter Change
Feature change	None	None
Editorial change	Revised descriptions in 3.4.10 Edge Handover and 3.5.8 Edge Handover .	None

03 (2025-03-28)

This issue includes the following changes.

Change Type	Change Description	Parameter Change
Feature change	None	None

Change Type	Change Description	Parameter Change
Editorial change	Revised descriptions in: • 2.1 Introduction • 3.2 Handover Preprocessing	None

02 (2025-02-07)

This issue includes the following changes.

Change Type	Change Description	Parameter Change
Feature change	None	None
Editorial change	Revised descriptions in 3.4.3 BQ Handover .	None

01 (2024-04-30)

GBSS26.1 01 (2024-04-30) introduces the following changes to GBSS25.1 01 (2023-03-09).

Change Type	Change Description	Parameter Change
Feature change	None	None
Editorial change	Revised descriptions in 6.4.3 Activation and 6.5.3 Activation .	None

2 Overview

2.1 Introduction

The GSM network comprises multiple cells with continuous coverage. The handover technique is introduced into the GSM system to enable the users who are in motion to continue with the current call without interruption, optimizing the network performance.

During a handover, the MS and BTS in service measure the conditions of uplink and downlink radio links respectively, record the measurement results into measurement reports (MRs), and then send the MRs to the BSC. The BSC determines whether to trigger a handover based on the MRs and the actual conditions of the radio network.

Huawei handover algorithms (handover algorithm I and handover algorithm II) involve measurement and MR reporting, MR processing, handover decision, and handover execution.

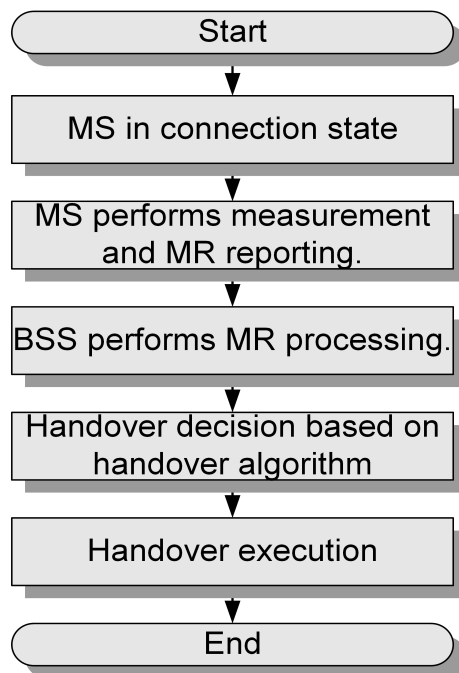
NOTE

Huawei handover algorithms apply to the handovers on TCHs as well as the handovers on SDCCHs.

You can determine the handover algorithm used in a cell by setting **HOCTRLSWITCH** (*BSC6900, BSC6910*).

Figure 2-1 shows the handover procedure based on Huawei handover algorithms (including handover algorithm I and handover algorithm II).

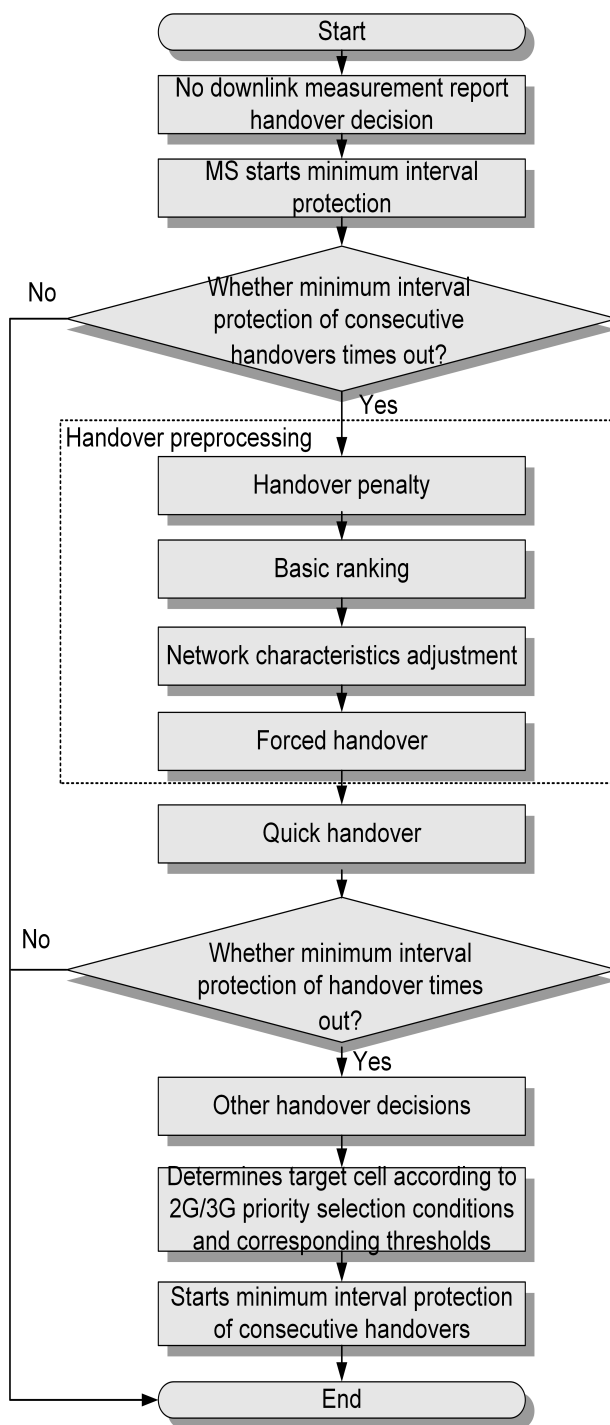
Figure 2-1 Handover procedure based on Huawei handover algorithms



Handover Decision Based on Handover Algorithm I

Figure 2-2 shows the procedure for handover decision based on handover algorithm I.

Figure 2-2 Procedure for handover decision based on handover algorithm I



In handover algorithm I, five types of handover decisions are defined:

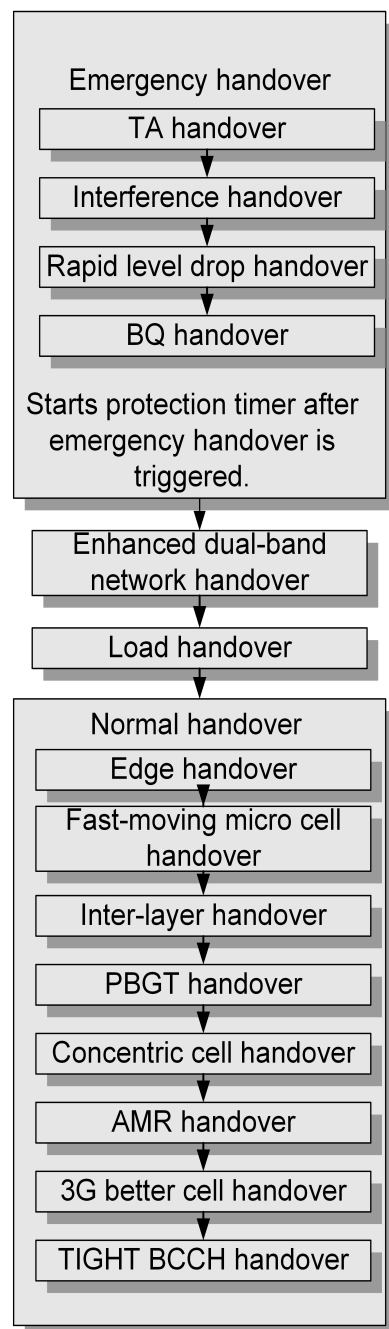
- Quick handover (including quick PBGT handover and frequency offset handover): Good and stable services can be provided when the voice quality deteriorates during the fast movement of an MS. Quick handover is mainly applicable in the railway scenario.
- Emergency handover: An emergency handover can ensure the call continuity when the radio condition severely deteriorates. Theoretically, an emergency

handover has a bigger deviation than other handovers in terms of the selection of the target cell. In a normal cell, frequent emergency handovers should be avoided.

- Enhanced dual-band network handover: In an enhanced dual-band network, the resources in the overlaid (OL) DCS1800 cell and underlaid (UL) GSM900 cell can be shared during the assignment and handover procedures. That is, the MSs in the high-traffic GSM900 cell can be handed over to the low-traffic DCS1800 cell to balance traffic.
- Load handover: Load handover enables the system load to be balanced among multiple cells so that the system performance can be ensured.
- Normal handover: Normal handover ensures good services when an MS is moving.

Figure 2-3 shows the handovers and their priorities in handover algorithm I.

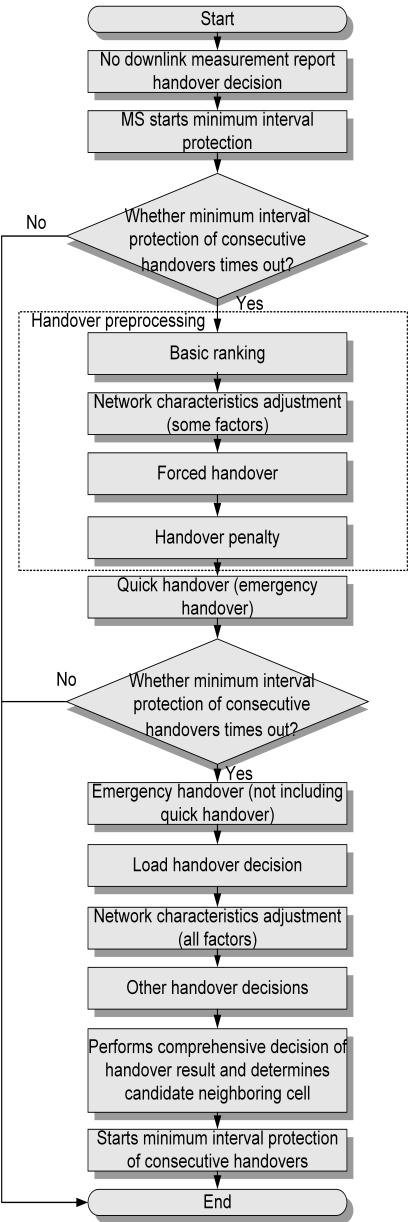
Figure 2-3 Handover decisions based on handover algorithm I



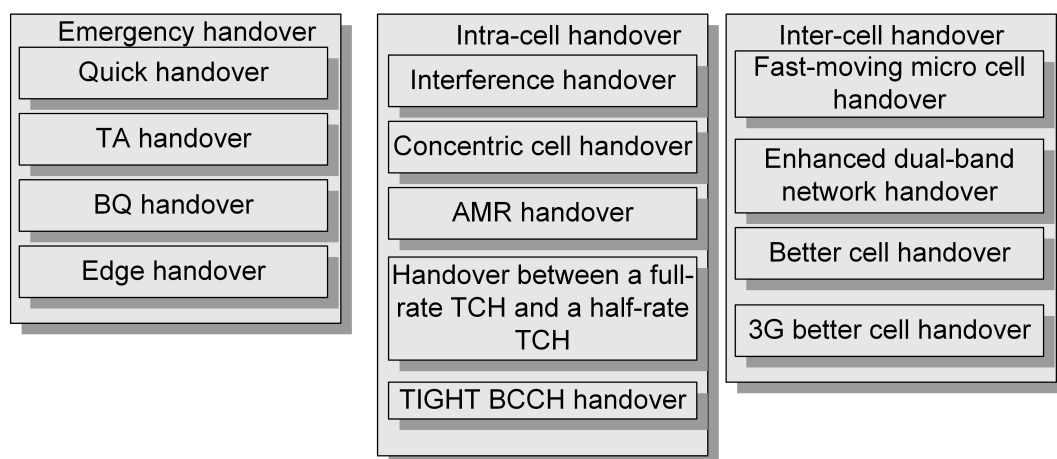
Handover Decision Based on Handover Algorithm II

Figure 2-4 shows the procedure for handover decision based on handover algorithm II.

Figure 2-4 Procedure for handover decision based on handover algorithm II



In handover algorithm II, three handover decisions are defined, as shown in [Figure 2-5](#).

Figure 2-5 Handover decisions based on handover algorithm II

Handover Execution

During the handover execution of handover algorithm I, when a candidate cell is fully loaded: If **HoCDCOverLodeHoEn (BSC6900, BSC6910)** is set to **YES**, the BSC allows an MS to be handed over to the candidate cell. If **HoCDCOverLodeHoEn (BSC6900, BSC6910)** is set to **NO**, the BSC does not allow an MS to be handed over to the candidate cell.

The BTS power lift for handover function determines whether the BTS of the serving cell transmits signals at the maximum power during a handover. The BSC maximizes the transmit power of the BTS before sending a handover command to the MS. The BSC maintains the BTS power during the handover to ensure the success of the handover. This function is controlled by the **HOPOWERBOOST (BSC6900, BSC6910)** parameter. (GBFD-117101 BTS power lift for handover)

Synchronous Handover and Asynchronous Handover

The BSC supports both synchronous handovers and asynchronous handovers. Generally, the BSC performs synchronous handovers for intra-BTS inter-cell or intra-cell handovers and performs asynchronous handovers for inter-BTS handovers. However, the BSC also performs asynchronous handovers for intra-BTS inter-cell handovers when the following conditions are met:

- **DIRMAGANSITEFLAG (BSC6900, BSC6910)** is set to **YES**. That is, a repeater is configured for the source cell or target cell.
- **FRAMEOFFSET (BSC6900, BSC6910)** is set to a value other than **0** or **255**. That is, frame offset is configured for the source cell or target cell.
- The source cell or target cell is a multi-site cell.

2.2 NEs Supporting the Feature

Table 2-1 NEs supporting the feature

Feature Name	BSC6900	BSC6910	GBTS	eGBTS
HUAWEI I Handover	√	√	√	√
Direct Retry	√	√	√	√
SDCCH Handover	√	√	√	√
Processing of Measurement Report	√	√	√	√
Pre-processing of Measurement Report	√	√	√	√
AMR FR/HR Dynamic Adjustment	√	√	√	√
Dynamic HR/FR Adaptation	√	√	√	√
Fast Move Handover	√	√	√	√
Chain Cell Handover	√	√	√	√
HUAWEI II Handover	√	√	√	√
BTS power lift for handover	√	√	√	√

 NOTE

√ indicates that the NE supports this feature. × indicates that the NE does not support this feature.

3 Technical Description

3.1 Processing of Measurement Report

The MR processing involves interpolation processing and filtering processing.

This section describes GBFD-110801 Processing of Measurement Report and GBFD-110802 Pre-processing of Measurement Report.

NE Selection for Measurement Report Processing

The processing can be performed either on the BSC side or on the BTS side.

- If ***BTSMESRPTPREPROC (BSC6900, BSC6910)*** is set to **BSC_Preprocessing(BSC preprocessing)**, the processing is performed on the BSC side.
- If ***BTSMESRPTPREPROC (BSC6900, BSC6910)*** is set to **BTS_Preprocessing(BTS preprocessing)**, the processing is performed on the BTS side. You can specify the content and reporting period of MRs by setting the parameters ***PRIMMESPT (BSC6900, BSC6910)***, ***BSMSPWRLEV (BSC6900, BSC6910)***, and ***MRPREPROC_FREQ (BSC6900, BSC6910)***. This implementation decreases signaling traffic over the Abis interface and traffic processed by the BSC.

Data Selection of Measurement Report

MRs can be classified into enhanced MRs and normal MRs. The value of ***MEASURETYPE (BSC6900, BSC6910)*** determines the MR type to be used.

In an MR, the TCH measurement of the serving cell is classified into FULL SET and SUB SET.

The BSC determines the TCH measurement based on the measurement result and the value of ***DLDTXUPDATESWITCH (BSC6900, BSC6910)***.

- When ***DLDTXUPDATESWITCH (BSC6900, BSC6910)*** is set to **NO(No)**:
 - If the previous MR indicates that the DTX is used, the SUB SET value should be used.

- If the previous MR indicates that the DTX is not used, the FULL SET value is used.
- When ***DLDTXUPDATESWITCH (BSC6900, BSC6910)*** is set to **YES(Yes)**, the measured value corresponding to the "dtx-downlink" element information (IE) in the measurement report is used.
- When ***BCCHIntraFreqHOOptimizeSw (BSC6900, BSC6910)*** is set to **ON(On)**, the BSC imposes penalties on the downlink level of a co-BCCH intra-frequency neighboring cell for the serving cell. Specifically, the BSC subtracts the value of ***BCCHIntraFreqHOREctThld (BSC6900, BSC6910)*** from the downlink level of this neighboring cell and then performs interpolation processing and filtering processing.

Measurement Report Interpolation

The neighboring cell indexes are found on the basis of the BCCH ARFCNs and BSICs provided by the MS. Then, the uplink and downlink measurement results are obtained from the MRs.

- If MRs are issued continuously, they are directly interposed in the measurement report list.
- If MRs are not issued continuously and the number of lost MRs is less than or equal to the value of ***MRMISSCOUNT (BSC6900, BSC6910)***, the BSC performs operations as follows:
 - When ***MrIntrplOptSwitch (BSC6900, BSC6910)*** is set to **NO(No)**: For the serving cell, handover algorithm I performs the linear interpolation for the MRs. The lowest values are applied to the interpolation of MRs by handover algorithm II according to the protocols; specifically, level 0 (–110 dBm) and quality level 7 are applied in the interpolation. For a neighboring cell, handover algorithms I and II use the lowest values stipulated in the protocols for the lost signal level; specifically, level 0 (–110 dBm) is used in the interpolation.
 - When ***MrIntrplOptSwitch (BSC6900, BSC6910)*** is set to **YES(Yes)**: For the serving cell, handover algorithms I and II perform the linear interpolation for the MRs. For a neighboring cell, handover algorithms I and II perform the linear interpolation for the MRs.
- If MRs are not issued continuously and the number of lost MRs is greater than the value of ***MRMISSCOUNT (BSC6900, BSC6910)***, the previous MRs are discarded. When new MRs are issued, calculation is performed again.

Measurement Report Filtering

Filtering is performed on MRs obtained continuously from the MR list. Averaging is performed on the following items:

- Uplink/downlink receive level
- Uplink/downlink receive quality
- Time advance (TA)
- Radio quality indication (RQI)
- BTS power
- Neighboring GSM cell level

- Common Pilot Channel (CPICH), received signal code power (RSCP), and E_c/N_o of the neighboring UMTS cell

The averaging minimizes the effect on the result of handover decision due to sudden changes in the measurement values.

Power control compensation needs to be performed for the downlink receive level of the serving cell by handover algorithm II. If you compare the receive level of the serving cell after power control with that of all BCCH TRXs of the neighboring cell, there is no mapping between them. If the cells overlap severely, handovers are easily triggered, causing ping-pong handovers. After the power control compensation is performed, the receive level of the serving cell can reflect the coverage condition of the BCCH TRX of the serving cell. The power control compensation of the serving cell is performed after the interpolation processing and before the filtering processing. In general, the compensation of power control is calculated as follows: Power control compensation = Downlink receive level of the serving cell + 2 x current BTS downlink transmit power control level indicated by **BS-POWER-LEVEL**

The number of consecutive MRs required for filtering is determined by the measurement object and channel type. For details, see [Table 3-1](#).

Table 3-1 Parameters related to the number of measurement reports

Object Type	Channel Type	Parameter Name	Parameter ID
Receive level of the serving cell	SDCCH	Filter Length for SDCCH Level	<i>SIGSTRFLTLEN</i> <i>(BSC6900, BSC6910)</i>
	TCH	Filter Length for TCH Level	<i>DATASTRFLTLEN</i> <i>(BSC6900, BSC6910)</i>
Quality of the serving cell	SDCCH	Filter Length for SDCCH Qual	<i>SIGQUAFLTLEN</i> <i>(BSC6900, BSC6910)</i>
	TCH	Filter Length for TCH Qual	<i>DATAQUAFLTLEN</i> <i>(BSC6900, BSC6910)</i>
TA of the serving cell	TCH	Filter Length for TA	<i>TAFLTLEN</i> <i>(BSC6900, BSC6910)</i>
	SDCCH	TA Filter Length for SDCCH Level	<i>TASIGSTRFLTLEN</i> <i>(BSC6900, BSC6910)</i>
Receive level of the neighboring cell	BCCH	Filter Length for Ncell RX_LEV	<i>NCELLFLTLEN</i> <i>(BSC6900, BSC6910)</i>
	SDCCH	NCell Filter Length for SDCCH Level	<i>NSIGSTRFLTLEN</i> <i>(BSC6900, BSC6910)</i>
Power of the BTS in the serving cell	TCH	Filter Length for TCH Level	<i>DATASTRFLTLEN</i> <i>(BSC6900, BSC6910)</i>
RQI	TCH	Filter Length for TCH Qual	<i>DATAQUAFLTLEN</i> <i>(BSC6900, BSC6910)</i>

If consecutive MRs are insufficient, the filtering fails. In this case, the handover decision is not performed.

3.2 Handover Preprocessing

Handover Penalty

According to the neighboring cell information in an MR and configured parameters, the system performs handover preprocessing and adjusts the priorities of neighboring cells.

The handover penalty is performed after a successful quick handover, TA handover, Bad Quality (BQ) handover, fast-moving micro cell handover, and overlaid subcell to underlaid subcell handover within an enhanced concentric cell, and after the handover failures.

When *IntraFreHOFailPenlEnSw* (**BSC6910**, **BSC6900**) is set to **ON(On)**, handover penalty is performed on co-BCCH neighboring cells of the serving cell. A larger number of handover failures indicates a longer penalty time.

- The period of penalty on a co-BCCH intra-frequency neighboring cell following the first-time handover failure is the product of values of *IntraFreFstHOFailPenlCoef* (**BSC6900**, **BSC6910**) and *UMPENALTYTIMER* (**BSC6900**, **BSC6910**).
- The period of penalty on this cell following the second-time handover failure is the product of values of *IntraFreScndHOFailPenlCoef* (**BSC6900**, **BSC6910**) and *UMPENALTYTIMER* (**BSC6900**, **BSC6910**).
- After a third-time handover failure, the ongoing call is prohibited from being handed over from the current channel of the serving cell to this neighboring cell. During the handover penalty period, this neighboring cell is ranked at the bottom of the neighboring cell list for the BQ handover, interference handover, no-DL-MR handover, and quick level drop handover; this cell is removed from the neighboring cell list for other types of handovers.

In handover algorithm II, in addition to the situations mentioned above, the handover penalty is also performed after a successful load handover and an interference handover.

NOTE

In the handover decision procedure of handover algorithm II, the handover penalty is performed after network characteristics adjustment and before emergency handover decision.

Handover penalty inheritance enables an MS to inherit its penalty state after the MS is successfully handed over to another cell.

- After a quick handover, TA handover, BQ handover, or load handover (in handover algorithm II) is successfully performed, the penalty level is subtracted from the actual receive level of the original cell during the penalty period. [Table 3-2](#) lists the parameters related to handover penalty.

Table 3-2 Parameters related to handover penalty

Handover Type	Parameter
Quick handover	<i>TIMEPUNISH (BSC6900, BSC6910)/ TIMEPUNISH (BSC6900, BSC6910) HOPUNISHVALUE (BSC6900, BSC6910)/HOPUNISHVALUE (BSC6900, BSC6910) QuickHoPunishSw (BSC6900, BSC6910)</i>
TA handover	<i>SSTAPUNISH (BSC6900, BSC6910) TIMETAPUNISH (BSC6900, BSC6910)</i>
BQ handover	<i>SSBQPUNISH (BSC6900, BSC6910) TIMEBQPUNISH (BSC6900, BSC6910)</i>
Load handover (handover algorithm II)	<i>LOADHOPENTIME (BSC6900, BSC6910) LOADHOPENVALUE (BSC6900, BSC6910) LoadHoPunishInheritSwitch (BSC6900, BSC6910)</i>

- After a fast-moving micro cell handover is successfully performed, penalty is performed on all the neighboring cells of the micro cell. Related parameters are *SPEEDPUNISH (BSC6900, BSC6910)* and *SPEEDPUNISHT (BSC6900, BSC6910)*.
- If an MS fails to initiate an intra-cell adaptive multirate (AMR) TCHF to TCHH handover, it cannot initiate another intra-cell AMR TCHF to TCHH handover within the duration specified by *TIMEAMRFHPUNISH (BSC6900, BSC6910)*.
- In handover algorithm II, after the interference handover is initiated, initiating this handover again is prohibited within the duration specified by *INTERFEREHOPENTIME (BSC6900, BSC6910)* regardless of whether the handover is successful or not.
- After an overlaid subcell to underlaid subcell handover within an enhanced concentric cell is successful, the handover from underlaid subcell to overlaid subcell is not allowed within the duration specified by *UTOOHOPENTIME (BSC6900, BSC6910)*.
- After an overlaid cell to underlaid cell handover in the enhanced dual-band network is successful, the handover from underlaid cell to overlaid cell is not allowed within the duration specified by *HOPENALTYTIME (BSC6900, BSC6910)*.
- After a handover fails, different penalties are performed on the target cell based on the causes:

- If a handover to a neighboring GSM or UMTS cell fails, the actual receive level of the target cell is subtracted by the value of ***FAILSIGSTRPUNISH (BSC6900, BSC6910)*** for neighboring cell ranking during the penalty.

NOTE

Based on the handover failure cause, the penalty time could be ***UMPENALTYTIMER (BSC6900, BSC6910)***, ***RSCPENALTYTIMER (BSC6900, BSC6910)***, or ***PENALTYTIMER (BSC6900, BSC6910)***.

- If an overlaid subcell to underlaid subcell handover within a concentric cell fails, the handover from overlaid subcell to underlaid subcell is not allowed within ***TIMEOTOUFAPUN (BSC6900, BSC6910)***.
- If an underlaid subcell to overlaid subcell handover within a concentric cell fails, the handover from underlaid subcell to overlaid subcell is not allowed within ***TIMEUTOOFAPUN (BSC6900, BSC6910)***.

Basic Ranking

Basic ranking is performed after handover penalty to generate a candidate cell list in descending order taking the following information into account: receive levels of the serving cell and neighboring cells carried in the MRs, hysteresis, usage of TCHs in the neighboring cells, and so on.

- In the case of non-directed retry, if an MS in an external BSC cell occupies an SDCCH and ***INRBSCSDHOEN (BSC6900, BSC6910)*** is set to **NO(No)**, this cell should be removed from the candidate cell list. Specifically, the handover to this external BSC cell is prohibited.
- In handover algorithm I, if the TCH usage of a neighboring GSM cell is 100% and the ***HoCDCOverLodeHoEn (BSC6900, BSC6910)*** parameter is set to **NO**, this neighboring cell is removed from the candidate cell list; specifically, the handover to this neighboring cell is prohibited. Handover algorithm II does not have this restriction on neighboring cells.
- If the downlink receive level of a neighboring GSM cell is less than the sum of ***HOCDCMINDWPWR (BSC6900, BSC6910)*** and ***MINOFFSET (BSC6900, BSC6910)***, the neighboring cell should be removed from the candidate cell list; that is, the handover to this neighboring cell is prohibited.

The BSC cannot obtain the uplink receive level of a neighboring GSM cell and needs to predict the uplink receive level. The predicted uplink receive level is restricted by ***ULRxlevBoundProtectionSw (BSC6900, BSC6910)***:

- When this parameter is set to **ON(On)**, the BSC performs border protection on the predicted uplink receive level of a neighboring cell and restricts the receive level within the range of 0 to 63 dB.
- When this parameter is set to **OFF(Off)**, the BSC does not perform border protection on the predicted uplink receive level of a neighboring cell and allows a receive level beyond the range of 0 to 63 dB.
- If the uplink receive level of a neighboring GSM cell is less than the sum of ***HOCDCMINUPPWR (BSC6900, BSC6910)*** and ***MINOFFSET (BSC6900, BSC6910)***, the neighboring cell should be removed from the candidate cell list; that is, the handover to this neighboring cell is prohibited.
- If a neighboring UMTS cell is a frequency division duplex (FDD) cell, the cell is processed according to ***FDDREP (BSC6900, BSC6910)***:

- If **FDDREP (BSC6900, BSC6910)** is set to **EcNo(Ec/No)**, and the Ec/No of a neighboring cell is less than **MINECNOTHRES (BSC6900, BSC6910)**, the neighboring cell should be removed from the candidate cell list; that is, the handover to this neighboring cell is prohibited.
- If **FDDREP (BSC6900, BSC6910)** is set to **RSCP(RSCP)**, and the received signal code power (RSCP) of a neighboring cell is less than **MINRSCPTHRES (BSC6900, BSC6910)**, the neighboring cell should be removed from the candidate cell list; that is, the handover to this neighboring cell is prohibited.
- If a neighboring UMTS cell is a time division duplex (TDD) cell and the RSCP after penalty is less than the **MINRSCPTHRES (BSC6900, BSC6910)**, the neighboring cell should be removed from the candidate cell list; that is, the handover to this neighboring cell is prohibited.
- Calculate the difference between the downlink receive level of the neighboring cells and the downlink receive level of the serving cell. Based on the difference, rank the neighboring cells in descending order.

Network Characteristics Adjustment

Factors considered during network characteristics adjustment have higher priorities than those considered during basic ranking.

During network characteristic adjustment, the following factors are considered based on their priorities from high to low:

- Whether the serving cell and a neighboring cell share the same BSC or MSC
- Load in a cell
- Layer where a cell is located

After performing basic ranking on all candidate cells based on their priorities, the BSC ranks these cells again to determine their comprehensive priorities.

After network characteristics adjustment, the BSC generates a final candidate cell list (including the serving cell and neighboring cells). The candidate cells are ranked based on their priorities from high to low. Then, the BSC starts the handover decision procedure.

If **HOPRIOMODEN (BSC6900, BSC6910)** is set to **NO(No)**: In handover algorithm II, the emergency handover decision is made before network characteristics adjustment and forced handover and after handover penalty (except load handover penalty) and basic ranking. Load handover penalty is performed before network characteristics adjustment and after an emergency handover. After the emergency handover decision, **LOADHOPENVALUE (BSC6900, BSC6910)** is subtracted from the signal level of the original cell within the time specified by **LOADHOPENTIME (BSC6900, BSC6910)** if the load handover is successful. After load handover penalty, the signal level of the target cell changes. Therefore, the network characteristics need to be readjusted.

If **HOPRIOMODEN (BSC6900, BSC6910)** is set to **YES(Yes)**: In handover algorithm II, all handover decisions are made after handover penalty, basic ranking, network characteristics adjustment, and forced handovers.

3.3 Forced Handover

If a forced handover is triggered, no subsequent handover decision is performed.

A forced handover is triggered in the following scenarios:

- If no TCH is available in the serving cell during the MS access process, the directed retry procedure is performed when ***DIRECTRYEN (BSC6900, BSC6910)*** is set to **YES(Yes)**.
- When a BTS is under maintenance, the MSs served by the BTS should be handed over to the cells controlled by another functional BTS. This ensures that no call drop occurs during the BTS maintenance.
- Due to network operation and maintenance, MSC speech version adjustment, and channel adjustment and preemption, the BSC needs to adjust radio resources, which will trigger forced handovers.

Direct Retry

When the MS initiates a call, after the BSC receives an ASSIGN REQ message from the MSC, the BSC determines an assignment mode based on the load of the serving cell.

NOTE

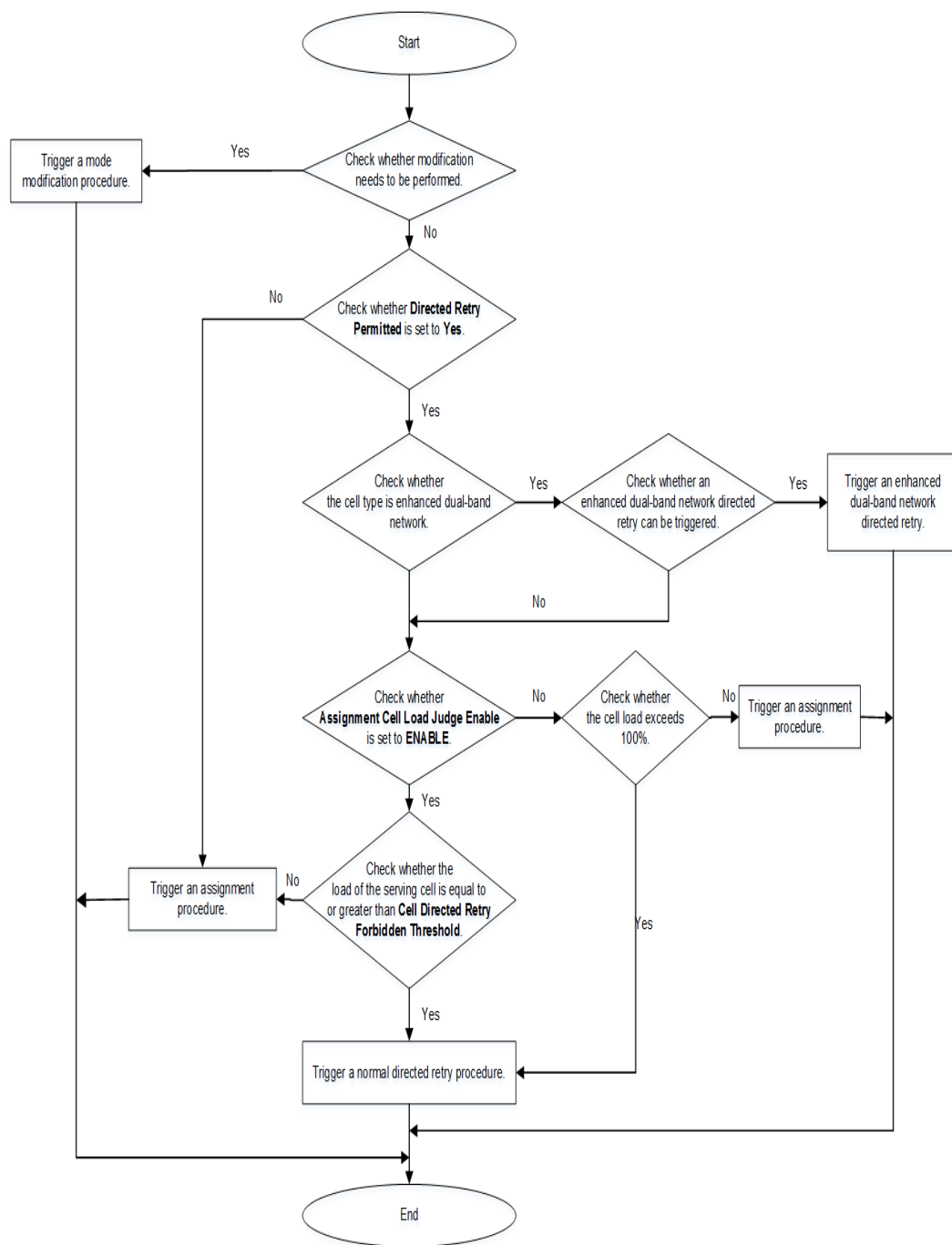
Assignment mode is categorized into normal assignment procedure, mode modification procedure, and directed retry procedure. The commands issued by the BSC vary according to the procedure.

- For a normal assignment procedure, the BSC activates a channel and issues a channel assignment command.
- For a mode modification procedure, the BSC issues a mode modification command.
- For a directed retry procedure, the BSC issues a handover command.

If the serving cell is so overloaded that new calls cannot be admitted or admitting new calls will affect ongoing services, the BSC triggers a directed retry procedure. By using the directed retry, the MS is handed over to the target cell and part of the traffic in the serving cell is distributed to the target cell. This avoids traffic congestion in the serving cell. (GBFD-110607 Direct Retry)

Figure 3-1 shows the procedure for a directed retry decision.

Figure 3-1 Procedure for a directed retry decision



As shown in [Figure 3-1](#), directed retry is categorized into enhanced dual-band network directed retry and normal directed retry.

- **Enhanced Dual-Band Network Directed Retry**

In an enhanced dual-band network, two cells form a group and the MS camps on one of the two cells. After the directed retry is triggered, the MS is handed over to the other cell. For details on the conditions of triggering an enhanced dual-band network directed retry, see [3.4.7 Enhanced Dual-Band Network Handover](#) and [3.5.6 Enhanced Dual-Band Network Handover](#).

- **Target Cell Selection in a Normal Directed Retry Procedure**

The target cell must have the highest priority in the candidate cell list after handover preprocessing. In addition, the target cell must meet the following conditions:

- The serving cell cannot be selected as the target cell.
- Load of the candidate neighboring cells $\leq$ **DTLOADTHRED (BSC6900, BSC6910)**
- In handover algorithm II, serving cell level $<$ receive level of neighboring cells $<$ serving cell level + **DRHOLEVRANGE (BSC6900, BSC6910)**
- In handover algorithm I, receive level of neighboring cells $\geq$ **MINPWRLEVDIRTRY (BSC6900, BSC6910)**

If **DRTAGCELLSEL (BSC6900, BSC6910)** is set to **YES(Yes)**, the MS can be handed over to multiple target cells by using directed retry. If **DRTAGCELLSEL (BSC6900, BSC6910)** is set to **NO(No)**, the MS can be handed over to only one target cell. The number of target cells is controlled by the **HOTRYCNT (BSC6900, BSC6910)** parameter.

3.4 Handover Decision Based on Handover Algorithm I

This section describes the feature GBFD-110601 HUAWEI I Handover.

According to the emergency condition of an MS in the network, the handover decision based on handover algorithm I is made in the following order: quick handover, emergency handover, enhanced dual-band network handover, load handover, and normal handover.

Handover decision based on handover algorithm I involves the following procedures:

- Determines whether the serving cell meets the trigger conditions.
- Selects corresponding candidate cells.

In handover algorithm I, **HOOPTSEL (BSC6900, BSC6910)** specifies the preferred candidate cell.

- When **HOOPTSEL (BSC6900, BSC6910)** is set to **Pre_2G_Cell(Preference for 2G cell)**:
A neighboring GSM cell is preferred. If the candidate cell list contains suitable neighboring UMTS cells but no suitable neighboring GSM cells, a neighboring UMTS cell is selected.
- When **HOOPTSEL (BSC6900, BSC6910)** is set to **Pre_3G_Cell(Preference for 3G cell)**:
A neighboring UMTS cell is preferred. If the candidate cell list contains suitable neighboring GSM cells but no suitable neighboring UMTS cells, a neighboring GSM cell is selected.
- When **HOOPTSEL (BSC6900, BSC6910)** is set to **Pre_2G_CellThres(Preference 2G cell by thresh)**:
If the receive level of a candidate GSM cell is lower than or equal to the threshold specified by **HOPRETH2G (BSC6900, BSC6910)**, a neighboring UMTS cell is preferred.

If the trigger conditions for an emergency handover are met and there is at least one candidate cell, the emergency handover timer **NEWURGHOMININTV (BSC6900, BSC6910)** is started. Another emergency handover decision can be performed only when **NEWURGHOMININTV (BSC6900, BSC6910)** times out.

3.4.1 Quick Handover

This section describes the features GBFD-510103 Chain Cell Handover and GBFD-510102 Fast Move Handover.

Quick handover aims to increase the handover success rate of an MS moving at a high speed and to ensure the call continuity and low call drop rate. Quick handover applies to the scenario where an MS moves fast along an urban backbone road, a selected route, or a high-speed railroad.

Quick Handover Types

Quick handover consists of frequency offset handover and fast move handover.

- Frequency offset handover
Whether the MS moves away from the serving cell is determined based on the frequency offset information provided by an MS moving at a high speed. Frequency offset handover decision is made according to the uplink/downlink receive level of the serving cell and the path loss of neighboring cells.
- Fast move handover
Fast move handover decision is made according to the path loss of neighboring cells.

For quick handover, the handover response speed is enhanced by:

- Accurately calculating the move speed of the MS
- Lifting the restriction on the interval between handover decisions
- Reducing the number of MRs for the handover decision
- Introducing the alpha filtering

Quick Handover Preparation

The preparation for quick handover involves the following aspects:

- Frequency offset is decoded from the MR.
Frequency offset of the MS is obtained from the uplink MR that the BTS sends to the BSC.
- Alpha filtering is performed on the MR. The value of **SCELLFILTER (BSC6900, BSC6910)** is used as the filter length for the serving cell, and the value of **NCELLFILTER (BSC6900, BSC6910)** is used as the filter length for a neighboring cell.
- If the number of MRs consecutively discarded in the serving cell is greater than the value of **IGNOREMRNUM (BSC6900, BSC6910)**, the BSC clears the MR queue for the serving cell and does not perform a handover decision.
- If the number of MRs consecutively discarded in a neighboring cell is greater than the value of **IGNOREMRNUM (BSC6900, BSC6910)**, the BSC clears the MR queue for the neighboring cell and does not perform a handover decision.

Trigger Conditions

During the handover decision, whether the trigger conditions for a frequency offset handover are met is determined first. When the BTS cannot send the frequency offset information or the reported frequency offset information is invalid, a fast move handover is triggered, provided that other conditions for a frequency offset handover are met.

If **QUICKHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**, the trigger conditions for a quick handover are as follows:

- The MS moves away from the serving cell (the frequency offset in the measurement result is a negative value) and the move speed of the MS is greater than **MOVESPEEDTHRES (BSC6900, BSC6910)**.
- The filtered uplink level of the serving cell is less than **HOUPTRIGE (BSC6900, BSC6910)**.
- The compensated downlink level of the serving cell is less than **HODOWNTRIGE (BSC6900, BSC6910)**.
- The difference between the path loss of a chain neighboring cell and that of the serving cell is greater than the predefined threshold. The predefined threshold is equal to the value of **HOOFFSET (BSC6900, BSC6910)** (in the **SET GCELLHOF** command for an internal neighboring cell or in the **MOD GEXT2GCELL** command for an external neighboring cell) plus the value of **ADJHOOFFSET (BSC6900, BSC6910)** for the serving cell.

The trigger conditions for a quick handover are as follows:

- If all the following three conditions are met, the handover decision is made as follows:
 - If the conditions for determining a frequency offset handover are met for a period longer than **AFCHOLASTTIME (BSC6900, BSC6910)** during **AFCHOSTATTIME (BSC6900, BSC6910)**, a frequency offset handover is triggered.
 - If **QUICKPBGTHOEN (BSC6900, BSC6910)** is set to **YES(Yes)** and the first condition is not met, a fast move handover is triggered.
 - If **QUICKPBGTHOEN (BSC6900, BSC6910)** is set to **NO(No)**, a fast move handover is not triggered.
- If all the preceding three conditions are not met, a quick handover is not triggered.
- When **MSLEVSTRQPBGT (BSC6900, BSC6910)** is set to **YES(Yes)**, the BSC triggers a fast handover only when an MS moves away from the BTS.

Target Cell Selection

The target cell must be a chain neighboring cell of the serving cell. The target cell can be selected by setting **ISCHAINNCELL (BSC6900, BSC6910)**. If **HODIRFORECASTEN (BSC6900, BSC6910)** is set to **YES**, a neighboring cell in the moving direction of the MS is selected preferentially.

To forecast the moving direction of the MS, when **HODIRFORECASTEN (BSC6900, BSC6910)** is set to **YES**, the direction of a chain neighboring cell (A or B) compared with the serving cell is specified by **CHAINNCELLTYPE (BSC6900, BSC6910)**. If the number of times that the MS is handed over to neighboring cells

in the same direction (B for example) is greater than or equal to **HODIRLASTTIME (BSC6900, BSC6910)** when the handover time reaches **HODIRSTATIME (BSC6900, BSC6910)**, the MS is inferred to be moving towards the B direction. Subsequently, the MS is preferentially handed over to the neighboring cell whose **CHAINNCELLTYPE (BSC6900, BSC6910)** is B.

Limitations

The limitations on quick handover are as follows:

- The serving cell cannot be selected as the target cell.
- The candidate cells for quick handover must be chain neighboring cells of the serving cell.
- If quick handover penalty is enabled, after a quick handover is successful, the penalty is performed on the original cell during the penalty time to prevent an immediate handover back to the original cell. Quick handover penalty is determined jointly by **QuickHoPunishSw (BSC6910, BSC6900)**, **TIMEPUNISH (BSC6900, BSC6910)**, and **HOPUNISHVALUE (BSC6900, BSC6910)**. The penalty time and penalty value are specified by **TIMEPUNISH (BSC6900, BSC6910)** and **HOPUNISHVALUE (BSC6900, BSC6910)**.

3.4.2 TA Handover

TA handover is a type of emergency handover. The TA handover decision is made according to the TA value reported by the MS.

The TA value of a normal cell ranges from 0 to 63 and that of an underlaid subcell of a dual-timeslot cell ranges from 0 to 219. The unit of TA values is bit. The transmission distance over the air interface corresponding to the delay of one bit is 553.5 m. When the TA value is 63, the transmission distance over the air interface is 35 km.

Trigger Conditions

TA handover is triggered when the following conditions are met:

- **TAHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**.
- Filtered TA value in the MR provided by the MS is greater than or equal to **TALIMIT (BSC6900, BSC6910)**.

The TA handover can be triggered only when the preceding two conditions are met simultaneously.

NOTE

From the perspective of the trigger conditions for a TA handover, TA can be regarded as a limitation to the size of a cell.

Target Cell Selection

The target cell must have the highest priority in the candidate cell list after handover preprocessing. In addition, the target cell must meet the following conditions:

- The serving cell cannot be selected as the target cell.
- If **TALIMIT (BSC6900, BSC6910)** of a co-site neighboring cell is less than or equal to the **TALIMIT (BSC6900, BSC6910)** of the serving cell, a handover to the neighboring cell is prohibited.

If the trigger conditions for a TA handover are met but the candidate GSM cells are not suitable, the following operations are performed:

- When a neighboring UMTS cell is available, if **INTERRATOUTBSCHOEN (BSC6900, BSC6910)** is set to **YES(Yes)** and the MS supports the GSM/UMTS inter-RAT handover, the GSM/UMTS inter-RAT handover is performed.
- When no neighboring UMTS cell is available, if **INTERRATOUTBSCHOEN (BSC6900, BSC6910)** is set to **NO(No)** or the MS does not support the GSM/UMTS inter-RAT handover, the decision on whether to initiate another type of emergency handover is made.

Limitations

After the TA handover is successful, the penalty is performed on the original cell. During **TIMETAPUNISH (BSC6900, BSC6910)**, **SSTAPUNISH (BSC6900, BSC6910)** is subtracted from the level of the original cell to prevent an immediate handover back to the original cell.

3.4.3 BQ Handover

BQ handover is a type of emergency handover in which the system makes the decision based on the uplink/downlink receive quality on the Um interface.

The receive quality is measured in bit error rate (BER). The BSC measures the quality of a radio link based on the quality level in the MR. A rise in BER may result from an excessively low signal power or channel interference.

Trigger Conditions

If **BQHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**, the BQ handover is triggered when either of the following conditions is met:

- The uplink receive quality is greater than or equal to the uplink receive quality threshold of the serving cell.
- The downlink receive quality is greater than or equal to the downlink receive quality threshold of the serving cell.

The BQ handover is triggered when either of the preceding conditions is met.

The parameters for specifying the uplink and downlink receive quality thresholds are as follows:

- For non-AMR calls, the parameter for specifying the uplink receive quality threshold is **ULQUALIMIT (BSC6900, BSC6910)**, and the parameter for specifying the downlink receive quality threshold is **DLQUALIMIT (BSC6900, BSC6910)**.
- For AMR FR calls, the parameter for specifying the uplink receive quality threshold is **ULQUALIMITAMRFR (BSC6900, BSC6910)** and the parameter for specifying the downlink receive quality threshold is **DLQUALIMITAMRFR (BSC6900, BSC6910)**.

- For AMR HR calls, the parameter for specifying the uplink receive quality threshold is **ULQUALIMITAMRHR (BSC6900, BSC6910)** and the parameter for specifying the downlink receive quality threshold is **DLQUALIMITAMRHR (BSC6900, BSC6910)**.

Target Cell Selection

The target cell that has the highest priority in the candidate cell list after handover preprocessing must meet the conditions listed below:

If the target cell is a neighboring cell:

- If **BqNCellAbsThresSw (BSC6900, BSC6910)** is set to **ON(On)**, filtered downlink receive level of the target cell > **HOTHRES (BSC6900, BSC6910)** of the neighboring cell + (**BQMARGIN (BSC6910, BSC6900)** – 64)
- If **BqNCellAbsThresSw (BSC6900, BSC6910)** is set to **FF(Off)**, filtered downlink receive level of the target cell > filtered downlink receive level of the serving cell after compensation + (**INTERCELLHYST (BSC6900, BSC6910)** of the serving cell configured for the neighboring cell – 64) – (**BQMARGIN (BSC6900, BSC6910)** – 64)

NOTE

If there is only one serving cell in the candidate cell list:

- If **INTRACELLSINUSEREN (BSC6900, BSC6910)** is set to **NO**, no intra-cell handover is allowed within the time specified by **BANTIME (BSC6900, BSC6910)** when multiple intra-cell handovers are performed consecutively.
- If **INTRACELLSINUSEREN (BSC6900, BSC6910)** is set to **YES**, no quality-related intra-cell handovers (such as BQ handover, interference handover, TCHH-to-TCHF handover, and tight BCCH handover) are allowed within the time specified by **BANTIME (BSC6900, BSC6910)** when the same MS consecutively performs multiple such types of handovers within a cell.

If the target cell is not a neighboring cell:

- An MS is handed over to the serving cell (priority: different frequency band channel > different frequency set channel > different TRX channel > different timeslot channel) when the following conditions are met:
 - INTRACELLHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**.
 - BADQUALHOOPTALLOW (BSC6900, BSC6910)** is set to **NO(No)**.
 - The serving cell is not in the intra-cell handover penalty state.
- An MS is not handed over to the serving cell if either of the following conditions is met:
 - Both **INTRACELLHOEN (BSC6900, BSC6910)** and **BADQUALHOOPTALLOW (BSC6900, BSC6910)** are set to **YES(Yes)**.
 - INTRACELLHOEN (BSC6900, BSC6910)** is set to **NO(No)**.

Limitations

If the trigger conditions for a BQ handover are met but the candidate GSM cells are not suitable, the following operations are performed:

- When a neighboring UMTS cell is available, if **INTERRATOUTBSCHOEN (BSC6900, BSC6910)** is set to **YES(Yes)** and the MS supports the GSM/UMTS inter-RAT handover, the GSM/UMTS inter-RAT handover is performed.

- When no neighboring UMTS cell is available, if **INTERRATOUTBSCHOEN (BSC6900, BSC6910)** is set to **NO(No)** or the MS does not support the GSM/UMTS inter-RAT handover, the decision on whether to initiate another type of emergency handover is made.

After the BQ handover is successful, the penalty is performed on the original cell. During **TIMEBQPUNISH (BSC6900, BSC6910)**, **SSBQPUNISH (BSC6900, BSC6910)** is subtracted from the level of the original cell to prevent an immediate handover back to the original cell.

3.4.4 Rapid Level Drop Handover

Rapid level drop handover is a type of emergency handover.

In edge handover and PBGT handover, the mean value filtering and P/N decision methods are not responsive to short-period rapid level drop. Therefore, to solve the rapid level drop problem, the finite impact response filtering can be performed on the original receive level. This filtering method is responsive to the rapid level drop based on the drop slope of the original receive level.

Trigger Conditions

If **RXQCKFALLHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**, the trigger conditions for a rapid level drop handover are as follows:

- Filtered uplink level < **ULEDGETHRES (BSC6900, BSC6910)**
- $A1 \times C(nt) + A2 \times C(nt - t) + A3 \times C(nt - 2t) + \dots + A8 \times C(nt - 7t) < B$
where,
 - A1 indicates the value of **FLTPARAA1 (BSC6900, BSC6910)**.
 - A2 indicates the value of **FLTPARAA2 (BSC6900, BSC6910)**.
 - A3 indicates the value of **FLTPARAA3 (BSC6900, BSC6910)**.
 - A4 indicates the value of **FLTPARAA4 (BSC6900, BSC6910)**.
 - A5 indicates the value of **FLTPARAA5 (BSC6900, BSC6910)**.
 - A6 indicates the value of **FLTPARAA6 (BSC6900, BSC6910)**.
 - A7 indicates the value of **FLTPARAA7 (BSC6900, BSC6910)**.
 - A8 indicates the value of **FLTPARAA8 (BSC6900, BSC6910)**.B indicates the value of **FLTPARAB (BSC6900, BSC6910)**.

Target Cell Selection

The target cell must have the highest priority in the candidate cell list after handover preprocessing. In addition, the target cell must meet the following conditions:

- The target cell has a higher priority than the serving cell.
- The serving cell cannot be selected as the target cell.

If the trigger conditions for a rapid level drop handover are met but the candidate 2G cells are not suitable, the following operations are performed:

- When a neighboring UMTS cell is available, if ***INTERRATOUTBSCHOEN (BSC6900, BSC6910)*** is set to **YES(Yes)** and the MS supports the GSM/UMTS inter-RAT handover, the GSM/UMTS inter-RAT handover is performed.
- When no neighboring UMTS cell is available, if ***INTERRATOUTBSCHOEN (BSC6900, BSC6910)*** is set to **NO(No)** or the MS does not support the GSM/UMTS inter-RAT handover, the decision on whether to initiate another type of emergency handover is made.

3.4.5 Interference Handover

In handover algorithm I, if ***INTERHOOPTALLOW (BSC6900, BSC6910)*** is set to **NO(No)**, interference handover is regarded as a type of emergency handover. If ***INTERHOOPTALLOW (BSC6900, BSC6910)*** is set to **YES(Yes)**, interference handover is not regarded as a type of emergency handover and the decision on interference handover is made after a decision on AMR handover is made.

Interference handover helps protect calls affected by interference and reduce network interference. It is applicable in scenarios with interference.

In handover algorithm I, the difference between interference handover and BQ handover is as follows: In BQ handover, the bad signal quality resulting from both coverage and interference is checked. In interference handover, the bad signal quality resulting from coverage is not checked.

Trigger Conditions

If ***INTERFHOEN (BSC6900, BSC6910)*** is set to **YES(Yes)**, the interference handover is triggered when either of the following conditions is met:

- The filtered value of uplink receive quality is greater than the specified receive quality threshold at the current uplink receive level.
- The filtered value of downlink receive quality is greater than the specified receive quality threshold at the current downlink receive level.

The interference handover is triggered if either of the previous conditions is met.

The parameters for specifying the uplink and downlink receive quality thresholds are as follows:

- For non-AMR FR calls, the parameter for specifying the receive quality threshold is ***RXQUALn***, where $1 \leq n \leq 12$.

NOTE

Non-AMR FR calls consist of FR, HR, EFR, and AMR HR calls.

- For AMR FR calls, the parameters for specifying the receive quality threshold are ***RXQUALn*** ($1 \leq n \leq 12$) and ***RXLEVOFF (BSC6900, BSC6910)***.
 - If $n=1$, the receive quality threshold is ***RXQUAL1 (BSC6900, BSC6910)***.
 - If $2 \leq n \leq 12$, and the filtered value of uplink and downlink receive quality - ***RXLEVOFF (BSC6900, BSC6910)*** > 0 , use the value of the filtered value of uplink receive quality - ***RXLEVOFF (BSC6900, BSC6910)***.
 - If $2 \leq n \leq 12$, and the filtered value of uplink and downlink receive quality - ***RXLEVOFF (BSC6900, BSC6910)*** ≤ 0 , use the value of the filtered value of uplink receive quality.

Target Cell Selection

In handover algorithm I, the target cell must have the highest priority in the candidate cell list. In addition, the target cell must meet the following conditions:

- If **INTRACELLHOEN (BSC6900, BSC6910)** is set to **YES(Yes)** and the intra-cell handover penalty timer expires, the serving cell can be selected as the target cell.
- If both **INTRACELLHOEN (BSC6900, BSC6910)** and **INTERHOOPTALLOW (BSC6900, BSC6910)** are set to **YES(Yes)**, only the serving cell can be selected as the target cell.
- If **INTRACELLHOEN (BSC6900, BSC6910)** is set to **YES(Yes)** and **INTERHOOPTALLOW (BSC6900, BSC6910)** is set to **NO(No)**, either the serving cell or a neighboring cell can be selected as the target cell. If **INTRACELLHOEN (BSC6900, BSC6910)** is set to **NO(No)**, the serving cell cannot be selected as the target cell.

NOTE

If **INTRACELLSINUSEREN (BSC6900, BSC6910)** is set to **NO(No)**, no intra-cell handover is allowed within the time specified by **BANTIME (BSC6900, BSC6910)** when multiple intra-cell handovers are performed consecutively. If **INTRACELLSINUSEREN (BSC6900, BSC6910)** is set to **YES(Yes)**, no quality-related intra-cell handovers (such as BQ handover, interference handover, TCHH-to-TCHF handover, and tight BCCH handover) are allowed within the time specified by **BANTIME (BSC6900, BSC6910)** when the same MS consecutively performs multiple such types of handovers.

- If the filtered level of a neighboring cell after handover penalty $\geq$ **HOTHRES (BSC6900, BSC6910)** of the neighboring cell + **INTELEVHOHYST (BSC6900, BSC6910)** – 64, this neighboring cell can serve as the target cell.

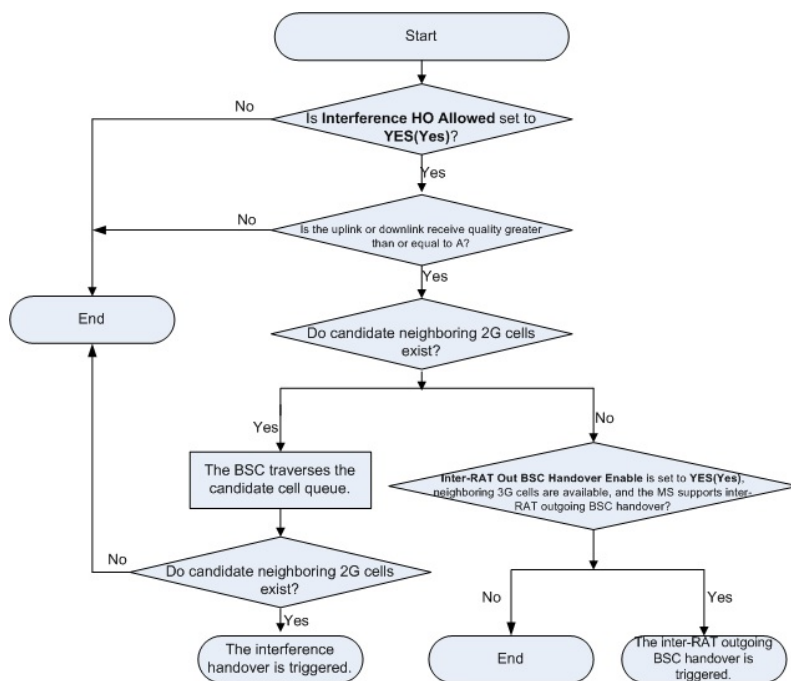
If the trigger conditions for an interference handover are met but the candidate 2G cells are not suitable, the following operations need to be performed:

- When a neighboring UMTS cell is available, if **INTERRATOUTBSCHOEN (BSC6900, BSC6910)** is set to **YES(Yes)** and the MS supports the GSM/UMTS inter-RAT handover, the GSM/UMTS inter-RAT handover is performed.
- When no neighboring UMTS cell is available, if **INTERRATOUTBSCHOEN (BSC6900, BSC6910)** is set to **NO(No)** or the MS does not support the GSM/UMTS inter-RAT handover, the decision on whether to initiate another type of emergency handover is made.

Interference Handover Process

[Figure 3-2](#) shows the interference handover process.

Figure 3-2 Interference handover process



NOTE

In Figure 3-2, "A" indicates the receive quality threshold under the current uplink or downlink signal level.

- For non-AMR FR calls, "A" ranges from **RXQUAL1 (BSC6900, BSC6910)** to **RXQUAL12 (BSC6900, BSC6910)**. Non-AMR FR calls consist of FR, HR, EFR, and AMR HR calls.
- For AMR FR calls, "A" equals **RXQUAL1 (BSC6900, BSC6910)** or the sum of a value from **RXQUAL2 (BSC6900, BSC6910)** to **RXQUAL12 (BSC6900, BSC6910)** plus **RXLEVOFF (BSC6900, BSC6910)**.

3.4.6 No-DL-MR Handover

No-DL-MR handover is performed on the basis of the uplink quality. The purpose is to ensure the call continuity and minimize the possibility of call drops.

No-DL-MR handover is generally caused by adverse radio environment on the uplink. In this case, the requirements of the filtering algorithm cannot be met, so other handover decisions cannot be performed.

Trigger Conditions

In handover algorithm I, the trigger conditions for a no-DL-MR handover are as follows:

- NODLMRHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**.
- There is no downlink information in the MR of the call.
- The filtered value of uplink quality is greater than or equal to **NODLMRHOQUALLIMIT (BSC6900, BSC6910)**.
- The number of lost downlink MRs is less than **NODLMRHOALLOWLIMIT (BSC6900, BSC6910)**.

- For TCH, the number of saved MRs with uplink quality value is greater than **DATAQUAFLTLEN (BSC6900, BSC6910)**. For SDCCH, the number of saved MRs with uplink quality value is greater than **SIGQUAFLTLEN (BSC6900, BSC6910)**.
In addition, the BSC triggers a no-DL-MR handover when all the preceding conditions are met and MRs of the serving cell keep lost for **NODLMRHOLASTTIME (BSC6900, BSC6910)** within **NODLMRHOSTATTIME (BSC6900, BSC6910)**.

Target Cell Selection

In handover algorithm I, the conditions for selecting the target cell are as follows:

- The ranked neighboring cells recorded in the last complete MR are saved as candidate cells.
- A neighboring cell is preferentially selected as the target cell. If **BADQUALHOOPTALLOW (BSC6900, BSC6910)** is set to **YES(Yes)**, only a neighboring cell can be selected as the target cell.
- If no neighboring cell is available, the serving cell is selected as the target cell. If **BADQUALHOOPTALLOW (BSC6900, BSC6910)** is set to **NO(No)**, the serving cell is selected as the target cell.

3.4.7 Enhanced Dual-Band Network Handover

Enhanced dual-band network handover is performed based on the traffic volume of the OL and UL cells and based on the receive level.

Enhanced dual-band network handover is classified into the following types:

- Handover due to high load in the UL cell
- Handover due to low load in the UL cell
- Handover due to MS movement to the border of the OL cell

Trigger Conditions for a Handover Due to High Load in the Underlaid Cell

The trigger conditions for a handover due to high load in the UL cell are as follows:

- The two cells are in the enhanced dual-band network and **OUTLOADHOENABLE (BSC6900, BSC6910)** is set to **YES(Yes)**.
- The MS supports the frequency band on which the OL cell operates.
- The handover due to high load in the UL cell is performed only on TCHs.
- The load in the UL cell is greater than or equal to **OUTGENOVERLDTHRED (BSC6900, BSC6910)**.
- The load in the OL cell is less than **INNERIOVERLDTHRED (BSC6900, BSC6910)**.
- The system traffic volume is less than or equal to **EDBSYSFLOWLEV (BSC6900, BSC6910)**.
- The receive level in the overlaid subcell is greater than or equal to **INTOINNREXLEVTHRED (BSC6900, BSC6910)**.

- The receive level in the underlaid subcell is greater than or equal to **INTOINNREXLEVTHRED (BSC6900, BSC6910)**.
- The receive level in the underlaid subcell minus the receive level of the optimal neighboring cell is greater than the sum of **ATCBTHRED (BSC6900, BSC6910)** and **ATCBHYST (BSC6900, BSC6910)**.

When all the preceding conditions are met, the handover due to high load in the UL cell is triggered.

If the load of the UL subcell in the cell is greater than or equal to **OUTSERIOVERLDTHRED (BSC6900, BSC6910)**, the handover margin is adjusted in a period of **OUTLOADHOPERIOD (BSC6900, BSC6910)** subtracted by **OUTLOADHOMODPERI (BSC6900, BSC6910)**. The step length for handover margin adjustment is specified by **OUTLOADHOSTEP (BSC6900, BSC6910)**.

Trigger Conditions for a Handover Due to Low Load in the Underlaid Cell

The trigger conditions for a handover due to low load in the UL cell are as follows:

- The load in the underlaid subcell is smaller than **OUTLOWLOADTHRED (BSC6900, BSC6910)**.
- The system traffic volume is less than or equal to **EDBSYSFLOWLEV (BSC6900, BSC6910)**.
- The current call is within the handover margin and the receive level is greater than or equal to **OUTINNREXLEVTHRED (BSC6900, BSC6910)**.

When all the preceding conditions are met, the handover due to low load in the underlaid subcell is triggered.

If the load of the UL subcell in the cell is less than **OUTLOWLOADTHRED (BSC6900, BSC6910)** for a specified period, the handover margin is adjusted in a period of **INNLOADHOPERI (BSC6900, BSC6910)**. The step length for handover margin adjustment is specified by **INNLOADHOSTEP (BSC6900, BSC6910)**.

Trigger Conditions for a Handover Due to MS Movement to the Border of the Overlaid Cell

The trigger conditions for a handover due to MS movement to the border of the OL cell are as follows:

- $SS(S) < Thdouter$
- $SS(u) - SS(n) < ATCB_THRD - ATCB_HYST$
where,
 - $SS(s)$: specifies the filtering compensated downlink receive level in the serving cell.
 - $Thdouter$: specifies **OUTINNREXLEVTHRED (BSC6900, BSC6910)**.
 - $SS(u)$: specifies the downlink level (power compensation is performed on the downlink level based on the measurement) of the UL subcell where the call is originated. If the $SS(u)$ value cannot be obtained, the decision of enhanced dual-band network handover is not performed and the decision condition is met by default.
 - $SS(n)$: The best neighboring cell is the one whose measured BCCH level is the highest among neighboring cells. $SS(n)$ is the level value of such an

optimum neighboring cell that is on the same frequency band and layer with the UL subcell but is not co-sited with the UL subcell. If such a neighboring cell is not available, the value of SS(n) is -110 dBm.

- ATCB_THRD: specifies **ATCBTHRED (BSC6900, BSC6910)**.
- ATCB_HYST: specifies **ATCBHYST (BSC6900, BSC6910)**.

Handover due to MS movement to the border of the OL cell is triggered if either of the preceding conditions is met.

NOTE

- In the Adapter distance To Cell Border (ATCB) handover algorithm, the border between the OL and UL subcells is determined according to the signal strength of the serving cell and that of neighboring cells. If $SS(s) = SS(n)$, the system considers that the MS is located at the border of the UL cell. If $SS(s) - SS(n) > ATCB_THRD$, the system considers that the MS is located in the coverage area of the OL subcell. The coverage area of the OL subcell is determined according to different networking and coverage conditions of the existing network. In addition, the OL subcell of the serving cell and the OL subcell of the neighboring cells will not overlap regardless of the distance between BTSs.
- The handover margin specifies the range of signal level. In the case of OL/UL load handover on the enhanced dual-band network, the MSs whose downlink levels are within the handover margin are handed over level by level.

Target Cell Selection

The requirements for target cell selection in the enhanced dual-band network are as follows:

- For the handover due to high load in the UL cell, the MS must be handed over to the OL cell.
- For the handover due to low load in the UL cell, the MS must be handed over to the UL cell.
- For the handover due to MS movement to the border of the OL cell, the MS is handed over to the neighboring cell that ranks first among neighboring cells. The MS should not be handed over to the cell that ranks after the serving cell. Generally, the target cell is the UL cell. The target cell can also be another neighboring cell.

Limitations

The limitations on the handover due to high load in the UL cell are as follows:

- If the cell where the call is located is on an enhanced dual-band network, **CELLINEXTP (BSC6900, BSC6910)** is set to **Extra(Extra)**.
- The **OUTLOADHOENABLE (BSC6900, BSC6910)** parameter is set.
- The maximum range of the handover margin is from 63 to **INTOINNREXLEVTHRED (BSC6900, BSC6910)**. The MS with the highest receive level is handed over first.

The limitations on the handover due to low load in the UL cell are as follows:

- If the cell where the call is located is on an enhanced dual-band network, **CELLINEXTP (BSC6900, BSC6910)** is set to **Inner(Inner)**.
- The **INNLOADHOEN (BSC6900, BSC6910)** parameter is set.

- The maximum range of the handover margin is from 63 to **OUTINNREXLEVTHRED (BSC6900, BSC6910)**. The MS with the lowest receive level is handed over first.

The limitations on the handover due to MS movement to the border of the OL cell are as follows:

- If the cell where the call is located is on an enhanced dual-band network, **CELLINEXTP (BSC6900, BSC6910)** is set to **Inner(Inner)**.

Impact of the Enhanced Dual-Band Network Handover on the Existing Algorithm

The impact of the enhanced dual-band network handover on the existing algorithm is as follows:

- On the enhanced dual-band network, the MS should not be handed over to a cell in the same UL/OL cell group when the load handovers between the OL cell and the UL cell (specified by **OUTLOADHOENABLE (BSC6900, BSC6910)** and **INNLOADHOEN (BSC6900, BSC6910)**) are allowed. This is to prevent a load handover of a normal cell from colliding with a load handover between the OL cell and the UL cell on the network.
- The PBGT handover algorithm may cause inter-cell handovers. Therefore, to avoid ping-pong handovers, PBGT handovers are restricted in either of the following conditions:

When **INNLOADHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**, PBGT handovers from the underlaid subcell to the overlaid subcell are prohibited.

When **OUTLOADHOENABLE (BSC6900, BSC6910)** is set to **YES(Yes)**, PBGT handovers from the overlaid subcell to the underlaid subcell are prohibited.

3.4.8 Load Handover

In the network, some cells carry heavy load whereas the overlapping upper-layer cells and the neighboring cells may carry light load. The load handover is required to balance the load of these cells.

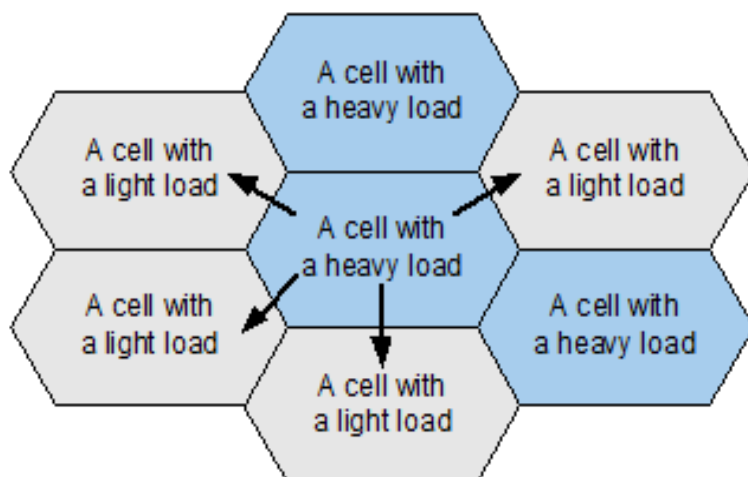
In a load handover procedure, some load in heavy-load cells is switched to light-load cells. Meanwhile, the load in neighboring cells is not switched to heavy-load cells.

Load handover can be performed between cells at different layers. [Figure 3-3](#) shows the details.

NOTE

For details about the inter-RAT load handover, see *Interoperability Between GSM and WCDMA Feature Parameter Description*.

Figure 3-3 Load handover between cells



To perform load sharing, increase **DLEDGETHRES (BSC6900, BSC6910)** to enable the edge handover area to shrink towards the center of the serving cell so that the load at the edge of a cell is switched to a neighboring cell with light load.

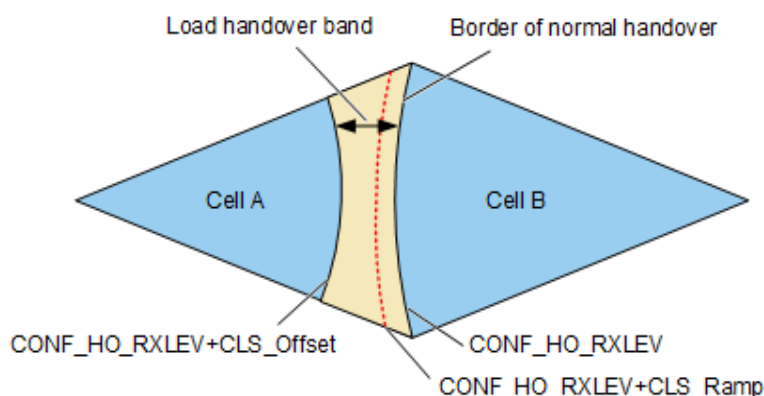
Whether a cell carries heavy load or light load is determined by the traffic volume in the cell, that is, whether the traffic volume (generally TCH usage) in the cell exceeds the preset threshold. The TCH usage is calculated using the following formula:

$$\text{TCH usage} = \frac{(\text{Number of occupied TCHFs} \times 2 + \text{Number of occupied TCHHs})}{(\text{Number of available TCHFs} \times 2 + \text{Number of available TCHHs})}$$
Where, number of available TCHHs is the sum of idle and occupied TCHHs and number of available TCHFs is the sum of idle and occupied TCHFs.

- If the traffic volume in a cell is greater than **TRIGTHRES (BSC6900, BSC6910)**, the load in this cell is heavy. The load handover algorithm needs to be enabled.
- If the traffic volume in a cell is less than **LoadAccThres (BSC6900, BSC6910)**, the load in this cell is light and the cell can receive load from the heavy-load cells.
 - Parameter for internal GSM cells: Load handover Load Accept Threshold **LoadAccThres (BSC6900, BSC6910)**
 - Parameter for external GSM cells: Load handover Load Accept Threshold **LoadAccThres (BSC6900, BSC6910)**
 - Parameter for external UMTS cells: Load handover Load Accept Threshold **LoadAccThres (BSC6900, BSC6910)**

Load handover may lead to many handovers. Therefore, the load of the system CPU should be considered before load handover is performed. In other words, the system traffic volume should be taken into account. In addition, to prevent too many MSs from being handed over at a time, load handover is performed step by step. In other words, the edge handover threshold is increased on the basis of **LOADHOSTEP (BSC6900, BSC6910)** (CLS_Ramp) and **LOADHOPERIOD (BSC6900, BSC6910)** (CLS_Period). When the increase in the edge handover threshold equals **LOADOFFSET (BSC6900, BSC6910)** (CLS_Offset), the edge handover threshold is not increased any more. For details, see [Figure 3-4](#).

Figure 3-4 Load handover



Trigger Conditions

If **LoadHoEn** (*BSC6900, BSC6910*) is set to **YES(Yes)**, the trigger conditions for a load handover are as follows:

- The CPU usage of the system is less than or equal to **SYSFLOWLEV** (*BSC6900, BSC6910*).
- The current load of the serving cell is greater than or equal to **TRIGTHRES** (*BSC6900, BSC6910*).

Target Cell Selection

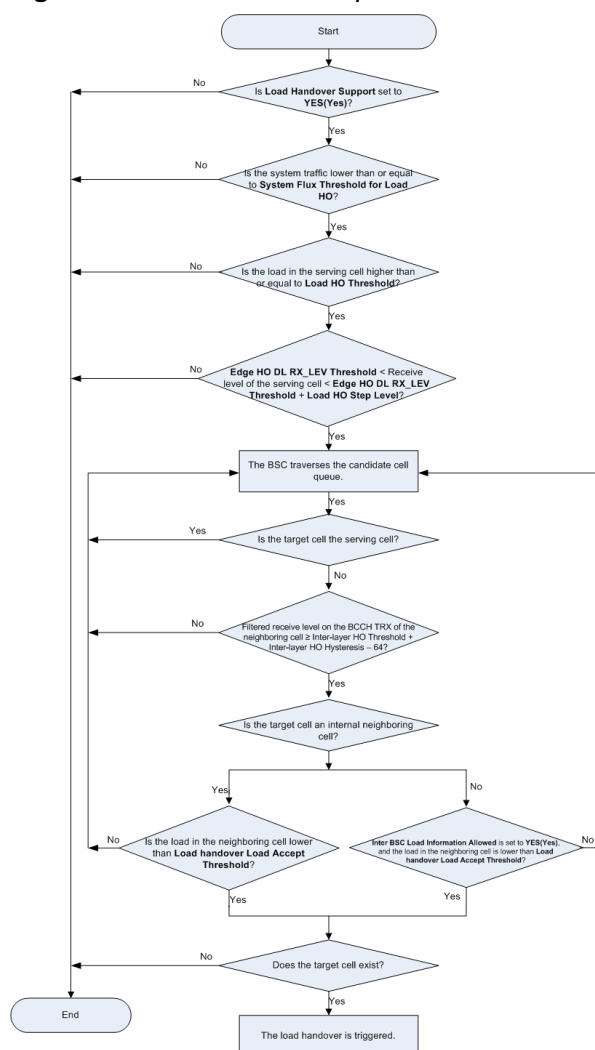
The conditions for selecting the target cell are as follows:

- Filtered receive level after handover penalty $\geq$ **HOTHRES** (*BSC6900, BSC6910*) + **INTELEVHOHYST** (*BSC6900, BSC6910*) - 64
- The serving cell cannot be selected as the target cell.
- If the target cell and the serving cell are in the same BSC, a load handover is performed when the current load of the target cell is less than **LoadAccThres** (*BSC6900, BSC6910*).
- If the target cell and the serving cell are not in the same BSC, a load handover is performed when the load of the target cell is less than **LoadAccThres** (*BSC6900, BSC6910*) and **OutBscLoadHoEn** (*BSC6900, BSC6910*) is set to **YES(Yes)**.

Load Handover Process

Figure 3-5 shows the load handover process.

Figure 3-5 Load handover process



Examples

The system assigns MSs to different load handover margins based on the downlink receive level. The load handover algorithm is used to hand over the MSs out of a cell step by step.

- The MSs in load handover margin 1 are handed over to the neighboring cells. Load handover margin 1 specifies the area where the downlink level ranges from **DLEDGETHRES (BSC6900, BSC6910)** to the sum of **DLEDGETHRES (BSC6900, BSC6910)** and **LOADHOSTEP (BSC6900, BSC6910)**.
- After a **LOADHOPERIOD (BSC6900, BSC6910)** elapses, the MSs in load handover margin 2 are handed over to the neighboring cells. Load handover margin 2 specifies the area where the downlink level ranges from **DLEDGETHRES (BSC6900, BSC6910)** to the sum of **DLEDGETHRES (BSC6900, BSC6910)** and (2 x **LOADHOSTEP (BSC6900, BSC6910)**).
- The load handover stops when the traffic volume in the cell is less than or equal to **TRIDTHRES (BSC6900, BSC6910)**.

The load handover is performed step by step to prevent call drops caused by a sudden increase in CPU load or the congestion in the target cell.

3.4.9 Enhanced Load Handover

Like the load handover, the enhanced load handover is used to balance load of cells in a network. Unlike the load handover, the enhanced load handover considers the handover quality and the load in the target cell before the handover is performed. In this way, the possibility of low level and congestion due to heavy load in the target cell after the handover is minimized. The enhanced load handover is applicable to the scenario where multiple base stations are located at the same place.

Trigger Conditions for an Enhanced Load Handover

If **LOADHOAD (BSC6900, BSC6910)** is set to **YES(Yes)**, the trigger conditions for an enhanced load handover are as follows:

- The CPU usage of the current system is less than or equal to **SYSFLOWLEV (BSC6900, BSC6910)**.
- The load of the serving cell is greater than **TRIGTHRES (BSC6900, BSC6910)**.
Here, the load of the serving cell is expressed in the percentage of the channels that are occupied. If the built-in PCU is used, the calculation method of the cell load depends on the setting of **LOADSTATYPE (BSC6900, BSC6910)**. The setting of the parameter determines whether the dynamic PDCHs that can be preempted are considered as occupied channels.
- When **LOADSTATYPE (BSC6900, BSC6910)** is set to **NODYNPDCH(Not count dynamic PDCH)**, the dynamic PDCHs that can be preempted are not considered in the cell load.
- When **LOADSTATYPE (BSC6900, BSC6910)** is set to **DYNPDCHASOCCUPY(Count dynamic PDCH(Occupy))**, the BSC considers the dynamic PDCHs that can be preempted as TCHs and calculates the load.
- When **LOADSTATYPE (BSC6900, BSC6910)** is set to **DYNPDCHASIDLE(Count dynamic PDCH(Idle))**, the dynamic PDCHs that can be preempted are considered as idle TCHs in the cell load.

The number of the dynamic PDCHs that can be preempted depends on the number of dynamic PDCHs and **DYNCHNPREEMPTLEV (BSC6900, BSC6910)**. The number of dynamic PDCHs is the total number of channels whose **CHTYPE (BSC6900, BSC6910)** is set to **FULLTCH**.

If the external PCU is used, the number of the dynamic PDCHs that can be preempted is always 0. The setting of **LOADSTATYPE (BSC6900, BSC6910)** is irrelevant to calculation of the cell load.

Target Cell Selection

When a candidate cell satisfying the following conditions is found and not a single MS within the range specified by **LOADHOUSRRATIO (BSC6900, BSC6910)** is handed over to the target cell, further search of the target cell is stopped and current traffic is handed over to the candidate cell. The conditions for selecting the target cell are as follows:

- The value of **LOADHOPBGMTMARGIN (BSC6900, BSC6910)** is not 0 and the path loss in the serving cell minus that in the target cell is greater than **LOADHOPBGMTMARGIN (BSC6900, BSC6910)**.

- The load of the target cell is less than **LoadAccThres (BSC6900, BSC6910)**.
- The receive level of the target cell is greater than **HOTHRES (BSC6900, BSC6910)** of the target cell + **INTELEVHOHYST (BSC6900, BSC6910)** of the handover from the serving cell to the target cell – 64.

Of all the MSs that are within the range specified by **LOADHOUSRRATIO (BSC6900, BSC6910)** and meet the preceding conditions, only one MS can initiate the handover at a time. This regulation prevents too many MSs from being handed over to the target cell at one time and avoids congestion in the target cell.

3.4.10 Edge Handover

Edge handover is performed on the basis of receive level. To trigger an edge handover, the receive level of the target cell should be at least one hysteresis value (specified by **INTERCELLHYST (BSC6900, BSC6910)** – 64) greater than the receive level of the serving cell.

Trigger Conditions

If **FRINGEHOEN (BSC6900, BSC6910)** is set to YES(Yes) and **EDOUTHOADEN (BSC6900, BSC6910)** is set to NO(No), the trigger conditions for an edge handover are as follows:

- Either of the following conditions is met:
 - Filtered downlink receive level of the serving cell after compensation < **DLEDGETHRES (BSC6900, BSC6910)**
 - Filtered uplink receive level of the serving cell after compensation < **ULEDGETHRES (BSC6900, BSC6910)**
- Receive level of a neighboring cell ≥ Downlink receive level of the serving cell after compensation + **INTERCELLHYST (BSC6900, BSC6910)** – 64

An edge handover is triggered when the P/N criterion is met, that is, when the previous conditions are met for **EDGELAST1 (BSC6900, BSC6910)** within **EDGESTAT1 (BSC6900, BSC6910)**.

If both **FRINGEHOEN (BSC6900, BSC6910)** and **EDOUTHOADEN (BSC6900, BSC6910)** are set to YES(Yes), the trigger conditions for an edge handover are as follows:

- Either of the following conditions is met:
 - Filtered downlink receive level of the serving cell after compensation < **DLEDGETHRES (BSC6900, BSC6910) + EDOUTHOOFFSET (BSC6900, BSC6910)** – 64
 - Filtered uplink receive level of the serving cell after compensation < **ULEDGETHRES (BSC6900, BSC6910) + EDOUTHOOFFSET (BSC6900, BSC6910)** – 64
- Receive level of a neighboring cell ≥ Downlink receive level of the serving cell after compensation + **INTERCELLHYST (BSC6900, BSC6910)** – 64

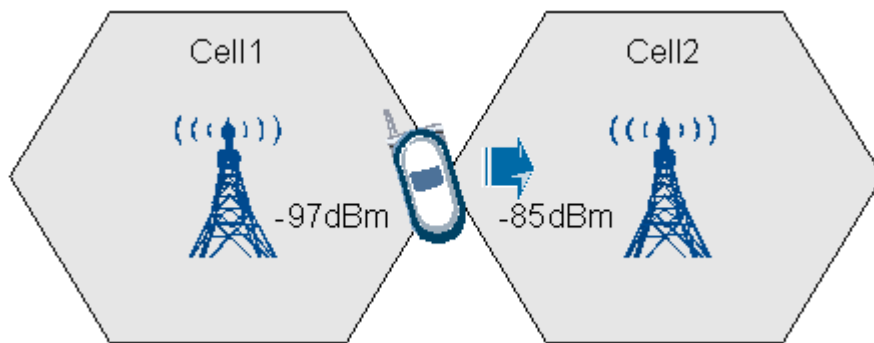
An edge handover is triggered when the P/N criterion is met, that is, when the previous conditions are met for **EDGELAST1 (BSC6900, BSC6910)** within **EDGESTAT1 (BSC6900, BSC6910)**.

 NOTE

If *FRINGEHOEN* (*BSC6900*, *BSC6910*) is set to **NO(No)**, *EDOUTHODEN* (*BSC6900*, *BSC6910*) and *EDOUTHOOFFSET* (*BSC6900*, *BSC6910*) are invalid.

Figure 3-6 shows an edge handover.

Figure 3-6 Edge handover



Target Cell Selection

The target cell must have the highest priority among the candidate cells. In addition, it must meet the following conditions:

- The serving cell cannot be selected as the target cell.
- After cells are ranked, the target cell must have a higher priority than the serving cell.

A cell becomes the target cell if the previous conditions are met for *EDGEADJLASTTIME* (*BSC6900*, *BSC6910*) within *EDGEADJSTATTIME* (*BSC6900*, *BSC6910*).

If the trigger conditions for an edge handover are met but the candidate GSM cells are not suitable, the following operations are performed:

- When a neighboring UMTS cell is available, if *INTERRATOUTBSCHOEN* (*BSC6900*, *BSC6910*) is set to **YES(Yes)** and the MS supports the GSM/UMTS inter-RAT handover, the GSM/UMTS inter-RAT handover is performed.
- When no neighboring UMTS cell is available, if *INTERRATOUTBSCHOEN* (*BSC6900*, *BSC6910*) is set to **NO(No)** or the MS does not support the GSM/UMTS inter-RAT handover, the decision on whether to initiate another type of emergency handover is made.

3.4.11 Fast-Moving Micro Cell Handover

Fast-moving micro cell handover is performed from a micro cell to a macro cell according to the relative speed of an MS so that the number of handovers can be minimized.

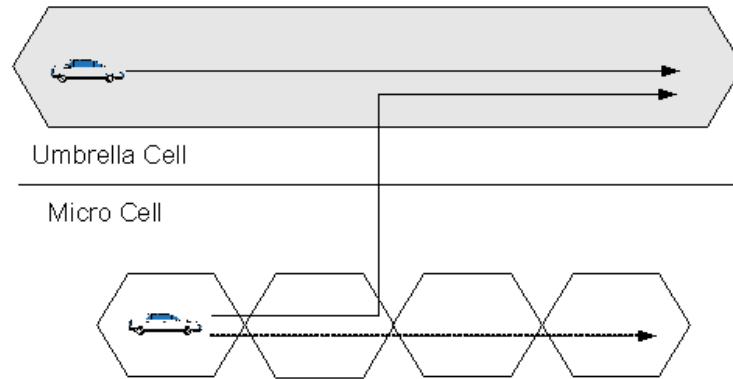
Fast-moving micro cell handover applies to the following scenarios:

- If an MS is moving fast in a micro cell, it is handed over to a macro cell.

- To prevent an MS that is moving fast in a macro cell from entering a micro cell, time penalty is performed on the micro cell so that the fast-moving MS camps on the macro cell.

Figure 3-7 shows a fast-moving micro cell handover.

Figure 3-7 Fast-moving micro cell handover



Trigger Conditions

If **QCKMVHOEN** (*BSC6900, BSC6910*) is set to **YES(Yes)**, the handover decision procedure for fast-moving micro cell handover is as follows:

- When the trigger conditions for an edge handover or PBGT handover are met, the fast-moving micro cell handover decision is started.
- When the period during which the MS camps on the serving cell is shorter than **QCKTIMETH** (*BSC6900, BSC6910*), the number of cells which the fast-moving MS passes through is incremented by one.

NOTE

The cells counted by the system must locate at a layer lower than layer 4. In other words, they must be non-Umbrella cells.

- When the number of cells that the MS passes in fast movement reaches **QCKSTATCNT** (*BSC6900, BSC6910*), the fast-moving micro cell handover is triggered if the number of cells that the MS passes in fast movement counted by the system is greater than or equal to **QCKTRUECNT** (*BSC6900, BSC6910*).

Target Cell Selection

In handover algorithm I, the target cell must have the highest priority in the candidate cell list. In addition, the target cell must meet the following conditions:

- The target cell must be at layer 4, that is, Umbrella cell.
- Filtered receive level of the target cell $\geq$ **HOTHRES** (*BSC6900, BSC6910*) + **INTELEVHOHYST** (*BSC6900, BSC6910*) - 64

Limitations

After the fast-moving micro cell handover is successful, the penalty is performed on all the neighboring micro cells. During **SPEEDPUNISHT** (*BSC6900, BSC6910*),

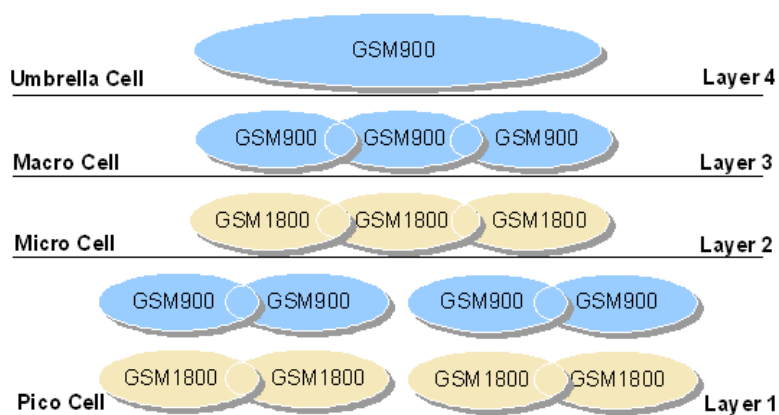
SDPUNVAL (BSC6900, BSC6910) is subtracted from the receive level of every neighboring micro cell.

Cell Layer and Cell Priority

With Huawei multiband handover algorithm, a proper traffic volume distribution can be realized among multiple frequency bands.

Huawei multiband handover algorithm divides cells into four layers, with 16 priorities at each layer. The ***LAYER (BSC6900, BSC6910)*** parameter specifies at which layer a cell is located. This algorithm is applicable to complex networking scenarios. [Figure 3-8](#) shows the cell layers.

Figure 3-8 Cell layers



In Huawei multiband handover algorithm, a GSM network covering a certain area is divided into four layers, which are:

- Layer 4: Umbrella cell. The umbrella cells are generally GSM900 cells having the wide coverage feature. It also implements fast MS connection.
- Layer 3: Macro cell. The macro cells are generally GSM900 cells, which are commonly used in current GSM system and serve majority of subscribers.
- Layer 2: Micro cell. The micro cells are generally DCS1800 cells having the small coverage feature. They enable capacity expansion.
- Layer 1: Pico cell. The Pico cells are generally DCS1800 cells, which are used in hot spots and blind spots.

The cell at the lower layer has a higher priority.

PRIOR (BSC6900, BSC6910) controls a handover between cells at the same layer. Each layer has 16 priorities, numbered 1-16 respectively. A low value indicates a high priority. If the cells at the same layer have different priorities, a cell with a smaller priority value has a higher priority. ***PRIOR (BSC6900, BSC6910)*** and ***LAYER (BSC6900, BSC6910)*** determine the cell priority. The priority affects the sequence of neighboring cells for handover.

3.4.12 Inter-Layer Handover

Inter-layer handover is a type of normal handover. It is used to enable the micro cells at low layers (the priority is high) to absorb traffic volume.

The GSM network is divided into several layers to balance the traffic volume flexibly and to meet the requirements of different network topologies. For details, see [3.4.11 Fast-Moving Micro Cell Handover](#). In Huawei handover algorithm I, both inter-layer handovers and PBGT handovers are better cell handovers. The inter-layer handover is performed between different layers, whereas the PBGT handover is performed at the same layer.

Trigger Conditions

If **LEVHOEN** (*BSC6900, BSC6910*) is set to **YES(Yes)**, the trigger conditions for an inter-layer handover are as follows:

- The layer at which the target cell is located has a higher priority than the layer at which the serving cell is located.
- Filtered downlink receive level of the target cell $\geq$ **HOTHRES** (*BSC6900, BSC6910*) + **INTELEVHOHYST** (*BSC6900, BSC6910*) – 64
- After basic ranking and comprehensive ranking during network characteristics adjustment, the target cell has a higher priority than the serving cell.

The inter-layer handover is triggered when the P/N criterion is met, that is, the previous conditions are met for **LEVLAST** (*BSC6900, BSC6910*) within **LEVSTAT** (*BSC6900, BSC6910*).

If **NCELLPUNEN** (*BSC6900, BSC6910*) is set to **NO(No)** for the candidate cell, an inter-layer handover is triggered when the previous conditions are met.

If **NCELLPUNEN** (*BSC6900, BSC6910*) is set to **YES(Yes)** for the candidate cell, the timer **NCELLPUNTM** (*BSC6900, BSC6910*) is started when the previous conditions are met. After this timer is started, the value of **NCELLPUNLEV** (*BSC6900, BSC6910*) is subtracted from the filtered receive level of the candidate cell. If the timer expires, an inter-layer handover is triggered. If the timer stops, no inter-layer handover is triggered. The timer stops in either of the following cases:

- If the filtered BCCH receive level of the candidate cell is less than the value of **NCELLPUNSTPTH** (*BSC6900, BSC6910*), the timer stops for the candidate cell.
- If the trigger conditions for a BQ handover are met, the timer for all handover candidate cells stops.

Target Cell Selection

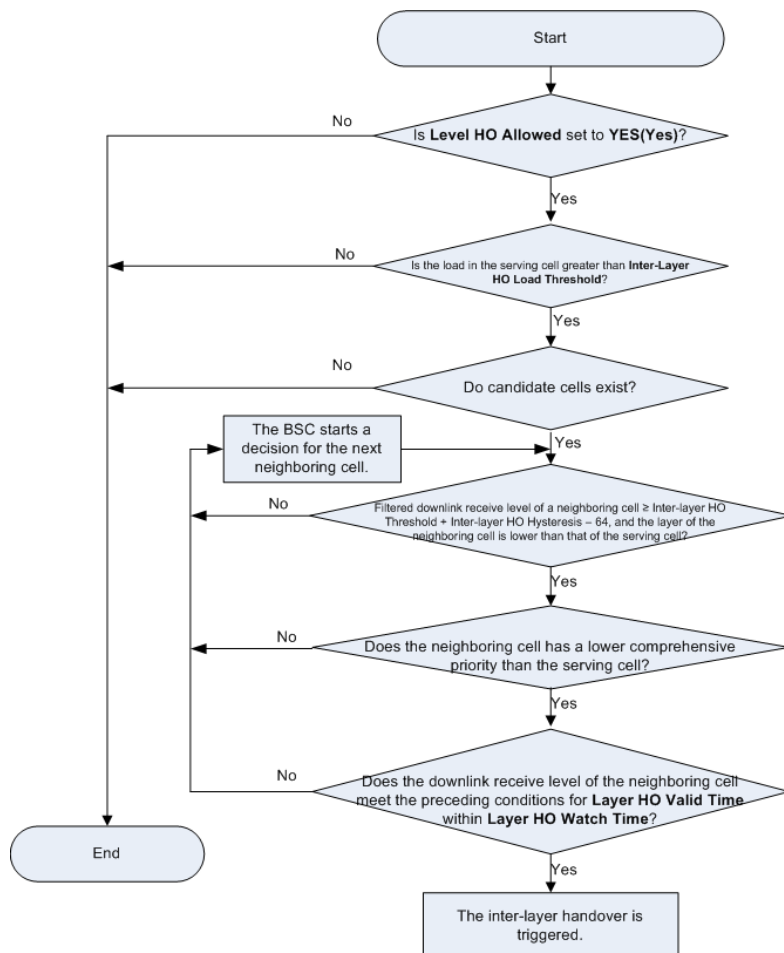
The requirements for target cell selection are as follows:

- The trigger conditions are met.
- The serving cell cannot be selected as the target cell.
- The target cell has the highest priority in the candidate cell list.

Inter-Layer Handover Process

Figure 3-9 shows the inter-layer handover process.

Figure 3-9 Inter-layer handover process



3.4.13 PBGT Handover

PBGT handover is a type of normal handover.

Trigger Conditions

If **PBGTHOEN** (*BSC6900, BSC6910*) is set to **YES(Yes)**, the trigger conditions for a PBGT handover are as follows:

- The target cell and the serving cell are at the same layer and have the same priority.
- The following condition is met for **PBGTLAST** (*BSC6900, BSC6910*) within **PBGTSTAT** (*BSC6900, BSC6910*):

$$(\text{MIN}(\text{MS_TXPWR_MAX}, P) - \text{RXLEV_DL} - \text{PWR_DIFF}) - (\text{MIN}(\text{MS_TXPWR_MAX}(n), P) - \text{RXLEV_NCELL}(n)) > \text{PBGT_HO_MARGIN}$$

where,

- RXLEV_DL: indicates the filtered downlink receive level of the serving cell.
- MS_TXPWR_MAX: indicates the maximum allowed transmit power of an MS in the serving cell.
- MS_TXPWR_MAX (n): indicates the maximum allowed transmit power of an MS in neighboring cell n.
- RxLev_NCELL (n): indicates the downlink receive level in neighboring cell n.
- PWR_DIFF: indicates the difference between the maximum downlink transmit power and the actual downlink transmit power in the serving cell. The difference is caused by power control.
- P: indicates the maximum transmit power of an MS.
- PBGT_HO_MARGIN: indicates the value of **PBGTMARGIN (BSC6900, BSC6910)** of the serving cell configured for neighboring cells minus 64.

If **NCELLPUNEN (BSC6900, BSC6910)** is set to **NO(No)** for the candidate cell, a PBGT handover is triggered when the previous conditions are met.

If **NCELLPUNEN (BSC6900, BSC6910)** is set to **YES(Yes)** for the candidate cell, the timer **NCELLPUNTM (BSC6900, BSC6910)** is started when the previous conditions are met. After this timer is started, the value of **NCELLPUNLEV (BSC6900, BSC6910)** is subtracted from the filtered receive level of the candidate cell. If the timer expires, a PBGT handover is triggered. If the timer stops, no PBGT handover is triggered. The timer stops in either of the following cases:

- If the filtered BCCH receive level of the candidate cell is less than the value of **NCELLPUNSTPTH (BSC6900, BSC6910)**, the timer stops for the candidate cell.

If the trigger conditions for a BQ handover are met, the timer for all handover candidate cells stops.

Target Cell Selection

The target cell must meet the following conditions:

- The target cell and the serving cell are at the same layer and have the same priority.
- The serving cell cannot be selected as the target cell.
- The target cell has the highest priority in the candidate cell list.

3.4.14 AMR Handover

This section describes the feature GBFD-115504 AMR FR/HR Dynamic Adjustment.

The AMR handover in handover I algorithm consists of the AMR TCHF-TCHH handover and AMR TCHH-TCHF handover algorithm. The AMR TCHF-TCHH handover is conducted based on cell load and RQI, whereas the AMR TCHH-TCHF handover is conducted based on RQI.

The conversion formula between RQI and C/I is $RQI = 2 \times C/I$.

The handover I algorithm also supports dynamic adjustment between TCHFs and TCHHs. For details, see *Dynamic Channel Conversion Feature Parameter Description*.

Trigger Conditions for an AMR TCHF-TCHH Handover

The trigger conditions for an AMR TCHF-TCHH handover are as follows:

- **INTRACELLFHHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**.
- The target call is an AMR call.
- The half-rate function must be enabled in the cell where the call is initiated.
- The full-rate speech version 3 and half-rate speech version 3 must be supported by the cell where the call is initiated.
- The type of channel specified by the MSC during a call can be changed during a handover.
- For AMR FR calls, if **AMRTCHHPRIORALLOW (BSC6900, BSC6910)** is set to **ON(On)**, TCHF-to-TCHH handover is triggered when the cell load is greater than or equal to the value of **AMRTCHHPRIORLOAD (BSC6900, BSC6910)** and the proportion of AMR HR users is less than the value of **ALLOWAMRHALFRATEUSERPERC (BSC6900, BSC6910)**.
- For AMR FR calls, if **AMRTCHHPRIORALLOW (BSC6900, BSC6910)** is set to **OFF(Off)**, TCHF-to-TCHH handover is triggered when the proportion of AMR HR users is less than the value of **ALLOWAMRHALFRATEUSERPERC (BSC6900, BSC6910)**.
- The call occupies the full-rate TCH, and the RQI/2 is greater than **INHOF2HTH (BSC6900, BSC6910)**.

For an AMR FR call, the AMR TCHF-TCHH handover can be performed if the preceding conditions are met for **INFHHOLAST (BSC6900, BSC6910)** within **INFHHOSTAT (BSC6900, BSC6910)**.

Trigger Conditions for an AMR TCHH-TCHF Handover

The trigger conditions for an AMR TCHH-TCHF handover are as follows:

- **INTRACELLFHHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**.
- The target call is an AMR call.
- The half-rate function must be enabled in the cell where the call is initiated.
- The full-rate speech version 3 and half-rate speech version 3 must be supported by the cell where the call is initiated.
- The type of channel specified by the MSC during a call can be changed during a handover.
- The call occupies the half-rate TCH, and the RQI/2 is less than **INHOH2FTH (BSC6900, BSC6910)**.

For an AMR HR call, the AMR TCHH-TCHF handover can be performed if the preceding conditions are met for **INFHHOLAST (BSC6900, BSC6910)** within **INFHHOSTAT (BSC6900, BSC6910)**.

 NOTE

If **INTRACELLSINUSEREN (BSC6900, BSC6910)** is set to **NO(No)**, no intra-cell handover is allowed within the time specified by **BANTIME (BSC6900, BSC6910)** when multiple intra-cell handovers are performed consecutively. If **INTRACELLSINUSEREN (BSC6900, BSC6910)** is set to **YES(Yes)**, no quality-related intra-cell handovers (such as BQ handover, interference handover, TCHH-to-TCHF handover, and tight BCCH handover) are allowed within the time specified by **BANTIME (BSC6900, BSC6910)** when the same MS consecutively performs multiple such types of handovers.

Target Cell Selection

The AMR handover is an intra-cell handover. Therefore, only the serving cell can be selected as the target cell.

3.4.15 SDCCH Handover

This section describes the feature GBFD-110608 SDCCH Handover.

SDCCH handover is a process in which the MS is handed over from an SDCCH to another SDCCH in an immediate assignment. SDCCH handover helps improve the access success rate of the MSs on the edge of the network, improving the network QoS.

The principle of SDCCH handover is the same as that of TCH handover. Regarding the procedure, an SDCCH handover involves measurement and MR reporting, MR processing, handover decision, and handover execution.

Whether an SDCCH handover can be performed is controlled by the **SIGCHANHOEN (BSC6900, BSC6910)** parameter. If an inter-BSC SDCCH handover is required, both **SIGCHANHOEN (BSC6900, BSC6910)** and **INRBSCSDHOEN (BSC6900, BSC6910)** need to be set to **YES(Yes)**.

The handover decision algorithm for SDCCH handover is different from that for TCH handover in the following ways:

- The following handover algorithms support SDCCH handover:
Quick handover, TA handover, BQ handover, rapid level drop handover, interference handover, no-DL-MR handover, Inter-layer handover, edge handover, and fast-moving micro cell handover
- The following handover algorithms do not support SDCCH handover:
Dual-band network handover, load handover, PBGT handover, AMR handover, better 3G cell handover, concentric cell handover, and tight BCCH handover

3.4.16 Other Handovers

Other handovers here refer to better 3G cell handover and tight BCCH handover.

Better 3G Cell Handover

See *Interoperability Between GSM and WCDMA Feature Parameter Description*.

Tight BCCH Handover

See *BCCH Dense Frequency Multiplexing Feature Parameter Description*.

3.5 Handover Decision Based on Handover Algorithm II

This section describes the feature GBFD-510501 HUAWEI II Handover.

Handover decision based on handover algorithm II is made in the following order: forced handover, emergency handover, intra-cell handover, and inter-cell handover.

Handover decision based on handover algorithm II involves the following procedures:

- Determines whether the serving cell meets the trigger conditions.
- Selects corresponding candidate cell list for each handover type.
- Performs the comprehensive decision and determining the candidate neighboring cells.

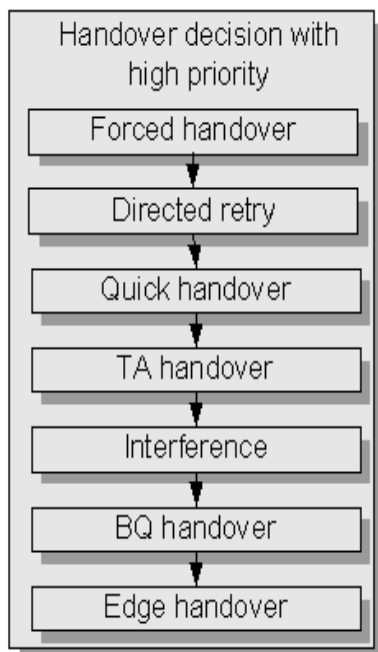
The procedure for performing comprehensive decision based on handover results and determining the candidate neighboring cells is as follows:

- The BSC selects a handover type with the highest priority from all the handovers that can be performed on each neighboring cell.

The handover priority is as follows:

Forced handover, emergency handover, and interference handover have a high priority. **Figure 3-10** shows the details.

Figure 3-10 Handovers with high priority



NOTE

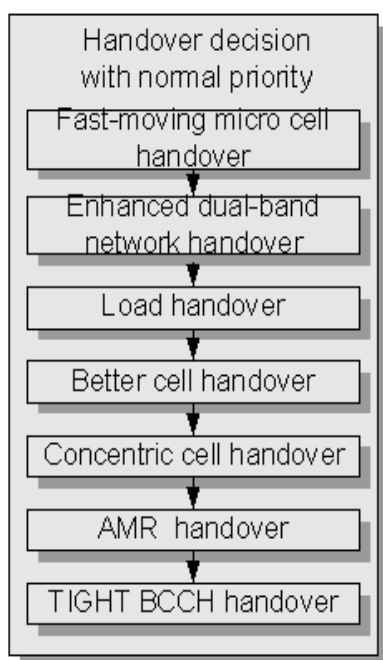
Quick handover is classified into frequency offset handover and fast move handover. Frequency offset handover has a higher priority than fast move handover.

Intra-cell handover (excluding interference handover) and inter-cell handover have a normal priority. **Figure 3-11** shows the details.

NOTE

AMR handover has the same priority as TCHF-TCHH handover.

Figure 3-11 Handovers with normal priority



- The BSC ranks the candidate cells according to the network characteristics adjustment algorithm and then generates the final candidate cell list. Every neighboring cell in the candidate cell list has its own handover decision. Neighboring GSM and UMTS cells are ranked separately.
- In handover algorithm II, ***HOOPTSEL (BSC6900, BSC6910)*** specifies whether a neighboring GSM or UMTS cell is preferred.
 - When ***HOOPTSEL (BSC6900, BSC6910)*** is set to **Pre_2G_Cell(Preference for 2G cell)**:
A neighboring GSM cell is preferred. If the candidate cell list contains suitable neighboring UMTS cells but no suitable neighboring GSM cells, a neighboring UMTS cell is selected.
 - When ***HOOPTSEL (BSC6900, BSC6910)*** is set to **Pre_3G_Cell(Preference for 3G cell)**:
A neighboring UMTS cell is preferred. If the candidate cell list contains suitable neighboring GSM cells but no suitable neighboring UMTS cells, a neighboring GSM cell is selected.
 - When ***HOOPTSEL (BSC6900, BSC6910)*** is set to **Pre_2G_CellThres(Preference 2G cell by thresh)**:
If the receive level of a candidate GSM cell is lower than or equal to ***HOPRETH2G (BSC6900, BSC6910)***, a neighboring UMTS cell is preferred.

 NOTE

When a neighboring UMTS cell is preferred among the candidate cells, the priority of 3G better-cell handover is the lowest.

If the trigger conditions for an emergency handover are met and there is at least one candidate cell, the emergency handover timer **NEWURGHOMININTV (BSC6900, BSC6910)** is started. Another emergency handover decision can be performed only when **NEWURGHOMININTV (BSC6900, BSC6910)** times out.

3.5.1 Quick Handover

This section describes the features GBFD-510103 Chain Cell Handover and GBFD-510102 Fast Move Handover.

Quick handover aims to increase the handover success rate of an MS moving at a high speed and to ensure the call continuity and low call drop rate. Quick handover applies to the scenario where an MS moves fast along an urban backbone road, a selected route, or a high-speed railroad.

Quick Handover Types

Quick handover consists of frequency offset handover and fast move handover.

- Frequency offset handover
Whether the MS moves away from the serving cell is determined based on the frequency offset information provided by an MS moving at a high speed. Frequency offset handover decision is made according to the uplink/downlink receive level of the serving cell and the path loss of neighboring cells.
- Fast move handover
Fast move handover decision is made according to the path loss of neighboring cells.

For quick handover, the handover response speed is enhanced by:

- Accurately calculating the move speed of the MS
- Lifting the restriction on the interval between handover decisions
- Reducing the number of MRs for the handover decision
- Introducing the alpha filtering

Quick Handover Preparation

The preparation for quick handover involves the following aspects:

- Frequency offset is decoded from the MR.
Frequency offset of the MS is obtained from the uplink MR that the BTS sends to the BSC.
- Alpha filtering is performed on the MR. The value of **SCELLFILTER (BSC6900, BSC6910)** is used as the filter length for the serving cell, and the value of **NCELLFILTER (BSC6900, BSC6910)** is used as the filter length for a neighboring cell.

- If the number of MRs consecutively discarded in the serving cell is greater than the value of **IGNOREMRNUM (BSC6900, BSC6910)**, the BSC clears the MR queue for the serving cell and does not perform a handover decision.
- If the number of MRs consecutively discarded in a neighboring cell is greater than the value of **IGNOREMRNUM (BSC6900, BSC6910)**, the BSC clears the MR queue for the neighboring cell and does not perform a handover decision.

Trigger Conditions

During the handover decision, whether the trigger conditions for a frequency offset handover are met is determined first. When the BTS cannot send the frequency offset information or the reported frequency offset information is invalid, a fast move handover is triggered, provided that other conditions for a frequency offset handover are met.

If **QUICKHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**, the trigger conditions for a quick handover are as follows:

- The MS moves away from the serving cell (the frequency offset in the measurement result is a negative value) and the move speed of the MS is greater than **MOVESPEEDTHRES (BSC6900, BSC6910)**.
- The filtered uplink level of the serving cell is less than **HOUPTRIGE (BSC6900, BSC6910)**.
- The compensated downlink level of the serving cell is less than **HODOWNTRIGE (BSC6900, BSC6910)**.
- The difference between the path loss of a chain neighboring cell and that of the serving cell is greater than the predefined threshold.

The threshold is equal to the value of **HOOFFSET** (in the **SET GCELLHOFAST** command for an internal neighboring cell or in the **MOD GEXT2GCELL** command for an external neighboring cell) plus the value of **ADJHOOFFSET** for the serving cell.

The trigger conditions for a quick handover are as follows:

- If all the last three conditions are met, the handover decision is performed as follows:
 - If the first condition is met for **AFCHOLASTTIME (BSC6900, BSC6910)** during **AFCHOSTATTIME (BSC6900, BSC6910)**, a frequency offset handover is triggered.
 - If **QUICKPBGTHOEN (BSC6900, BSC6910)** is set to **YES(Yes)** and the first condition is not met, a fast move handover is triggered.
 - If **QUICKPBGTHOEN (BSC6900, BSC6910)** is set to **NO(No)**, a fast move handover is not triggered.
- If all the last three conditions are not met, a quick handover is not triggered.
- When **MSLEVSTRQPBGT (BSC6900, BSC6910)** is set to **YES(Yes)**, the BSC triggers a fast handover only when an MS moves away from the BTS.

Target Cell Selection

The target cell must be a chain neighboring cell of the serving cell. The target cell can be selected by setting **ISCHAINNCELL (BSC6900, BSC6910)**. If

HODIRFORECASTEN (BSC6900, BSC6910) is set to **YES**, a neighboring cell in the moving direction of the MS is selected preferentially.

To forecast the moving direction of the MS, when **HODIRFORECASTEN (BSC6900, BSC6910)** is set to **YES**, the direction of a chain neighboring cell (A or B) compared with the serving cell is specified by **CHAINNCELLTYPE (BSC6900, BSC6910)**. If the number of times that the MS is handed over to neighboring cells in the same direction (B for example) is greater than or equal to **HODIRLASTTIME (BSC6900, BSC6910)** when the handover time reaches **HODIRSTATIME (BSC6900, BSC6910)**, the MS is inferred to be moving towards the B direction. Subsequently, the MS is preferentially handed over to the neighboring cell whose **CHAINNCELLTYPE (BSC6900, BSC6910)** is B.

Limitations

The limitations on quick handover are as follows:

- The serving cell cannot be selected as the target cell.
- The candidate cells for quick handover must be chain neighboring cells of the serving cell.
- If quick handover penalty is enabled, after a quick handover is successful, the penalty is performed on the original cell during the penalty time to prevent an immediate handover back to the original cell. Quick handover penalty is determined jointly by **QuickHoPunishSw (BSC6910, BSC6900)**, **TIMEPUNISH (BSC6900, BSC6910)**, and **HOPUNISHVALUE (BSC6900, BSC6910)**. The penalty time and penalty value are specified by **TIMEPUNISH (BSC6900, BSC6910)** and **HOPUNISHVALUE (BSC6900, BSC6910)**.

3.5.2 TA Handover

TA handover is a type of emergency handover. The TA handover decision is made according to the TA value reported by the MS.

The TA value of a normal cell ranges from 0 to 63 and that of an underlaid subcell of a dual-timeslot cell ranges from 0 to 219. The unit of TA values is bit. The transmission distance over the air interface corresponding to the delay of one bit is 553.5 m. When the TA value is 63, the transmission distance over the air interface is 35 km.

Trigger Conditions

TA handover is triggered when the following conditions are met:

- **TAHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**.
- Filtered TA value in the MR provided by the MS is greater than or equal to **TALIMIT (BSC6900, BSC6910)**.

The TA handover can be triggered only when the preceding two conditions are met simultaneously.

NOTE

From the perspective of the trigger conditions for a TA handover, TA can be regarded as a limitation to the size of a cell.

Target Cell Selection

The target cell must have the highest priority in the candidate cell list after handover preprocessing. In addition, the target cell must meet the following conditions:

- The serving cell cannot be selected as the target cell.
- If **TALIMIT (BSC6900, BSC6910)** of a co-site neighboring cell is less than or equal to the **TALIMIT (BSC6900, BSC6910)** of the serving cell, a handover to the neighboring cell is prohibited.
- In handover algorithm II, a cell becomes the target cell for TA handover if the previous conditions are met for **TALASTTIME (BSC6900, BSC6910)** within **TASTATTIME (BSC6900, BSC6910)**.

If the trigger conditions for a TA handover are met but the candidate GSM cells are not suitable, the following operations are performed:

- When a neighboring UMTS cell is available, if **INTERRATOUTBSCHOEN (BSC6900, BSC6910)** is set to **YES(Yes)** and the MS supports the GSM/UMTS inter-RAT handover, the GSM/UMTS inter-RAT handover is performed.
- When no neighboring UMTS cell is available, if **INTERRATOUTBSCHOEN (BSC6900, BSC6910)** is set to **NO(No)** or the MS does not support the GSM/UMTS inter-RAT handover, the decision on whether to initiate another type of emergency handover is made.

Limitations

After the TA handover is successful, the penalty is performed on the original cell. During **TIMETAPUNISH (BSC6900, BSC6910)**, **SSTAPUNISH (BSC6900, BSC6910)** is subtracted from the level of the original cell to prevent an immediate handover back to the original cell.

3.5.3 BQ Handover

BQ handover is a type of emergency handover in which the system makes the decision based on the uplink/downlink receive quality on the Um interface.

The receive quality is measured in bit error rate (BER). The BSC measures the quality of a radio link based on the quality level in the MR. A rise in BER may result from an excessively low signal power or channel interference.

Trigger Conditions

If **BQHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**, the BQ handover is triggered when either of the following conditions is met:

- The uplink receive quality is greater than or equal to the uplink receive quality threshold of the serving cell.
- The downlink receive quality is greater than or equal to the downlink receive quality threshold of the serving cell.

The BQ handover is triggered when either of the preceding conditions is met.

The parameters for specifying the uplink and downlink receive quality thresholds are as follows:

- For non-AMR calls, the parameter for specifying the uplink receive quality threshold is **ULQUALIMIT (BSC6900, BSC6910)**, and the parameter for specifying the downlink receive quality threshold is **DLQUALIMIT (BSC6900, BSC6910)**.
- For AMR FR calls, the parameter for specifying the uplink receive quality threshold is **ULQUALIMITAMRFR (BSC6900, BSC6910)** and the parameter for specifying the downlink receive quality threshold is **DLQUALIMITAMRFR (BSC6900, BSC6910)**.
- For AMR HR calls, the parameter for specifying the uplink receive quality threshold is **ULQUALIMITAMRHR (BSC6900, BSC6910)** and the parameter for specifying the downlink receive quality threshold is **DLQUALIMITAMRHR (BSC6900, BSC6910)**.

Target Cell Selection

The target cell that has the highest priority in the candidate cell list after handover preprocessing must meet the conditions listed below:

- In handover algorithm II, the serving cell cannot be selected as the target cell.
- The receive level of target cell must meet the following conditions:
 - The serving cell and the target cell work in the same frequency band.
Filtered downlink receive level of the target cell > Filtered downlink receive level of the serving cell after compensation + (**INTERCELLHYST (BSC6900, BSC6910)** of the serving cell configured for the neighboring cell – 64) – (**BQMARGIN (BSC6900, BSC6910)** – 64)
 - The serving cell works in GSM900 and the target cell works in DCS1800.
Filtered downlink receive level of the target cell ≥ **HOTHRES (BSC6900, BSC6910)** + (**LEVHOHYST (BSC6900, BSC6910)** – 64)
 - The serving cell works in DCS1800 and the target cell works in GSM900.
There is no constraint on the target cell in this condition.
- In handover algorithm II, a cell becomes the target cell for BQ handover if the previous conditions are met for **BQLASTTIME (BSC6900, BSC6910)** within **BQSTATTIME (BSC6900, BSC6910)**.

If the trigger conditions for a BQ handover are met but the candidate GSM cells are not suitable, the following operations are performed:

- When a neighboring UMTS cell is available, if **INTERRATOUTBSCHOEN (BSC6900, BSC6910)** is set to **YES(Yes)** and the MS supports the GSM/UMTS inter-RAT handover, the GSM/UMTS inter-RAT handover is performed.
- When no neighboring UMTS cell is available, if **INTERRATOUTBSCHOEN (BSC6900, BSC6910)** is set to **NO(No)** or the MS does not support the GSM/UMTS inter-RAT handover, the decision on whether to initiate another type of emergency handover is made.

Limitations

After the BQ handover is successful, the penalty is performed on the original cell. During **TIMEBQPUNISH (BSC6900, BSC6910)**, **SSBQPUNISH (BSC6900, BSC6910)** is subtracted from the level of the original cell to prevent an immediate handover back to the original cell.

3.5.4 Interference Handover

Interference handover is a type of intra-cell handover in handover algorithm II.

Interference handover helps protect calls affected by interference and reduce network interference. It is applicable in scenarios with interference.

In handover algorithm II, the BQ handover is a type of emergency handover and the interference handover is a type of intra-cell handover.

Trigger Conditions

If both **INTRACELLHOEN (BSC6900, BSC6910)** and **INTERFHOEN (BSC6900, BSC6910)** are set to **YES(Yes)**, the trigger conditions for an interference handover are as follows:

- The filtered value of uplink receive quality is greater than the specified receive quality threshold at the current uplink receive level.
- The filtered value of downlink receive quality is greater than the specified receive quality threshold at the current downlink receive level.

In handover algorithm II, the interference handover is triggered when either of the previous conditions is met for **INTERFERELASTIME (BSC6900, BSC6910)** within **INTERFERESTATTIME (BSC6900, BSC6910)**.

The parameters for specifying the uplink and downlink receive quality thresholds are as follows:

- For non-AMR FR calls, the parameter for specifying the receive quality threshold is **RXQUALn**, where $1 \leq n \leq 12$. If $2 \leq n \leq 12$, the receive quality threshold is equal to the sum of **RXQUALn**, **FHGAINOFFSET**, and quality offset.

NOTE

Non-AMR FR calls consist of FR, HR, EFR, and AMR HR calls.

- For AMR FR calls, the parameters for specifying the receive quality threshold are **RXQUALn** ($1 \leq n \leq 12$) and **RXLEVOFF (BSC6900, BSC6910)**.
 - If $n=1$, the receive quality threshold is **RXQUAL1 (BSC6900, BSC6910)**.
 - If $2 \leq n \leq 12$, and the filtered value of uplink and downlink receive quality - **RXLEVOFF (BSC6900, BSC6910)** > 0 , use the value of the filtered value of uplink receive quality - **RXLEVOFF (BSC6900, BSC6910)**.
 - If $2 \leq n \leq 12$, and the filtered value of uplink and downlink receive quality - **RXLEVOFF (BSC6900, BSC6910)** ≤ 0 , use the value of the filtered value of uplink receive quality.

Target Cell Selection

In handover algorithm II, only the serving cell can be selected as the target cell.

Limitations

If **INTRACELLSINUSEREN (BSC6900, BSC6910)** is set to **NO(No)**, no intra-cell handover is allowed within the time specified by **BANTIME (BSC6900, BSC6910)**

when multiple intra-cell handovers are performed consecutively. If **INTRACELLSINUSEREN (BSC6900, BSC6910)** is set to **YES(Yes)**, no quality-related intra-cell handovers (such as BQ handover, interference handover, TCHH-to-TCHF handover, and tight BCCH handover) are allowed within the time specified by **BANTIME (BSC6900, BSC6910)** when the same MS consecutively performs multiple such types of handovers.

3.5.5 No-DL-MR Handover

No-DL-MR handover is performed on the basis of the uplink quality. The purpose is to ensure the call continuity and minimize the possibility of call drops.

No-DL-MR handover is generally caused by adverse radio environment on the uplink. In this case, the requirements of the filtering algorithm cannot be met, so other handover decisions cannot be performed.

Trigger Conditions

In handover algorithm II, the trigger conditions for a no-DL-MR handover are as follows:

- **NODLMRHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**.
- There is no downlink information in the MR of the call.
- The number of lost downlink MRs is less than **NODLMRHOALLOWLIMIT (BSC6900, BSC6910)**.
- For TCH, the number of saved MRs with uplink quality value is greater than **DATAQUAFLTLEN (BSC6900, BSC6910)**. For SDCCH, the number of saved MRs with uplink quality value is greater than **SIGQUAFLTLEN (BSC6900, BSC6910)**.
- The filtered value of uplink receive quality is greater than or equal to **ULQUALIMIT (BSC6900, BSC6910)**.
 - For AMR FR calls, the corresponding parameter is **ULQUALIMITAMRFR (BSC6900, BSC6910)**.
 - For AMR HR calls, the corresponding parameter is **ULQUALIMITAMRHR (BSC6900, BSC6910)**.

In addition, the BSC triggers a no-DL-MR handover when all the preceding conditions are met and MRs of the serving cell keep lost for **NODLMRHOLASTTIME (BSC6900, BSC6910)** within **NODLMRHOSTATTIME (BSC6900, BSC6910)**.

Target Cell Selection

In handover algorithm II, the conditions for selecting the target cell are as follows:

- The ranked neighboring cells recorded in the last complete MR are saved as candidate cells.
- Only a neighboring cell can serve as the target cell.
- Receive level in neighboring cell > Filtered receive level in serving cell + (**INTERCELLHYST (BSC6900, BSC6910)** – 64) – (**BQMARGIN (BSC6900, BSC6910)** – 64)

- A neighboring cell serves as the target cell if the previous conditions are met for **ULBQLASTTIME** (*BSC6900, BSC6910*) within **ULBQSTATTIME** (*BSC6900, BSC6910*).

3.5.6 Enhanced Dual-Band Network Handover

Enhanced dual-band network handover is performed based on the traffic volume of the OL and UL cells and based on the receive level.

Enhanced dual-band network handover is classified into the following types:

- Handover due to high load in the UL cell
- Handover due to low load in the UL cell
- Handover due to MS movement to the border of the OL cell

Trigger Conditions for a Handover Due to High Load in the Underlaid Cell

The trigger conditions for a handover due to high load in the UL cell are as follows:

- The two cells are in the enhanced dual-band network and **OUTLOADHOENABLE** (*BSC6900, BSC6910*) is set to **YES(Yes)**.
- The MS supports the frequency band on which the OL cell operates.
- The handover due to high load in the UL cell is performed only on TCHs.
- The load in the UL cell is greater than or equal to **OUTGENOVERLDTHRED** (*BSC6900, BSC6910*).
- The load in the OL cell is less than **INNERIOVERLDTHRED** (*BSC6900, BSC6910*).
- The system traffic volume is less than or equal to **EDBSYSFLOWLEV** (*BSC6900, BSC6910*).
- The current call is within the handover margin, and the receive level in the overlaid subcell is greater than or equal to **INTOINNRXLEVTHRED** (*BSC6900, BSC6910*).
- The receive level in the underlaid subcell minus the receive level of the optimal neighboring cell is greater than the sum of **ATCBTHRED** (*BSC6900, BSC6910*) and **ATCBHYST** (*BSC6900, BSC6910*).

When all the preceding conditions are met, the handover due to high load in the UL cell is triggered.

If the load of the UL subcell in the cell is greater than or equal to **OUTSERIOVERLDTHRED** (*BSC6900, BSC6910*), the handover margin is adjusted in a period of **OUTLOADHOPERIOD** (*BSC6900, BSC6910*) subtracted by **OUTLOADHOMODPERI** (*BSC6900, BSC6910*). The step length for handover margin adjustment is specified by **OUTLOADHOSTEP** (*BSC6900, BSC6910*).

Trigger Conditions for a Handover Due to Low Load in the Underlaid Cell

The trigger conditions for a handover due to low load in the UL cell are as follows:

- The load in the underlaid subcell is smaller than **OUTLOWLOADTHRED** (*BSC6900, BSC6910*).

- The system traffic volume is less than or equal to **EDBSYFLOWLEV** (*BSC6900, BSC6910*).
- The receive level in the overlaid subcell is greater than or equal to **OUTINNREXLEVTHRED** (*BSC6900, BSC6910*).

When all the preceding conditions are met, the handover due to low load in the underlaid subcell is triggered.

If the load of the UL subcell in the cell is less than **OUTLOWLOADTHRED** (*BSC6900, BSC6910*) for a specified period, the handover margin is adjusted in a period of **INNLOADHOPERI** (*BSC6900, BSC6910*). The step length for handover margin adjustment is specified by **INNLOADHOSTEP** (*BSC6900, BSC6910*).

Trigger Conditions for a Handover Due to MS Movement to the Border of the Overlaid Cell

The trigger conditions for a handover due to MS movement to the border of the OL cell are as follows:

- $SS(S) < Thdouter$
- $SS(u) - SS(n) < ATCB_THRD - ATCB_HYST$
where,
 - $SS(s)$: specifies the filtering compensated downlink receive level in the serving cell.
 - $Thdouter$: specifies **OUTINNREXLEVTHRED** (*BSC6900, BSC6910*).
 - $SS(u)$: specifies the downlink level (power compensation is performed on the downlink level based on the measurement) of the UL subcell where the call is originated. If the $SS(u)$ value cannot be obtained, the decision of enhanced dual-band network handover is not performed and the decision condition is met by default.
 - $SS(n)$: The best neighboring cell is the one whose measured BCCH level is the highest among neighboring cells. $SS(n)$ is the level value of such an optimum neighboring cell that is on the same frequency band and layer with the UL subcell but is not co-sited with the UL subcell. If such a neighboring cell is not available, the value of $SS(n)$ is -110 dBm.
 - $ATCB_THRD$: specifies **ATCBTHRED** (*BSC6900, BSC6910*).
 - $ATCB_HYST$: specifies **ATCBHYST** (*BSC6900, BSC6910*).

Handover due to MS movement to the border of the OL cell is triggered if either of the preceding conditions is met.

NOTE

- In the Adapter distance To Cell Border (ATCB) handover algorithm, the border between the OL and UL subcells is determined according to the signal strength of the serving cell and that of neighboring cells. If $SS(s) = SS(n)$, the system considers that the MS is located at the border of the UL cell. If $SS(s) - SS(n) > ATCB_THRD$, the system considers that the MS is located in the coverage area of the OL subcell. The coverage area of the OL subcell is determined according to different networking and coverage conditions of the existing network. In addition, the OL subcell of the serving cell and the OL subcell of the neighboring cells will not overlap regardless of the distance between BTSs.
- The handover margin specifies the range of signal level. In the case of OL/UL load handover on the enhanced dual-band network, the MSs whose downlink levels are within the handover margin are handed over level by level.

Target Cell Selection

The requirements for target cell selection in the enhanced dual-band network are as follows:

- For the handover due to high load in the UL cell, the MS must be handed over to the OL cell.
- For the handover due to low load in the UL cell, the MS must be handed over to the UL cell.
- For the handover due to MS movement in the OL subcell, the MS must be handed over to the UL cell.

Limitations

The limitations on the handover due to high load in the UL cell are as follows:

- If the cell where the call is located is on an enhanced dual-band network, **CELLINEXTP** (*BSC6900, BSC6910*) is set to **Extra(Extra)**.
- The **OUTLOADHOENABLE** (*BSC6900, BSC6910*) parameter is set.
- The maximum range of the handover margin is from 63 to **INTOINNREXLEVTHRED** (*BSC6900, BSC6910*). The MS with the highest receive level is handed over first.

The limitations on the handover due to low load in the UL cell are as follows:

- If the cell where the call is located is on an enhanced dual-band network, **CELLINEXTP** (*BSC6900, BSC6910*) is set to **Inner(Inner)**.
- The **INNLOADHOEN** (*BSC6900, BSC6910*) parameter is set.
- The maximum range of the handover margin is from 63 to **OUTINNREXLEVTHRED** (*BSC6900, BSC6910*). The MS with the lowest receive level is handed over first.

The limitations on the handover due to MS movement to the border of the OL cell are as follows:

- If the cell where the call is located is on an enhanced dual-band network, **CELLINEXTP** (*BSC6900, BSC6910*) is set to **Inner(Inner)**.

Impact of the Enhanced Dual-Band Network Handover on the Existing Algorithm

The impact of the enhanced dual-band network handover on the existing algorithm is as follows:

On the enhanced dual-band network, the MS should not be handed over to a cell in the same UL/OL cell group when the load handovers between the OL cell and the UL cell (specified by **OUTLOADHOENABLE** (*BSC6900, BSC6910*) and **INNLOADHOEN** (*BSC6900, BSC6910*)) are allowed. This is to prevent a load handover of a normal cell from colliding with a load handover between the OL cell and the UL cell on the network.

3.5.7 Load Handover

In handover algorithm II, load handover is not decided independently.

In handover algorithm II, the load handover decision is made in network adjustment phase. The selection of target cell should be processed by the better-cell handover. That is, the load handover is triggered when the trigger conditions for a load handover and better-cell handover are met simultaneously.

For the trigger conditions for a load handover, see [3.4.8 Load Handover](#). For the trigger conditions for a better-cell handover, see [3.5.10 Better-Cell Handover](#).

The ranges of load handover margins are defined as follows:

- Load handover margin 1 specifies the area where the neighboring cell receive level minus the downlink receive level of the serving cell ranges from (*INTERCELLHYST (BSC6900, BSC6910)* – 64 – *LOADHOSTEP (BSC6900, BSC6910)*) to *INTERCELLHYST (BSC6900, BSC6910)*.
- Load handover margin 2 specifies the area where the neighboring cell receive level minus the downlink receive level of the serving cell ranges from (*INTERCELLHYST (BSC6900, BSC6910)* – 64 – *LOADHOSTEP (BSC6900, BSC6910)* × 2) to (*INTERCELLHYST (BSC6900, BSC6910)* – 64).

When *LoadHoEn (BSC6900, BSC6910)* is set to YES(Yes) and *LoadHoPunishInheritSwitch (BSC6900, BSC6910)* is set to ON(On), the load handover supports handover penalty inheritance.

3.5.8 Edge Handover

Edge handover is performed on the basis of receive level.

To trigger an edge handover, the receive level of the target cell should be at least one hysteresis value (specified by *INTERCELLHYST (BSC6900, BSC6910)* – 64) greater than the receive level of the serving cell.

Trigger Conditions

If *FRINGEHOEN (BSC6900, BSC6910)* is set to YES(Yes), the trigger conditions for an edge handover are as follows:

- Either of the following conditions is met:
 - Filtered downlink receive level of the serving cell after compensation < *DLEDGETHRES (BSC6900, BSC6910)*
 - Filtered uplink receive level of the serving cell after compensation < *ULEDGETHRES (BSC6900, BSC6910)*
- Receive level of a neighboring cell ≥ Downlink receive level of the serving cell after compensation + *INTERCELLHYST (BSC6900, BSC6910)* – 64

An edge handover is triggered when the P/N criterion is met, that is, when the previous conditions are met for *EDGELAST (BSC6900, BSC6910)* within *EDGESTAT (BSC6900, BSC6910)*.

If both *FRINGEHOEN (BSC6900, BSC6910)* and *EDOUTHODEN (BSC6900, BSC6910)* are set to YES(Yes), the trigger conditions for an edge handover are as follows:

- Either of the following conditions is met:
 - Filtered downlink receive level of the serving cell after compensation < *DLEDGETHRES (BSC6900, BSC6910)* + (*EDOUTHOOFFSET (BSC6900, BSC6910)* – 64)

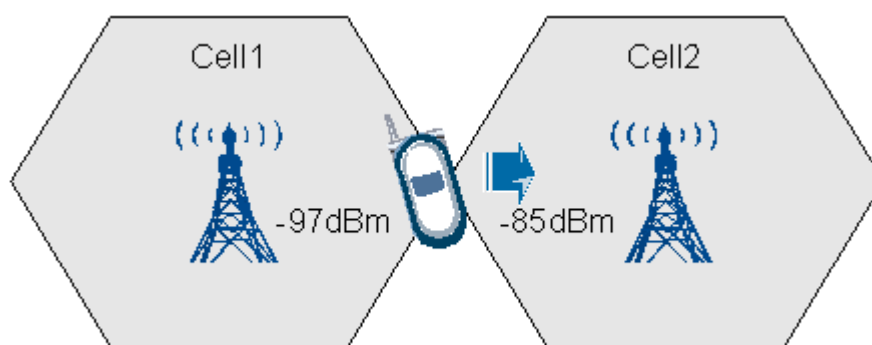
- Filtered uplink receive level of the serving cell after compensation < ***ULEDGETHRES (BSC6900, BSC6910) + (EDOUTHOOFFSET (BSC6900, BSC6910) - 64)***
- Receive level of a neighboring cell ≥ Downlink receive level of the serving cell after compensation + ***INTERCELLHYST (BSC6900, BSC6910) - 64***

An edge handover is triggered when the P/N criterion is met.

NOTE

If ***FRINGEHOEN (BSC6900, BSC6910)*** is set to **NO(No)**, ***EDOUTHOADEN (BSC6900, BSC6910)*** and ***EDOUTHOOFFSET (BSC6900, BSC6910)*** are invalid.

Figure 3-12 Edge handover



Target Cell Selection

The target cell must have the highest priority among the candidate cells. In addition, it must meet the following conditions:

- The serving cell cannot be selected as the target cell.
- After cells are ranked, the target cell must have a higher priority than the serving cell.

A cell becomes the target cell if the previous conditions are met for ***EDGEADJLASTTIME (BSC6900, BSC6910)*** within ***EDGEADJSTATTIME (BSC6900, BSC6910)***.

If the trigger conditions for an edge handover are met but the candidate GSM cells are not suitable, the following operations are performed:

- When a neighboring UMTS cell is available, if ***INTERRATOUTBSCHOEN (BSC6900, BSC6910)*** is set to **YES(Yes)** and the MS supports the GSM/UMTS inter-RAT handover, the GSM/UMTS inter-RAT handover is performed.
- When no neighboring UMTS cell is available, if ***INTERRATOUTBSCHOEN (BSC6900, BSC6910)*** is set to **NO(No)** or the MS does not support the GSM/UMTS inter-RAT handover, the decision on whether to initiate another type of emergency handover is made.

3.5.9 Fast-Moving Micro Cell Handover

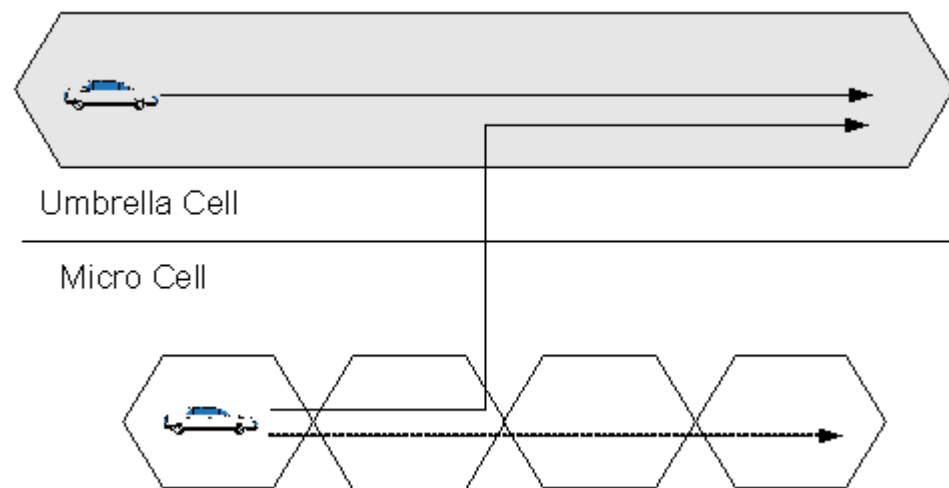
Fast-moving micro cell handover is performed from a micro cell to a macro cell according to the relative speed of an MS so that the number of handovers can be minimized.

Fast-moving micro cell handover applies to the following scenarios:

- If an MS is moving fast in a micro cell, it is handed over to a macro cell.
- To prevent an MS that is moving fast in a macro cell from entering a micro cell, time penalty is performed on the micro cell so that the fast-moving MS camps on the macro cell.

Figure 3-13 shows a fast-moving micro cell handover.

Figure 3-13 Fast-moving micro cell handover



Trigger Conditions

If **QCKMVHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**, the handover decision procedure for fast-moving micro cell handover is as follows:

- When the trigger conditions for an edge handover or PBGT handover are met, the fast-moving micro cell handover decision is started.
- When the period during which the MS camps on the serving cell is shorter than **QCKTIMETH (BSC6900, BSC6910)**, the number of cells which the fast-moving MS passes through is incremented by one.

NOTE

The cells counted by the system must locate at a layer lower than layer 4. In other words, they must be non-Umbrella cells.

- When the number of cells that the MS passes in fast movement reaches **QCKSTATCNT (BSC6900, BSC6910)**, the fast-moving micro cell handover is triggered if the number of cells that the MS passes in fast movement counted

by the system is greater than or equal to *QCKTRUECNT (BSC6900, BSC6910)*.

Target Cell Selection

In handover algorithm II, the target cell must have the highest priority in the candidate cell list. In addition, the target cell must meet the following conditions:

- The target cell must be at layer 4, that is, Umbrella cell.
- Filtered receive level of the target cell ≥ *HOTHRES (BSC6900, BSC6910)* + *INTELEVHOHYST (BSC6900, BSC6910)* – 64
- A neighboring cell serves as the target cell if the previous conditions are met for *HCSLASTTIME (BSC6900, BSC6910)* within *HCSSTATIME (BSC6900, BSC6910)*.

Limitations

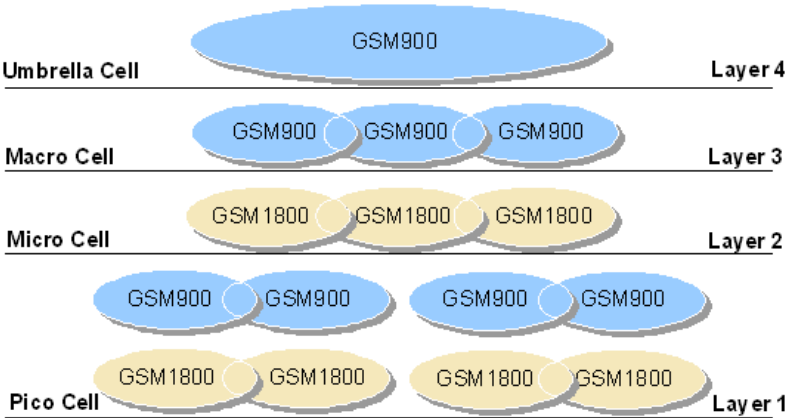
After the fast-moving micro cell handover is successful, the penalty is performed on all the neighboring micro cells. During *SPEEDPUNISHT (BSC6900, BSC6910)*, *SPEEDPUNISH (BSC6900, BSC6910)* is subtracted from the receive level of every neighboring micro cell.

Cell Layer and Cell Priority

With Huawei multiband handover algorithm, a proper traffic volume distribution can be realized among multiple frequency bands.

Huawei multiband handover algorithm divides cells into four layers, with 16 priorities at each layer. The *LAYER (BSC6900, BSC6910)* parameter specifies at which layer a cell is located. This algorithm is applicable to complex networking scenarios. [Figure 3-14](#) shows the cell layers.

Figure 3-14 Cell layers



In Huawei multiband handover algorithm, a GSM network covering a certain area is divided into four layers, which are:

- Layer 4: Umbrella cell. The umbrella cells are generally GSM900 cells having the wide coverage feature. It also implements fast MS connection.

- Layer 3: Macro cell. The macro cells are generally GSM900 cells, which are commonly used in current GSM system and serve majority of subscribers.
- Layer 2: Micro cell. The micro cells are generally DCS1800 cells having the small coverage feature. They enable capacity expansion.
- Layer 1: Pico cell. The Pico cells are generally DCS1800 cells, which are used in hot spots and blind spots.

The cell at the lower layer has a higher priority.

PRIOR (BSC6900, BSC6910) controls a handover between cells at the same layer. Each layer has 16 priorities, numbered 1-16 respectively. A low value indicates a high priority. If the cells at the same layer have different priorities, a cell with a smaller priority value has a higher priority. **PRIOR (BSC6900, BSC6910)** and **LAYER (BSC6900, BSC6910)** determine the cell priority. The priority affects the sequence of neighboring cells for handover.

3.5.10 Better-Cell Handover

Better-cell handover is an optimization of inter-layer handover and PBGT handover in handover algorithm I. Better-cell handover is specific to handover algorithm II.

Trigger Conditions

If **BETTERCELLHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**, the trigger conditions for a better-cell handover are as follows:

- If **PATHLOSSHOEN (BSC6900, BSC6910)** is set to **YES(Yes)**, path loss is considered during the sorting of the serving cell and handover candidate cells. If **PATHLOSSHOEN (BSC6900, BSC6910)** is set to **NO(No)**, path loss is not considered during the sorting of the serving cell and handover candidate cells.
- If the serving cell is in the candidate cell list, the priorities of the handover candidate cells must be higher than the priority of the serving cell after the sorting.
- If the serving cell is not in the candidate cell list, the handover candidate cells must meet the following conditions:

- **LoadHoEn (BSC6900, BSC6910)** is set to **NO(No)**, or **LoadHoEn (BSC6900, BSC6910)** is set to **YES(Yes)** but the load in the serving cell is less than **TRIGTHRES (BSC6900, BSC6910)**.
- If the serving cell and the handover candidate cells are at the same layer and have the same priority, the following conditions must be met:

$$SS_DL(N) - SS_DL(S) > H - 64$$

$SS_DL(N)$ indicates the filtered BCCH receive level of a target cell.
 $SS_DL(S)$ indicates the compensated receive level on the downlink TCH after filtration. H indicates **INTERCELLHYST (BSC6900, BSC6910)** of the serving cell over the neighboring cell.

Negative handover is a type of better-cell handover. When **INTERCELLHYST (BSC6900, BSC6910)** is set to a value less than 64, the value of $H - 64$ is negative. In this case, even if the downlink receive level in a candidate cell is less than the downlink receive level in the serving cell, a negative handover to the candidate cell can be performed.

- If the serving cell and the handover candidate cells are at different layers but have the same priority, the following conditions must be met:

$$SS_DL(N) \geq T_layer(N) + H_layer(S) - 64$$

$$SS_DL(N)$$
 indicates the filtered BCCH receive level of a target cell.

$$T_layer(N)$$
 indicates **HOTHRES (BSC6900, BSC6910)** of the target cell.

$$H_layer(S)$$
 indicates **INTELEVHOHYST (BSC6900, BSC6910)** of the serving cell.
- If **LoadHoEn (BSC6900, BSC6910)** is set to **YES(Yes)** and the load in the serving cell is greater than **TRIGTHRES (BSC6900, BSC6910)**, better-cell handover can be triggered.
- A cell becomes the target cell if the previous conditions are met for **BETTERCELLLASTTIME (BSC6900, BSC6910)** within **BETTERCELLSTATTIME (BSC6900, BSC6910)**.

If both **EDOUTHOADEN (BSC6900, BSC6910)** and **NCELLPUNEN (BSC6900, BSC6910)** are set to **NO(No)** for the target cell, a better-cell handover is triggered when the previous conditions are met.

If **EDOUTHOADEN (BSC6900, BSC6910)** is set to **YES(Yes)** for the target cell, a better-cell handover is triggered when the previous conditions and the following condition are met: Filtered downlink receive level of the serving cell after power control compensation $\leq$ **HOTHRES (BSC6900, BSC6910)** – (**LEVHOHYST (BSC6900, BSC6910)** + **EDOUTHOOFFSET (BSC6900, BSC6910)** – 64) If **NCELLPUNEN (BSC6900, BSC6910)** is set to **YES(Yes)** for the target cell, the timer **NCELLPUNTM (BSC6900, BSC6910)** is started. After this timer is started, the value of **NCELLPUNLEV (BSC6900, BSC6910)** is subtracted from the filtered receive level of the candidate cell. If this timer expires, a better-cell handover is triggered. If this timer stops, no better-cell handover is triggered. The timer stops when the filtered BCCH receive level of the target cell is less than the value of **NCELLPUNSTPTH (BSC6900, BSC6910)**.

Target Cell Selection

The requirements for target cell selection are as follows:

- The serving cell cannot be selected as the target cell.
- The target cell has the highest priority in the candidate cell list.

3.5.11 SDCCH Handover

This section describes the feature GBFD-110608 SDCCH Handover.

SDCCH handover is a process in which the MS is handed over from an SDCCH to another SDCCH in an immediate assignment. SDCCH handover helps improve the access success rate of the MSs on the edge of the network, improving the network QoS.

The principle of SDCCH handover is the same as that of TCH handover. Regarding the procedure, an SDCCH handover involves measurement and MR reporting, MR processing, handover decision, and handover execution.

Whether an SDCCH handover can be performed is controlled by the **SIGCHANHOEN (BSC6900, BSC6910)** parameter. If an inter-BSC SDCCH handover

is required, both **SIGCHANHOEN (BSC6900, BSC6910)** and **INRBSCSDHOEN (BSC6900, BSC6910)** need to be set to **YES(Yes)**.

The handover decision algorithm for SDCCH handover is different from that for TCH handover in the following ways:

- The following handover algorithms support SDCCH handover:
Quick handover, TA handover, BQ handover, interference handover, no-DL-MR handover, edge handover, and fast-moving micro cell handover
- The following handover algorithms do not support SDCCH handover:
Enhanced dual-band network handover, load handover, better-cell handover, handover between a full-rate TCH and a half-rate TCH, better 3G cell handover, concentric cell handover, and tight BCCH handover

3.5.12 Other Handovers

Other handovers here refer to better 3G cell handover and tight BCCH handover.

Better 3G Cell Handover

See *Interoperability Between GSM and WCDMA Feature Parameter Description*.

Tight BCCH Handover

See *BCCH Dense Frequency Multiplexing Feature Parameter Description*.

3.6 Optimization of Co-BCCH Intra-frequency Neighboring Cell Handover Decision

If **BCCHIntraFreqHOOptimizeSw (BSC6900, BSC6910)** is set to **ON(On)**, calls carried by the BCCH TRX can be handed over to a co-BCCH intra-frequency neighboring cell only when the following condition is met:

The result of the level obtained by uplink auxiliary measurement on the neighboring cell minus that on the serving cell is greater than the value of **BCCHIntraFreqHOHyst (BSC6900, BSC6910)**.

Uplink auxiliary measurement is controlled by **BCCHIntraFreqHOULAssMeaSw (BSC6900, BSC6910)**.

The BSC sends the BTS a specific command instructing the BTS to start uplink auxiliary measurement when both of the following conditions are met:

- **BCCHIntraFreqHOULAssMeaSw (BSC6900, BSC6910)** is set to **ON(On)** for both the serving cell and co-BCCH intra-frequency neighboring cells
- The number of consecutive MRs containing the same neighboring cell is greater than or equal to the value of **ULAssistMeasStartThld (BSC6900, BSC6910)**.

After receiving the command, the BTS starts uplink auxiliary measurement on the serving cell and co-BCCH intra-frequency neighboring cell, and then reports the measurement result to the BSC.

The BTS needs to detect different TSCs and considers a TSC valid only when the detection result is greater than the value of ***TSCChkThld (BSC6900, BSC6910)***.

When the number of consecutive MRs not containing the same neighboring cell is greater than or equal to the value of ***ULAssistMeasStopThld (BSC6900, BSC6910)***, the BSC sends the serving cell and the BTS a specific command instructing the BTS to stop uplink auxiliary measurement. The BTS stops uplink auxiliary measurement upon receiving the command.

 NOTE

If ***BCCHIntraFreqHOULAssMeasSw (BSC6900, BSC6910)*** is set to **ON**, signaling traffic of the RSL accommodating the BCCH increases. In this situation, set ***MPMODE (BSC6900, BSC6910)*** to **MODE1_1** or **MODE2_1** if the Abis interface uses TDM transmission.

4 Related Features

NOTE

The following sub-features do not have feature IDs: quick handover, TA handover, BQ handover, interference handover, handover due to no downlink measurement report, edge handover, fast-moving micro cell handover, enhanced dual-band network handover, load handover, better cell handover, handover between a full-rate TCH and a half-rate TCH, better 3G cell handover, concentric cell handover, and tight BCCH handover

4.1 GBFD-110601 HUAWEI I Handover

Prerequisite Features

Feature ID	Feature Name	Description
GBFD-110801	Processing of Measurement Report	This feature requires either of the features.
GBFD-110802	Pre-processing of Measurement Report	

Mutually Exclusive Features

Feature ID	Feature Name	Description
GBFD-510501	HUAWEI II Handover	-
GBFD-113702	BCCH Carrier Frequency Hopping	Mutually exclusive with the uplink auxiliary measurement function
GBFD-510104	Multi-site Cell	Mutually exclusive with the uplink auxiliary measurement function

Impacted Features

Feature ID	Feature Name	Description
GBFD-191201	Uplink CoMP (Joint Rx)	When both the Uplink CoMP (Joint Rx) feature and uplink auxiliary measurement are enabled, if TRX resources are insufficient, uplink auxiliary measurement preempts TRX resources for uplink CoMP coordinated cells. As a result, uplink CoMP gains decrease.

4.2 GBFD-510501 HUAWEI II Handover

Prerequisite Features

Feature ID	Feature Name	Description
GBFD-110801	Processing of Measurement Report	This feature requires either of the features.
GBFD-110802	Pre-processing of Measurement Report	

Mutually Exclusive Features

Feature ID	Feature Name	Description
GBFD-110601	HUAWEI I Handover	-
GBFD-113702	BCCH Carrier Frequency Hopping	Mutually exclusive with the uplink auxiliary measurement function
GBFD-510104	Multi-site Cell	Mutually exclusive with the uplink auxiliary measurement function

Impacted Features

Feature ID	Feature Name	Description
GBFD-191201	Uplink CoMP (Joint Rx)	When both the Uplink CoMP (Joint Rx) feature and uplink auxiliary measurement are enabled, if TRX resources are insufficient, uplink auxiliary measurement preempts TRX resources for uplink CoMP coordinated cells. As a result, uplink CoMP gains decrease.

4.3 GBFD-110607 Direct Retry

Prerequisite Features

Feature ID	Feature Name	Description
GBFD-110601	HUAWEI I Handover	This feature requires either of the features.
GBFD-510501	HUAWEI II Handover	

Mutually Exclusive Features

None.

Impacted Features

Feature ID	Feature Name	Description
GBFD-111005	Enhanced Channel Assignment Algorithm	If channel allocation fails, directed retry can be triggered using a forced handover. This increases the assignment success rate.

4.4 GBFD-110608 SDCCH Handover

Prerequisite Features

Feature ID	Feature Name	Description
GBFD-110601	HUAWEI I Handover	This feature requires either of the features.
GBFD-510501	HUAWEI II Handover	

Mutually Exclusive Features

None.

Impacted Features

None.

4.5 GBFD-110801 Processing of Measurement Report

Prerequisite Features

None

Mutually Exclusive Features

Feature ID	Feature Name	Description
GBFD-110802	Pre-processing of Measurement Report	-

Impacted Features

None

4.6 GBFD-110802 Pre-processing of Measurement Report

Prerequisite Features

None

Mutually Exclusive Features

Feature ID	Feature Name	Description
GBFD-110801	Processing of Measurement Report	-

Feature ID	Feature Name	Description
MRFD-211703	2.0MHz Central Frequency point separation between GSM and UMTS mode(GSM)	When both Multi-site Cell and Pre-processing of Measurement Report are enabled, power control cannot be performed in a timely manner for the MS performing an inter-subsite handover. The receive level of the MS becomes very low, and mute voice may occur. In this case, 2.0MHz Central Frequency point separation between GSM and UMTS mode(GSM) cannot be enabled when these two features have been enabled.

Impacted Features

Feature ID	Feature Name	Description
GBFD-200202	User-level Precise Power Control	The neighboring cell handover compensation function of User-level Precise Power Control does not take effect when Pre-processing of Measurement Report is enabled.

4.7 GBFD-510102 Fast Move Handover

Prerequisite Features

Feature ID	Feature Name	Description
GBFD-110601	HUAWEI I Handover	This feature requires either of the features.
GBFD-510501	HUAWEI II Handover	
GBFD-510103	Chain Cell Handover	-

Mutually Exclusive Features

None.

Impacted Features

None.

4.8 GBFD-510103 Chain Cell Handover

Prerequisite Features

None.

Mutually Exclusive Features

None.

Impacted Features

None.

4.9 GBFD-117101 BTS power lift for handover

Prerequisite Features

Feature ID	Feature Name	Description
GBFD-110601	HUAWEI I Handover	This feature requires either of the features.
GBFD-510501	HUAWEI II Handover	

Mutually Exclusive Features

None.

Impacted Features

None.

4.10 GBFD-115504 AMR FR/HR Dynamic Adjustment

Prerequisite Features

Feature ID	Feature Name	Description
GBFD-110601	HUAWEI I Handover	This feature requires either of the features.
GBFD-510501	HUAWEI II Handover	
GBFD-115502	AMR HR	-

Mutually Exclusive Features

None

Impacted Features

Feature ID	Feature Name	Description
GBFD-200201	1.2 MHz Networking for BCCH TRXs	AMR FR/HR Dynamic Adjustment feature and the channel group handover function of 1.2 MHz Networking for BCCH TRXs feature are mutually exclusive. Ping-pong handovers will occur if both features are enabled. You are not advised to enable both features at the same time.

5 Network Impact

5.1 HUAWEI I Handover and HUAWEI II Handover

System Capacity

None

Network Performance

Either the HUAWEI I Handover feature or the HUAWEI II Handover feature must be enabled to increase the handover success rate.

5.2 Direct Retry

System Capacity

With the Direct Retry feature, a heavy-traffic cell can share its traffic with neighboring cells.

Network Performance

This feature improves the immediate assignment success rate and assignment success rate, but increases the number of handovers in heavy-traffic areas.

5.3 SDCCH Handover

System Capacity

The SDCCH Handover feature increases the CPU load because the BSC needs to process SDCCH measurement reports and make handover decisions.

Network Performance

This feature reduces the SDCCH call drop rate and enhances Um-interface security, but increases the number of handovers.

5.4 Processing of Measurement Report

System Capacity

None

Network Performance

None

5.5 Pre-processing of Measurement Report

System Capacity

None

Network Performance

None

5.6 Fast Move Handover

System Capacity

None

Network Performance

The Fast Move Handover feature enables the BSC to hand over calls to a better cell based on the path loss, increasing the handover success rate and decreasing the call drop rate.

5.7 Chain Cell Handover

System Capacity

None

Network Performance

When a fast-moving MS travels in chain cells along an urban backbone road, expressway, or high-speed railway, the Chain Cell Handover feature increases the handover success rate, improves voice quality, and decreases the call drop rate.

5.8 BTS power lift for handover

System Capacity

None

Network Performance

The BTS power lift for handover feature increases the handover success rate and decreases the call drop rate in areas with poor signal coverage.

This feature increases the probability that the BTS transmits signals with full power. The larger the probability that the BTS transmits signals with full power, the stronger the co-channel or adjacent channel interference to other cells, and the greater the fluctuation of the downlink HQL.

5.9 AMR FR/HR Dynamic Adjustment

System Capacity

The AMR FR/HR Dynamic Adjustment feature enables automatic TCHF-to-TCHH handovers when traffic is heavy, improving system capacity.

Network Performance

- When traffic is heavy: The AMR FR/HR Dynamic Adjustment feature enables automatic TCHF-to-TCHH handovers, which increases the number of handovers, deteriorates voice quality, but improves system capacity.
- When traffic is light: The AMR FR/HR Dynamic Adjustment feature enables automatic TCHH-to-TCHF handovers, which increases the number of handovers but improves voice quality.

6 Engineering Guidelines

6.1 When to Use Handover

6.1.1 HUAWEI I Handover

It is recommended that the HUAWEI I Handover feature be used.

6.1.2 HUAWEI II Handover

It is recommended that the HUAWEI II Handover feature be used in a single-band and dual-band hybrid network.

6.1.3 Direct Retry

It is recommended that the Direct Retry feature be used in cells where the assignment success rate is low due to high load.

6.1.4 SDCCH Handover

When authentication is enabled on a network, SDCCH immediate assignment takes an extensive amount of time. This increases the probability of handovers due to MS mobility and a high interception risk over the Um interface. If authentication is not enabled on the network, the SDCCH handover feature is not recommended.

6.1.5 Processing of Measurement Report

Either the Processing of Measurement Report feature or the Pre-processing of Measurement Report feature must be enabled. The Processing of Measurement Report feature is recommended.

6.1.6 Pre-processing of Measurement Report

It is recommended that the Pre-processing of Measurement Report feature be used when the BSC load is high.

6.1.7 Fast Move Handover

It is recommended that the Fast Move Handover feature be used in chain cells along high-speed railways or expressways.

6.1.8 Chain Cell Handover

It is recommended that the Chain Cell Handover feature be used in chain cells along high-speed railways or expressways.

6.1.9 BTS power lift for handover

It is recommended that the BTS power lift for handover feature be used in scenarios where the handover success rate is low due to coverage problems.

6.1.10 AMR FR/HR Dynamic Adjustment

It is recommended that the AMR FR/HR Dynamic Adjustment feature be used in scenarios where traffic is heavy and the proportion of half-rate services is high.

6.2 Required Information

6.2.1 HUAWEI I Handover

None

6.2.2 HUAWEI II Handover

Before deploying the HUAWEI II Handover feature, collect the frequency planning information about the cell.

6.2.3 Direct Retry

Before deploying the Direct Retry feature, collect the following information:

- Assignment success rate in the cell
- Proportion of assignment failures due to lack of channels
- Traffic volume in the cell
- Mean Duration of Busy State for Channels

6.2.4 SDCCH Handover

Before deploying the SDCCH Handover feature, collect information about whether authentication is enabled on the network from the CN.

6.2.5 Processing of Measurement Report

None

6.2.6 Pre-processing of Measurement Report

Before deploying the Pre-processing of Measurement Report feature, collect the CPU usage of BSC boards. If the overall CPU load on the BSC is high and more functions or capacity expansion is required, enable this feature so that the BTS preprocesses measurement reports.

6.2.7 Fast Move Handover

Before deploying the Fast Move Handover feature, collect information about the environment within the cell.

6.2.8 Chain Cell Handover

Before deploying the Chain Cell Handover feature, collect information about the environment within the cell.

6.2.9 BTS power lift for handover

Before deploying the BTS power lift for handover feature, collect the following information:

- Handover success rate in the cell
- Number of handover failures due to weak uplink or downlink signal strength
- Signal quality in the cell

6.2.10 AMR FR/HR Dynamic Adjustment

Before deploying the AMR FR/HR Dynamic Adjustment feature, collect the traffic volume and the proportion of half-rate services in the cell.

6.3 Network Planning

None

6.4 Deploying HUAWEI I Handover

6.4.1 Deployment Requirements

Table 6-1 Deployment requirements

Aspect	Requirement
Related features	See 4 Related Features .
BSC	None
BTS	None
GSM networking	None

Aspect	Requirement
MS	None
MSC	None
License	None
Others	None

6.4.2 Precautions

- After a quick handover, BQ handover, or TA handover is complete, a handover penalty is imposed on the original cell during the specified penalty period to prevent an immediate handover back to the original cell.
- A BQ handover can be an intra- or inter-cell handover. If **Bad Quality Handover Optimization Allowed** is set to **YES(Yes)**, only an inter-cell BQ handover is allowed.
- If **TA Threshold** for a co-site neighboring cell is lower than or equal to that for the serving cell, a TA handover to the neighboring cell is not allowed.
- An interference handover can be an intra- or inter-cell handover. If **Interference Handover Optimization Allowed** is set to **YES(Yes)**, only an intra-cell interference handover is allowed. However, if intra-cell handovers have occurred several consecutive times, an intra-cell handover is prohibited within a specified period.
- On an enhanced dual-band network, an MS cannot be handed over to a cell in the same group when overlaid/underlaid load handovers, specified by **Load HO From UL Subcell to OL Subcell Allowed** and **Load HO of OL Subcell to UL Subcell Enabled**, respectively, are allowed. This prevents conflicts between a load handover on a common network and an overlaid/underlaid handover on an enhanced dual-band network.
- A PBGT handover to a cell in the same group is not allowed if it is triggered between two cells on an enhanced dual-band network.
- After the fast-moving micro cell handover is successful, the penalty is performed on all the neighboring micro cells.
- An SDCCH handover can be a quick handover, TA handover, BQ handover, quick level drop handover, interference handover, no-DL-MR handover, edge handover, and fast-moving micro cell handover. It cannot be an enhanced dual-band network handover, load handover, inter-layer handover, PBGT handover, AMR handover, better 3G cell handover, concentric cell handover, or tight BCCH handover.
- When **BCCHIntraFreqHOULAssMeaSw (BSC6900, BSC6910)** is set to **ON(On)**, the intra-frame bit offset must be set to **0**. Specifically,
 - a. When Soft-Synchronization Network is enabled, **SYMOFFSET (BSC6900, BSC6910)** must be set to **0**.
 - b. When BTS GPS Synchronization or Clock over IP support 1588v2 is enabled, **GSM_FBOFFSET** must be set to **0**.

Otherwise, the uplink auxiliary measurement result becomes less accurate, and the handover success rate decreases and call drop rate increases.

- It is recommended that the value of **BQMARGIN (BSC6900, BSC6910)** be differentiated for co-BCCH intra-frequency neighboring cells. Specifically, the value of this parameter for co-BCCH intra-frequency neighboring cells must be 1 dB to 5 dB greater than that for inter-frequency neighboring cells. A larger difference indicates a higher probability that MSs are handed over to co-BCCH intra-frequency neighboring cells.

6.4.3 Activation

Using MML Commands

- Step 1** Run the **SET GCELLHOBASIC** command with **Current HO Control Algorithm** set to **HOALGORITHM1(Handover algorithm I)**.
- Step 2** Run the **SET GCELLHOEMG** command with **Current HO Control Algorithm** set to **HOALGORITHM1(Handover algorithm I)**.
- Step 3** Run the **SET GCELLHOFITPEN** command with **Current HO Control Algorithm** set to **HOALGORITHM1(Handover algorithm I)**.
- Step 4** Run the **SET GCELLHOIUO** command with **Current HO Control Algorithm** set to **HOALGORITHM1(Handover algorithm I)**.
- Step 5** Run the **SET GCELLHOAD** command with **Max. Num. of HO Attempts** set according to onsite requirements.
- Step 6** (Optional) To improve the handover and power control performance when the BCCH TRX participates in baseband frequency hopping, run the **SET GCELLCCCH** command with **PWRC** set to **YES(Yes)** and run the **SET GCELLTHEXT** command with **Power Control and Handover CMP CON Sw** set to **YES(Yes)**.
- Step 7** Activate the quick handover algorithm by performing the following steps:
 - Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
 - Run the **MOD G2GNCELL** command with its parameters set as follows:
 - Set **Chain Neighbor Cell** to **YES(Yes)**.
 - Set **Chain Neighbour Cell Type** and **Quick Handover Offset for Neighbor Cell** according to onsite requirements.
 - Run the **SET GCELLHOBASIC** command with **Quick Handover Enable** set to **YES(Yes)**. If a fast move handover is required, set **Quick PBGT Handover Enable** to **YES(Yes)**.

NOTE

Quick handovers consist of fast move handovers and frequency offset handovers. **Quick Handover Enable** controls whether to allow fast move handovers and frequency offset handovers. **Quick PBGT Handover Enable** further controls whether to allow fast move handovers.

To enable the BSC to accurately make handover decisions, you are advised to set **Quick PBGT Handover Enabled** to **YES(Yes)** in high-speed railway scenarios or chain cells along the highway. In other scenarios, set **Quick PBGT Handover Enabled** to **NO(No)**.

- Run the **SET GCELLHOFAST** command with its parameters set as follows:
 - Set **Handover Direction Forecast Enable** to **YES(Yes)**.
 - Set **Quick Handover Up Trigger Level**, **Quick Handover Down Trigger Level**, **Quick Move Speed Threshold**, **Serving Cell Filter Length MR Number**, **Neighbor Cell Filter Length MR Number**, **Ignore Measurement Report Number**, **Quick Handover Punish Time**, **Quick Handover Punish Value**, **Quick Handover Offset**, **Handover Direction Forecast Statistic Times**, **Handover Direction Forecast Last Times**, **Frequency Shift Handover Statistic Time**, and **Frequency Shift Handover Duration** according to onsite requirements.

 NOTE

Run the **MOD GEXT2GCELL** command with **Quick Handover Offset**, **Quick Handover Punish Time**, and **Quick Handover Punish Value** set to appropriate values for an external neighboring GSM cell.

- (Optional) Run the **SET GCELLOTHEXT** command with **UL Frequency Adjust Switch**, **DL Frequency Adjust Switch**, and **UL Frequency Adjust Value** set based on the frequency offset reported by the BTS.

Step 8 Activate the TA handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **TA HO Allowed** set to **YES(Yes)**.
- Run the **SET GCELLHOEMG** command with **TA Threshold** set according to onsite requirements.
- Run the **SET GCELLHOFITPEN** command with **Penalty Level after TA HO** and **Penalty Time after TA HO** set according to onsite requirements.

Step 9 Activate the BQ handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **BQ HO Allowed** set to **YES(Yes)**.
- (Optional) For a non-AMR call, run the **SET GCELLHOEMG** command with **DL Qual. Threshold** and **UL Qual. Threshold** set according to onsite requirements.
- (Optional) For an AMR FR or AMR HR call, run the **SET GCELLAMRQUL** command with **UL Qual. Limit for AMR FR**, **DL Qual. Limit for AMR FR**, **UL Qual. Limit for AMR HR**, and **DL Qual. Limit for AMR HR** set according to onsite requirements.
- Run the **SET GCELLHOFITPEN** command with **Penalty Level after BQ HO** and **Penalty Time after BQ HO** set according to onsite requirements.

Step 10 Activate the rapid level drop handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **Rx_Level_Drop HO Allowed** set to **YES(Yes)** and **Edge HO UL RX_LEV Threshold** set according to onsite requirements.
- Run the **SET GCELLHOEMG** command with the following parameters set according to onsite requirements: **Filter Parameter A1**, **Filter Parameter A2**, **Filter Parameter A3**, **Filter Parameter A4**, **Filter Parameter A5**, **Filter Parameter A6**, **Filter Parameter A7**, **Filter Parameter A8**, and **Filter Parameter B**.

Step 11 Activate the interference handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with its parameters set as follows:
 - Set **Interference HO Allowed** to **YES(Yes)**.
 - Set **Intracell HO Allowed** and **Inter-layer HO Threshold** according to onsite requirements.

 NOTE

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with **Inter-layer HO Threshold** set according to onsite requirements.

- Run the **SET GCELLHOAD** command with **Forbidden time after MAX Times** set according to onsite requirements.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **Adjacent Cell Inter-layer HO Hysteresis** set according to onsite requirements.
- Run the **SET GCELLAMRQUL** command with the following parameters set according to onsite requirements: **Interfere HO Qual. Thresh 1 for Non-AMR FR**, **Interfere HO Qual. Thresh 2 for Non-AMR FR**, **Interfere HO Qual. Thresh 3 for Non-AMR FR**, **Interfere HO Qual. Thresh 4 for Non-AMR FR**, **Interfere HO Qual. Thresh 5 for Non-AMR FR**, **Interfere HO Qual. Thresh 6 for Non-AMR FR**, **Interfere HO Qual. Thresh 7 for Non-AMR FR**, **Interfere HO Qual. Thresh 8 for Non-AMR FR**, **Interfere HO Qual. Thresh 9 for Non-AMR FR**, **Interfere HO Qual. Thresh 10 for Non-AMR FR**, **Interfere HO Qual. Thresh 11 for Non-AMR FR**, **Interfere HO Qual. Thresh 12 for Non-AMR FR**, and **Interfere HO Qual. Thresh Offset for AMR FR**.

Step 12 Activate the no-DL-MR handover algorithm by performing the following steps:

- Run the **SET GCELLHOEMG** command with its parameters set as follows:
 - Set **No DL Mr.HO Allowed** to **YES(Yes)**.
 - Set **No DL Mr.UL Qual HO Limit**, **Cons.No DL Mr.HO Allowed Limit**, **No DL Mr. HO Statistic Time**, and **No DL Mr. HO Last Time** according to onsite requirements.
- Run the **SET GCELLHOFITPEN** command with **Filter Length for TCH Qual** and **Filter Length for SDCCH Qual** set according to onsite requirements.

Step 13 Activate the enhanced dual-band network handover algorithm by performing the following steps:

- Run the **MOD GCELL** command with its parameters set as follows:
 - Set **Cell IUO Type** to **EDB_cell(Enhanced Double Frequency Cell)**.
 - Set **Cell Inner/Extra Property** and **Same Group Cell Index Type** according to onsite requirements.
 - Set **Same Group Cell Index** or **Same Group Cell Name** to an appropriate value.
- Run the **SET GCELLHOEDBPARA** command with the following parameters set according to onsite requirements: **Load HO Allowed**, **UL Subcell General Overload Threshold**, **Inner Cell Serious OverLoad Thred**, **Incoming OL Subcell HO Level TH**, **UL Subcell Load Hierarchical HO Periods**, **MOD Step LEN of UL Load HO Period**, **Step Length of UL Subcell Load HO**, **UL Subcell Lower Load Threshold**, **Subcell HO Allowed Flow Control Level**,

Outgoing OL Subcell HO Level TH, OL Subcell Load Diversity HO Period, Step Length of OL Subcell Load HO, Distance Between Boundaries of Subcells, and Distance Hysteresis Between Boundaries.

Step 14 Activate the load handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with its parameters set as follows:
 - Set **Load Handover Support** to **YES(Yes)**.
 - Set **Edge HO DL RX_LEV Threshold** and **Inter-layer HO Threshold** according to onsite requirements.
- Run the **SET GCELLHOAD** command with the following parameters set according to onsite requirements: **System Flux Threshold for Load HO**, **Load HO Threshold**, **Load handover Load Accept Threshold**, **Load HO Bandwidth**, **Load HO Step Period**, and **Load HO Step Level**.

 NOTE

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with **Load handover Load Accept Threshold** set according to onsite requirements. If a cell is configured with external neighboring UMTS cells, run the **MOD GEXT3GCELL** command with **Load handover Load Accept Threshold** set according to onsite requirements.

- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **Adjacent Cell Inter-layer HO Hysteresis** set according to onsite requirements.

Step 15 Activate the edge handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with its parameters set as follows:
 - Set **Edge HO Allowed** to **YES(Yes)**.
 - Set **Edge HO UL RX_LEV Threshold**, **Edge HO DL RX_LEV Threshold**, **Handover Algorithm I Edge HO Watch Time**, and **Handover Algorithm I Edge HO Valid Time** according to onsite requirements.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **Adjacent Cell Inter-layer HO Hysteresis** set according to onsite requirements.
- (Optional) To set different edge handover thresholds for different neighboring cells, perform the following steps:
- Run the **SET GCELLHOBASIC** command with **Enhanced Outgoing Cell Handover Allowed** set to **YES(Yes)**.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **Adjacent Cell Inter-layer HO Hysteresis** set according to onsite requirements.

Step 16 Activate the fast-moving micro cell handover algorithm by performing the following steps:

- Run the **SET GCELLBASICPARA** command with **Layer of The Cell** set according to onsite requirements.
- Run the **SET GCELLHOBASIC** command with **MS Fast Moving HO Allowed** set to **YES(Yes)** and **Inter-layer HO Threshold** set according to onsite requirements.
- Run the **SET GCELLHOAD** command with the following parameters set according to onsite requirements: **MS Fast-moving Watch Cells**, **MS Fast-moving Valid Cells**, **MS Fast-moving Time Threshold**, **Penalty on MS Fast Moving HO**, and **Penalty Time on Fast Moving HO**.

 NOTE

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with the following parameters set according to onsite requirements: **Layer of the cell**, **Inter-layer HO Threshold**, **Penalty Time on Fast Moving HO**, and **Penalty on Fast Moving HO**.

- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **Adjacent Cell Inter-layer HO Hysteresis** set according to onsite requirements.

Step 17 Activate the inter-layer handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **Level HO Allowed** set to **YES(Yes)** and **Inter-layer HO Threshold** set according to onsite requirements.
- Run the **SET GCELLBASICPARA** command with **Layer of The Cell** and **Cell Priority** set according to onsite requirements.

 NOTE

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with **Layer of the cell** and **Cell Priority** set according to onsite requirements.

- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with the following parameters set according to onsite requirements: **Adjacent Cell Inter-layer HO Hysteresis**, **Layer HO Watch Time**, and **Layer HO Valid Time**.

 NOTE

To set different inter-layer handover conditions for different neighboring cells, do as follows:

- Set **Neighboring Cell Penalty Switch** to **YES(Yes)**.
- Set **Penalty Stop Level Threshold**, **Penalty Timer Length**, and **Level Penalty Value on Neighboring Cell** according to onsite requirements.

Step 18 Activate the PBGT handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **PBGT HO Allowed** set to **YES(Yes)**.
- Run the **SET GCELLBASICPARA** command with **Layer of The Cell** and **Cell Priority** set according to onsite requirements.

 NOTE

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with **Layer of the cell** and **Cell Priority** set according to onsite requirements.

- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with its parameters set as follows:
 - Set **Current HO CTRL Algorithm in Source Cell** to **HOALGORITHM1(Handover algorithm I)**.
 - Set **Adjacent Cell Inter-layer HO Hysteresis**, **PBGT Watch Time**, and **PBGT Valid Time** according to onsite requirements.

 NOTE

To set different PBGT handover conditions for different neighboring GSM cells, do as follows:

- Set **Neighboring Cell Penalty Switch** to **YES(Yes)**.
- Set **Penalty Stop Level Threshold**, **Penalty Timer Length**, and **Penalty Stop Level Threshold** according to onsite requirements.

Step 19 Activate the AMR handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with its parameters set as follows:
 - Set **Intracell F-H HO Allowed** to **YES(Yes)**.
 - Set **Intracell F-H HO Stat Time**, **Intracell F-H HO Last Time**, **F2H HO Threshold**, and **H2F HO Threshold** according to onsite requirements.
- Run the **SET GTRXDEV** command with **TCH Rate Adjust Allow** set to **YES(Yes)**.
- Run the **SET GCELLCCACCESS** command with at least **FULL_RATE_VER1(Full-rate VER 1)**, **FULL_RATE_VER3(Full-rate VER 3)**, and **HALF_RATE_VER3(Half-rate VER 3)** selected under **Speech Version**.
- Run the **SET BSCBASIC** command with **Um Interface Tag**, **Abis Interface Tag**, and **A Interface Tag** set to **GSM_PHASE_2Plus**.
- Run the **SET GCELLCHMGAD** command with **AMR TCH/H Prior Allowed** and **AMR TCH/H Prior Cell Load Threshold** set according to onsite requirements.

Step 20 Activate the capacity- and quality-based TCHF-TCHH handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with its parameters set as follows:
 - Set **Intracell F-H HO Allowed** to **HOBAS ECAPANDQUAL(HOBASECAPANDQUAL)**.
 - Set **AMR F-H Ho Allowed** and **Non-AM R F-H Ho Allowed** to **YES(Yes)**.
 - Set the following parameters according to onsite requirements:
AMR F-H Traffic Threshold, **F-H Ho Duration**, **F-H Ho Period**, **AMR F-H Ho Pathloss Adjust Step**, **AMR F-H Ho ATCB Adjust Step**, **AMR F-H Ho Pathloss Threshold**, **F-H Pathloss Offset Overlay**, **AMR F-H Ho ATCB Threshold**, **F-H ATCB Offset Overlay**, **AMR F-H Ho Qual. Threshold**, **F-H Ho Stat. Time**, **F-H Ho Last Time**, **AMR H-F Traffic Threshold**, **H-F Ho Duration**, **AMR H-F Ho ATCB Threshold**, **H-F ATCB Offset Overlay**, **AMR H-F Ho Pathloss Threshold**, **H-F Pathloss Offset Overlay**, **AMR H-F Ho Qual.**

Threshold, H-F Ho Stat. Time, H-F Ho Last Time, Non-AMR F-H Traffic Threshold, Non-AMR F-H Ho Pathloss Adjust Step, Non-AMR F-H Ho ATCB Adjust Step, Non-AMR F-H Ho Pathloss Threshold, Non-AMR F-H Ho ATCB Threshold, Non-AMR H-F Ho Qual. Threshold, Non-AMR H-F Traffic Threshold, Non-AMR H-F Ho ATCB Threshold, Non-AMR H-F Ho Pathloss Threshold, and Non-AMR H-F Ho Qual. Threshold

- Run the **SET GTRXDEV** command with **TCH Rate Adjust Allow** set to **YES(Yes)**.
- Run the **SET GCELLCCACCESS** command with **FULL_RATE_VER3(Full-rate VER 3)** and **HALF_RATE_VER3(Half-rate VER 3)** selected under **Speech Version**.
- Run the **SET BSCBASIC** command with **A Interface Tag** set to **GSM_PHASE_2Plus**.

Step 21 Activate the SDCCH handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **SDCCH HO Allowed** set to **YES(Yes)**.
- Run the **SET GCELLHOCTRL** command with **Inter-BSC SDCCH HO ALLOWed** set according to onsite requirements.

----End

MML Command Examples

//Deploying HUAWEI I Handover

```
SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1;  
SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1;  
SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1;  
SET GCELLHOIUO: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1;  
SET GCELLHOAD: IDTYPE=BYID, CELLID=0, HOTRYCNT=3;
```

//Activating the quick handover algorithm for the BSC6900

```
MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,  
NCELLTYPE=HANDOVERNCELL, ISCHAINNCELL=YES,  
CHAINNCELLTYPE=QUICK_HO_NCELL_TYPE_A, ADJHOOFFSET=1;  
  
SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,  
QUICKHOEN=YES, QUICKPBGTHOEN=YES;  
  
SET GCELLHOFAST: IDTYPE=BYID, CELLID=0, HOUPTRIGE=50, HODOWNTRIGE=50,  
MOVESPEEDTHRES=35, SCELLFILTER=4, NCELLFILTER=4, IGNOREMRNUM=1,  
TIMEPUNISH=10, HOPUNISHVALUE=63, HOOFFSET=68, MSLEVSTRQPBGT=NO,  
HODIRFORECASTEN=YES, HODIRSTATIME=3, HODIRLASTTIME=2,  
AFCHOSTATTIME=4, AFCHOLASTTIME=3;  
  
SET GCELLOTHEXT: IDTYPE=BYID, CELLID=0, FREQADJ=YES, DLFREQADJ=YES,  
FREQADJVALUE=36671;
```

//Activating the quick handover algorithm for the BSC6910

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, ISCHAINNCELL=YES,
CHAINNCELLTYPE=QUICK_HO_NCELL_TYPE_A, ADJHOOFFSET=1;

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
QUICKHOEN=YES, QUICKPBGTHOEN=YES;

SET GCELLHOFAST: IDTYPE=BYID, CELLID=0, HOUPTRIGE=50, HODOWNTRIGE=50,
MOVESPEEDTHRES=35, SCELLFILTER=4, NCELLFILTER=4, IGNOREMRNUM=1,
TIMEPUNISH=10, HOPUNISHVALUE=63, HOOFFSET=68, MSLEVSTRQPBGT=NO,
HODIRFORECASTEN=YES, HODIRSTATIME=3, HODIRLASTTIME=2,
AFCHOSTATTIME=4, AFCHOLASTTIME=3;

SET GCELLOTHEXT: IDTYPE=BYID, CELLID=0, FREQADJ=YES, DLFREQADJ=YES,
FREQADJVALUE=36671;

//Activating the TA handover algorithm

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
TAHOEN=YES;

SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
TALIMIT=125;

SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
SSTAPUNISH=40, TIMETAPUNISH=10;

//Activating the BQ handover algorithm

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
BQHOEN=YES;

SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
DLQUALIMIT=55, ULQUALIMIT=60;

SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
SSBQPUNISH=63, TIMEBQPUNISH=10;

//Activating the rapid level drop handover algorithm

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
RXQCKFALLHOEN=YES, ULEDGETHRES=10;

SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
FLTPARAA1=10, FLTPARAA2=10, FLTPARAA3=10, FLTPARAA4=10, FLTPARAA5=10,
FLTPARAA6=10, FLTPARAA7=10, FLTPARAA8=10, FLTPARAB=0;

//Activating the interference handover algorithm for the BSC6900

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
INTRACELLHOEN=YES, INTERFHOEN=YES, HOTHRES=2;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, BANTIME=20;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

SET GCELLAMRQUL: IDTYPE=BYID, CELLID=0, RXQUAL1=70, RXQUAL2=60,
RXQUAL3=59, RXQUAL4=58, RXQUAL5=57, RXQUAL6=56, RXQUAL7=55,
RXQUAL8=54, RXQUAL9=53, RXQUAL10=52, RXQUAL11=51, RXQUAL12=50;

//Activating the interference handover algorithm for the BSC6910

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
INTRACELLHOEN=YES, INTERFHOEN=YES, HOTHRES=2;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, BANTIME=20;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

SET GCELLAMRQUL: IDTYPE=BYID, CELLID=0, RXQUAL1=70, RXQUAL2=60,
RXQUAL3=59, RXQUAL4=58, RXQUAL5=57, RXQUAL6=56, RXQUAL7=55,
RXQUAL8=54, RXQUAL9=53, RXQUAL10=52, RXQUAL11=51, RXQUAL12=50;

//Activating the no-DL-MR handover algorithm

SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
NODLMRHOEN=YES, NODLMRHOQUALLIMIT=50, NODLMRHOALLOWLIMIT=8,
NODLMRHOSTATTIME=1, NODLMRHOLASTTIME=1;

SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
DATAQUAFLTLEN=4, SIGQUAFLTLEN=2;

//Activating the enhanced dual-band network handover algorithm

MOD GCELL: IDTYPE=BYID, CELLID=0, IUOTP=EDB_cell, CELLINEXTP=Inner,
DBLFREQADJIDTYPE=BYID, DBLFREQADJCID=1;

SET GCELLHOEDBPARA: IDTYPE=BYID, CELLID=0, OUTASSOPTEN=YES;

//Activating the load handover algorithm for the BSC6900

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
LoadHoEn=YES, DLEDGETHRES=20, HOTHRES=25;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, SYSFLOWLEV=10, TRIGTHRES=90,
LoadAccThres=80, LOADOFFSET=25, LOADHOPERIOD=10, LOADHOSTEP=5;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

//Activating the load handover algorithm for the BSC6910

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
LoadHoEn=YES, DLEDGETHRES=20, HOTHRES=25;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, SYSFLOWLEV=10, TRIGTHRES=90,
LoadAccThres=80, LOADOFFSET=25, LOADHOPERIOD=10, LOADHOSTEP=5;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

//Activating the edge handover algorithm for the BSC6900

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1, FRINGEHOEN=YES, ULEDGETHRES=10, DLEDGETHRES=20, EDGESTAT1=6, EDGELAST1=4;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

//Activating the edge handover algorithm for the BSC6910

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1, FRINGEHOEN=YES, ULEDGETHRES=10, DLEDGETHRES=20, EDGESTAT1=6, EDGELAST1=4;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

//Activating the fast-moving micro cell handover algorithm for the BSC6900

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, LAYER=3;

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1, QCKMVHOEN=YES, HOTHRES=25;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, QCKSTATCNT=3, QCKTRUECNT=2, QCKTIMETH=15, SPEEDPUNISH=30, SPEEDPUNISHT=4;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

//Activating the fast-moving micro cell handover algorithm for the BSC6910

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, LAYER=3;

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1, QCKMVHOEN=YES, HOTHRES=25;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, QCKSTATCNT=3, QCKTRUECNT=2, QCKTIMETH=15, SPEEDPUNISH=30, SPEEDPUNISHT=4;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

//Activating the inter-layer handover algorithm for the BSC6900

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1, LEVHOEN=YES, HOTHRES=25;

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, LAYER=3, PRIOR=PRIOR-3;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67, LEVSTAT=6, LEVLAST=4;

//Activating the inter-layer handover algorithm for the BSC6910

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1, LEVHOEN=YES, HOTHRES=25;

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, LAYER=3, PRIOR=PRIOR-3;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67,
LEVSTAT=6, LEVLAST=4;

//Activating the PBGT handover algorithm for the BSC6900

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
PBGTHOEN=YES;

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, LAYER=3, PRIOR=PRIOR-3;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67, PBGTSTAT=6, PBGTLAST=4;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
NCELLTYPE=HANDOVERNCELL, SRCHOCTRLSWITCH=HOALGORITHM1,
PBGTMARGIN=68, PBGTSTAT=6, PBGTLAST=4;

//Activating the PBGT handover algorithm for the BSC6910

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
PBGTHOEN=YES;

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, LAYER=3, PRIOR=PRIOR-3;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67,
PBGSTAT=6, PBGLAST=4;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL,
SRCHOCTRLSWITCH=HOALGORITHM1, PBGTMARGIN=68, PBGTSTAT=6,
PBGLAST=4;

//Activating the AMR handover algorithm

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
INTRACELFHHOEN=YES, INFHHOEN=10, INFHHOEN=8, INHO2H2H=25,
INHO2H2H=12;

SET GTRXDEV: TRXID=0, TCHAJFLAG=YES;

SET GCELLCCACCESS: IDTYPE=BYID, CELLID=0,
VOICEVER=FULL_RATE_VER1-1&FULL_RATE_VER2-0&FULL_RATE_VER3-1&HALF_R
ATE_VER1-1&HALF_RATE_VER2-0&HALF_RATE_VER3-1&FULL_RATE_VER5-0;

SET BSCBASIC: AVer=GSM_PHASE_2Plus, UmVer=GSM_PHASE_2Plus,
AbisVer=GSM_PHASE_2Plus;

SET GCELLCHMGAD: IDTYPE=BYID, CELLID=0, AMRTCHHPRIORALLOW=ON,
AMRTCHHPRIORLOAD=55;

//Activating the capacity- and quality-based TCHF-TCHH handover algorithm

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH= HOALGORITHM1,
INTRACELFHHOEN=HOBASECAPANDQUAL, AMRFULLTOHALFHODURATION=5,
AMRFULLTOHALFHOTHRESH=75, FULLTOHALFHODURATION=5,

```

FULLTOHALFHOPERIOD=1, AMRFULLTOHALFHOPATHADJSTEP=1,
AMRFULLTOHALFHOOATCBADJSTEP=2, AMRFULLTOHALFHOPATHTHRESH=100,
FULLTOHALFHOPATHOFFSET=10, AMRFULLTOHALFHOOATCBTHRESH=20,
FULLTOHALFHOOATCBOFFSET=20, AMRFULLTOHALFHOOQUALTHRESH=0,
FULLTOHALFHOOATTIME=6, FULLTOHALFHOOLASTTIME=4,
AMRHALFTOFULLHOOTHRESH=45, AMRHALFTOFULLHOQUALALLOW=YES,
HALFTOFULLHODURATION=5, AMRHALFTOFULLHOATCBTHRESH=4,
HALFTOFULLATCBOFFSET=20, AMRHALFTOFULLHOPATHTHRESH=108,
HALFTOFULLHOPATHOFFSET=10, AMRHALFTOFULLHOQUALTHRESH=3,
HALFTOFULLHOATTIME=6, HALFTOFULLHOOLASTTIME=4,
NOAMRFULLTOHALFHOOALLOW=YES, NOAMRFULLTOHALFHTHRESH=85,
NOAMRFULLTOHALFHOPATHADJSTEP=1,
NOAMRFULLTOHALFHOOATCBADJSTEP=2,
NOAMRFULLTOHALFHOPATHTHRESH=95,
NOAMRFULLTOHALFHOOATCBTHRESH=30,
NOAMRFULLTOHALFHOOQUALTHRESH=0, NOAMRHALFTOFULLTHRESH=25,
NOAMRHALFTOFULLHOQUALALLOW=YES,
NOAMRHALFTOFULLHOATCBTHRESH=14,
NOAMRHALFTOFULLHOPATHTHRESH=103,
NOAMRHALFTOFULLHOQUALTHRESH=2;

SET GTRXDEV: TRXID=0, TCHAJFLAG=YES;

SET GCELLCCACCESS: IDTYPE=BYID, CELLID=0,
VOICEVER=FULL_RATE_VER1-1&FULL_RATE_VER2-0&FULL_RATE_VER3-1&HALF_R
ATE_VER1-1&HALF_RATE_VER2-0&HALF_RATE_VER3-1&FULL_RATE_VER5-0;

SET BSCBASIC: AVer=GSM_PHASE_2;

//Activating the SDCCH handover algorithm

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH= HOALGORITHM1,
SIGCHANHOEN=YES;

SET GCELLHOCTRL: IDTYPE=BYID, CELLID=0, INRBSCSDHOEN=YES;

```

Using the MAE-Deployment

NOTE

When configuring the feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. Configure the parameters of a single object before a batch modification. Perform a batch modification before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-2](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-2 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLHOBASIC	BSC6900/BSC6910	Current HO Control Algorithm	HOCTRLSWITCH	Yes
2	GCELLHOEMG	BSC6900/BSC6910	Current HO Control Algorithm	HOCTRLSWITCH	Yes
3	GCELLHOFITPEN	BSC6900/BSC6910	Current HO Control Algorithm	HOCTRLSWITCH	Yes
4	GCELLHOIUO	BSC6900/BSC6910	Current HO Control Algorithm	HOCTRLSWITCH	Yes
5	GCELLHOAD	BSC6900/BSC6910	Max. Num. of HO Attempts	HOTRYCNT	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
6	GCELLCCH Configure this MO if you need to improve the handover and power control performance when the BCCH TRX participates in baseband frequency hopping.	BSC6900/ BSC6910	PWRC	CANPC	Yes
7	GCELLOTH EXT Configure this MO if you need to improve the handover and power control performance when the BCCH TRX participates in baseband frequency hopping.	BSC6900/ BSC6910	Power Control and Handover CMP CON Sw	PCHOCMP CON	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
8	G2GNCELL Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
9	G2GNCELL Configure this MO to activate the quick handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
10	G2GNCELL Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	Chain Neighbor Cell	ISCHAINNCELL	No
			Chain Neighbour Cell Type	CHAINNCELLTYPE	
			Quick Handover Offset for Neighbor Cell	ADJHOOFFSET	
11	GCELLHOBASIC Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	Quick Handover Enable	QUICKHOPEN	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification on Center
12	GCELLHOB ASIC Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	Quick PBGT Handover Enable	QUICKPBGTHOEN	Yes
13	GCELLHOF AST Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	Handover Direction Forecast Enable	HODIRFORECASTEN	Yes
			Quick Handover Up Trigger Level	HOUPTRIGGER	
			Quick Handover Down Trigger Level	HODOWNTRIGE	
			Quick Move Speed Threshold	MOVESPEEDTHRES	
			Serving Cell Filter Length MR Number	SCCELLFILTER	
			Neighbor Cell Filter Length MR Number	NCELLFILTER	
			Ignore Measurement Report Number	IGNOREMRNUM	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Quick Handover Punish Time	TIMEPUNISH	
			Quick Handover Punish Value	HOPUNISHVALUE	
			Quick Handover Offset	HOOFFSET	
			Handover Direction Forecast Statistic Times	HODIRSTATIME	
			Handover Direction Forecast Last Times	HODIRLASTTIME	
			Frequency Shift Handover Statistic Time	AFCHOSTATIME	
			Frequency Shift Handover Duration	AFCHOLASTTIME	
14	GCELLOTH EXT Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	UL Frequency Adjust Switch	FREQADJ	Yes
			DL Frequency Adjust Switch	DLFREQADJ	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification on Center
			UL Frequency Adjust Value	FREQADJVALUE	
15	GCELLHOBASIC Configure this MO to activate the TA handover algorithm.	BSC6900/ BSC6910	TA HO Allowed	TAHOEN	Yes
16	GCELLHOEMG Configure this MO to activate the TA handover algorithm.	BSC6900/ BSC6910	TA Threshold	TALIMIT	Yes
17	GCELLHOFITPEN Configure this MO to activate the TA handover algorithm.	BSC6900/ BSC6910	Penalty Level after TA HO	SSTAPUNISH	Yes
			Penalty Time after TA HO	TIMETAPUNISH	
18	GCELLHOBASIC Configure this MO to activate the BQ handover algorithm.	BSC6900/ BSC6910	BQ HO Allowed	BQHOEN	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
19	GCELLHOE MG Configure this MO to activate the BQ handover algorithm.	BSC6900/ BSC6910	DL Qual. Threshold	DLQUALIMIT	Yes
			UL Qual. Threshold	ULQUALIMIT	
20	GCELLAMR QUL Configure this MO to activate the BQ handover algorithm.	BSC6900/ BSC6910	UL Qual. Limit for AMR FR	ULQUALIMITAMRFR	Yes
			DL Qual. Limit for AMR FR	DLQUALIMITAMRFR	
			UL Qual. Limit for AMR HR	ULQUALIMITAMRHR	
			DL Qual. Limit for AMR HR	DLQUALIMITAMRHR	
21	GCELLHOFI TPEN Configure this MO to activate the BQ handover algorithm.	BSC6900/ BSC6910	Penalty Level after BQ HO	SSBQPUNISH	Yes
			Penalty Time after BQ HO	TIMEBQPUNISH	
22	GCELLHOB ASIC Configure this MO to activate the rapid level drop handover algorithm.	BSC6900/ BSC6910	Rx_Level_Drop HO Allowed set to YES(Yes)	RXQCKFALLHOEN	Yes
			Edge HO UL RX_LEV Threshold	ULEDGETHRES	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
23	GCELLHOE MG Configure this MO to activate the rapid level drop handover algorithm.	BSC6900/ BSC6910	Filter Parameter A1	FLTPARAA1	Yes
			Filter Parameter A2	FLTPARAA2	
			Filter Parameter A3	FLTPARAA3	
			Filter Parameter A4	FLTPARAA4	
			Filter Parameter A5	FLTPARAA5	
			Filter Parameter A6	FLTPARAA6	
			Filter Parameter A7	FLTPARAA7	
			Filter Parameter A8	FLTPARAA8	
			Filter Parameter B	FLTPARAB	
24	GCELLHOB ASIC Configure this MO to activate the interference handover algorithm.	BSC6900/ BSC6910	Interference HO Allowed	INTERFHON	Yes
			Intracell HO Allowed	INTRACELLHOEN	
			Inter-layer HO Threshold	HOTHRES	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
25	GCELLHOAD Configure this MO to activate the interference handover algorithm.	BSC6900/ BSC6910	Forbidden time after MAX Times	BANTIME	Yes
26	G2GNCELL Configure this MO to activate the interference handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
27	G2GNCELL Configure this MO to activate the interference handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
28	G2GNCELL Configure this MO to activate the interference handover algorithm.	BSC6900/ BSC6910	Adjacent Cell Inter-layer HO Hysteresis	INTELEVH OHYST	No

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
29	GCELLAMRQUL Configure this MO to activate the interference handover algorithm.	BSC6900/ BSC6910	Interfere HO Qual. Thresh 1 for Non-AMR FR	RXQUAL1	Yes
			Interfere HO Qual. Thresh 2 for Non-AMR FR	RXQUAL2	
			Interfere HO Qual. Thresh 3 for Non-AMR FR	RXQUAL3	
			Interfere HO Qual. Thresh 4 for Non-AMR FR	RXQUAL4	
			Interfere HO Qual. Thresh 5 for Non-AMR FR	RXQUAL5	
			Interfere HO Qual. Thresh 6 for Non-AMR FR	RXQUAL6	
			Interfere HO Qual. Thresh 7 for Non-AMR FR	RXQUAL7	
			Interfere HO Qual. Thresh 8 for Non-AMR FR	RXQUAL8	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Interfere HO Qual. Thresh 9 for Non-AMR FR	RXQUAL9	
			Interfere HO Qual. Thresh 10 for Non-AMR FR	RXQUAL10	
			Interfere HO Qual. Thresh 11 for Non-AMR FR	RXQUAL11	
			Interfere HO Qual. Thresh 12 for Non-AMR FR	RXQUAL12	
			Interfere HO Qual. Thresh Offset for AMR FR	RXLEVOFF	
30	GCELLHOE MG Configure this MO to activate the no-DL-MR handover algorithm.	BSC6900/ BSC6910	No DL Mr.HO Allowed	NODLMRHOEN	Yes
			No DL Mr.UL Qual HO Limit	NODLMRHOQUALLIMIT	
			Cons.No DL Mr.HO Allowed Limit	NODLMRHOALLOWLIMIT	
			No DL Mr. HO Statistic Time	NODLMRHOSTATTIME	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			No DL Mr. HO Last Time	NODLMRHOLASTTIME	
31	GCELLHOFITPEN Configure this MO to activate the no-DL-MR handover algorithm.	BSC6900/ BSC6910	Filter Length for TCH Qual	DATAQUAFILTLEN	Yes
			Filter Length for SDCCH Qual	SIGQUAFLTLEN	
32	GCELL Configure this MO to activate the enhanced dual-band network handover algorithm.	BSC6900/ BSC6910	Cell IUO Type	IUOTP	Yes
			Cell Inner/Extra Property	CELLINEXTP	
			Same Group Cell Index Type	DBLFREQA DJIDTYPE	
			Same Group Cell Index Type	DBLFREQA DJIDTYPE	
			Same Group Cell Index	DBLFREQA DJCID	
			Same Group Cell Name	DBLFREQA DJCNAME	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
33	GCELLHOE DBPARA Configure this MO to activate the enhanced dual-band network handover algorithm.	BSC6900/ BSC6910	Load HO Allowed	OUTLOAD HOENABLE	Yes
			UL Subcell General Overload Threshold	OUTGENOVERLDTTHRE	
			Inner Cell Serious OverLoad Thred	INNEROVERLDTTHRE	
			Subcell HO Allowed Flow Control Level	EDBSYSFLOWLEV	
			Incoming OL Subcell HO Level TH	INTOINNR EXLEVTHRE	
			Step Length of UL Subcell Load HO	OUTLOAD HOSTEP	
			UL Subcell Load Hierarchical HO Periods	OUTLOAD HOPERIOD	
			MOD Step LEN of UL Load HO Period	OUTLOAD HOMODPERI	
			Step Length of UL Subcell Load HO	OUTLOAD HOSTEP	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			UL Subcell Lower Load Threshold	OUTLOWLOADTHREDD	
			Subcell HO Allowed Flow Control Level	EDBSYSFLOWLEV	
			Outgoing OL Subcell HO Level TH	OUTINNREXLEVTHREDD	
			OL Subcell Load Diversity HO Period	INNLOADHOPERI	
			Step Length of OL Subcell Load HO	INNLOADHOSTEP	
			Distance Between Boundaries of Subcells	ATCBTHREDD	
			Distance Hysteresis Between Boundaries	ATCBHYST	
34	GCELLHOBASIC Configure this MO to activate the load handover algorithm.	BSC6900/ BSC6910	Load Handover Support	LoadHoEn	Yes
			Edge HO DL RX_LEV Threshold	DLEDGETHRES	
			Inter-layer HO Threshold	HOTHRES	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
35	GCELLHOAD Configure this MO to activate the load handover algorithm.	BSC6900/ BSC6910	System Flux Threshold for Load HO	SYSFLOWLEV	Yes
			Load HO Threshold	TRIDTHRES	
			Load handover Load Accept Threshold	LoadAccThres	
			Load HO Bandwidth	LOADOFFSET	
			Load HO Step Period	LOADHOPERIOD	
			Load HO Step Level	LOADHOSTEP	
36	G2GNCELL Configure this MO to activate the load handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
37	G2GNCELL Configure this MO to activate the load handover algorithm.	BSC6910	Is External Cell	ISNCELL	No

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
38	G2GNCELL Configure this MO to activate the load handover algorithm.	BSC6900/ BSC6910	Adjacent Cell Inter-layer HO Hysteresis	INTELEVH OHYST	No
39	GCELLHOBASIC Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Edge HO Allowed	FRINGEHOEN	
			Edge HO UL RX_LEV Threshold	ULEDGETHRES	
			Edge HO DL RX_LEV Threshold	DLEDGETHRES	
			Handover Algorithm 1 Edge HO Watch Time	EDGESTAT1	
			Handover Algorithm 1 Edge HO Valid Time	EDGELAST1	
40	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighborin g Cell Type	QUERYTYPE	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
41	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
42	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Adjacent Cell Inter-layer HO Hysteresis	INTELEVH OHYST	No
43	GCELLHOB ASIC Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Enhanced Outgoing Cell Handover Allowed	EDOUTHODEN	Yes
44	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
45	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6910	Is External Cell	ISNCELL	No

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
46	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Adjacent Cell Inter-layer HO Hysteresis	INTELEVH OHYST	No
47	GCELLBASICPARA Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	Layer of The Cell	LAYER	Yes
48	GCELLHOBASIC Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	MS Fast Moving HO Allowed	QCKMVHONEN	Yes
			Inter-layer HO Threshold	HOTHRES	
49	GCELLHOAD Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	MS Fast-moving Watch Cells	QCKSTATCNT	Yes
			MS Fast-moving Valid Cells	QCKTRUECNT	
			MS Fast-moving Time Threshold	QCKTIMETH	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Penalty on MS Fast Moving HO	SPEEDPUNISH	
			Penalty Time on Fast Moving HO	SPEEDPUNISHT	
50	G2GNCELL Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
51	G2GNCELL Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
52	G2GNCELL Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	Adjacent Cell Inter-layer HO Hysteresis	INTELEVH OHYST	No

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
53	GCELLHOBASIC Configure this MO to activate the inter-layer handover algorithm.	BSC6900/ BSC6910	Level HO Allowed	LEVHOEN	Yes
			Inter-layer HO Threshold	HOTHRES	
54	GCELLBASICPARA Configure this MO to activate the inter-layer handover algorithm.	BSC6900/ BSC6910	Layer of The Cell	LAYER	Yes
			Cell Priority	PRIOR	
55	G2GNCELL Configure this MO to activate the inter-layer handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
56	G2GNCELL Configure this MO to activate the inter-layer handover algorithm.	BSC6910	Is External Cell	ISNCELL	No

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
57	G2GNCELL Configure this MO to activate the inter-layer handover algorithm.	BSC6900/ BSC6910	Adjacent Cell Inter-layer HO Hysteresis	INTELEVH OHYST	No
			Layer HO Watch Time	LEVSTAT	
			Layer HO Valid Time	LEVLAST	
58	GCELLHOBASIC Configure this MO to activate the PBGT handover algorithm.	BSC6900/ BSC6910	Quick PBGT Handover Enable	PBGTHOEN	Yes
59	GCELLBASICPARA Configure this MO to activate the PBGT handover algorithm.	BSC6900/ BSC6910	Layer of The Cell	LAYER	Yes
			Cell Priority	PRIOR	
60	G2GNCELL Configure this MO to activate the PBGT handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification on Center
61	G2GNCELL Configure this MO to activate the PBGT handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
62	G2GNCELL Configure this MO to activate the PBGT handover algorithm.	BSC6900/ BSC6910	Current HO CTRL Algorithm in Source Cell	SRCHOCTRLSWITCH	No
			Adjacent Cell Inter-layer HO Hysteresis	INTELEVH OHYST	
			PBGT Watch Time	PBGTSTAT	
			PBGT Valid Time	PBGTLAST	
63	GCELLHOBASIC Configure this MO to activate the AMR handover algorithm.	BSC6900/ BSC6910	Intracell F-H HO Allowed	INTRACELL FHHOEN	Yes
			Intracell F-H HO Stat Time	INFHHOSTAT	
			Intracell F-H HO Last Time	INFHHOLAST	
			F2H HO Threshold	INHOF2HTH	
			H2F HO Threshold	INHOH2FTH	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
64	GTRXDEV Configure this MO to activate the AMR handover algorithm.	BSC6900/ BSC6910	TCH Rate Adjust Allow	TCHAJFLAG	Yes
65	GCELLCCACCESS Configure this MO to activate the AMR handover algorithm.	BSC6900/ BSC6910	Speech Version	VOICEVER	Yes
66	BSCBASIC Configure this MO to activate the AMR handover algorithm.	BSC6900/ BSC6910	Um Interface Tag	UmVer	Yes
			Abis Interface Tag	AbisVer	
			A Interface Tag	Aver	
67	GCELLCHMGAD Configure this MO to activate the AMR handover algorithm.	BSC6900/ BSC6910	AMR TCH/H Prior Allowed	AMRTCHHPRIORALLOW	Yes
			AMR TCH/H Prior Cell Load Threshold	AMRTCHHPRIORLOAD	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification on Center
68	GCELLHOBASIC Configure this MO to activate the capacity- and quality-based TCHF-TCHH handover algorithm.	BSC6900/ BSC6910	Intracell F-H HO Allowed	INTRACELLFHHOEN	Yes
			AMR F-H Ho Allowed	AMRFULLTOHALFHOALLOW	
			Non-AMR F-H Ho Allowed	NOAMRFULLTOHALFHOTALLOW	
			AMR F-H Traffic Threshold	AMRFULLTOHALFHO THRESH	
			F-H Ho Duration	FULLTOHALFHODURATION	
			F-H Ho Period	FULLTOHALFHOPERIOD	
			AMR F-H Ho Pathloss Adjust Step	AMRFULLTOHALFHO PATHADJUSTEP	
			AMR F-H Ho ATCB Adjust Step	AMRFULLTOHALFHO ATCBADJUSTEP	
			AMR F-H Ho Pathloss Threshold	AMRFULLTOHALFHO PATHTHRESH	
			F-H Pathloss Offset Overlay	FULLTOHALFHOPATHOFFSET	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			AMR F-H Ho ATCB Threshold	AMRFULLTOHALFHOTCBTHRESH	
			F-H ATCB Offset Overlay	FULLTOHALFHOTCB OFFSET	
			AMR F-H Ho Qual. Threshold	AMRFULLTOHALFHOTQUALTHRESH	
			F-H Ho Stat. Time	FULLTOHALFHOTSTAT TIME	
			F-H Ho Last Time	FULLTOHALFHOTLAST TIME	
			AMR H-F Traffic Threshold	AMRHALFTOFULLHOT HRESH	
			H-F Ho Duration	HALFTOFULLHODURATION	
			AMR H-F Ho ATCB Threshold	AMRHALFTOFULLHOATCBTHRESH	
			H-F ATCB Offset Overlay	HALFTOFULLATCBOFFSET	
			AMR H-F Ho Pathloss Threshold	AMRHALFTOFULLHOPATHTHRESH	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			H-F Pathloss Offset Overlay	HALFTOFULLHOPATHOFFSET	
			AMR H-F Ho Qual. Threshold	AMRHALFTOFULLHOQUALTHRESH	
			H-F Ho Stat. Time	HALFTOFULLHOSTATTIME	
			H-F Ho Last Time	HALFTOFULLHOLASTTIME	
			Non-AMR F-H Traffic Threshold	NOAMRFULLTOHALFTHRESH	
			Non-AMR F-H Ho Pathloss Adjust Step	NOAMRFULLTOHALFHOPATHADJUSTSTEP	
			Non-AMR F-H Ho ATCB Adjust Step	NOAMRFULLTOHALFHOATCBADJUSTSTEP	
			Non-AMR F-H Ho Pathloss Threshold	NOAMRFULLTOHALFHOPATHTHRESH	
			Non-AMR F-H Ho ATCB Threshold	NOAMRFULLTOHALFHOATCBTHRESH	
			Non-AMR F-H Ho Qual. Threshold	NOAMRFULLTOHALFHOQUALTHRESH	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Non-AMR H-F Traffic Threshold	NOAMRHALLFTOFULLTHRESH	
			Non-AMR H-F Ho ATCB Threshold	NOAMRHALLFTOFULLHOATCBTHRESH	
			Non-AMR H-F Ho Pathloss Threshold	NOAMRHALLFTOFULLHOPATHTHRESH	
			Non-AMR H-F Ho Qual. Threshold	NOAMRHALLFTOFULLHOQUALTHRESH	
69	GTRXDEV Configure this MO to activate the capacity- and quality-based TCHF-TCHH handover algorithm.	BSC6900/ BSC6910	TCH Rate Adjust Allow	TCHAJFLAG	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
70	GCELLCACCESS Configure this MO to activate the capacity- and quality-based TCHF-TCHH handover algorithm.	BSC6900/ BSC6910	Speech Version	VOICEVER	Yes
71	BSCBASIC Configure this MO to activate the capacity- and quality-based TCHF-TCHH handover algorithm.	BSC6900/ BSC6910	A Interface Tag	Aver	Yes
72	GCELLHOBASIC Configure this MO to activate the SDCCH handover algorithm.	BSC6900/ BSC6910	SDCCH HO Allowed	SIGCHANHOEN	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
73	GCELLHOC TRL Configure this MO to activate the SDCCH handover algorithm.	BSC6900/BSC6910	Inter-BSC SDCCH HO Allowed	INRBSCSD HOEN	Yes

6.4.4 Activation Observation

Step 1 Run the **LST GCELLHOBASIC** command to query the value of **Current HO Control Algorithm**.

Expected result: The value of **Current HO Control Algorithm** is **HOALGORITHM1(Handover algorithm I)**.

Step 2 Trace BSSAP messages over the A interface.

Expected result: The BSC sends a Handover Performed message to the MSC. The cause value contained in the Handover Performed message varies depending on the handover type. The following table lists the cause value for each handover type.

Table 6-3 Cause value for each handover type

Handover Type	Cause Value
Quick handover	better-cell
TA handover	distance
BQ handover	uplink-quality or downlink-quality
Rapid level drop handover	uplink-strength
Interference handover	uplink-quality or downlink-quality
No-DL-MR handover	uplink-quality
Enhanced dual-band network handover	better-cell or traffic
Load handover	traffic
Edge handover	uplink-strength or downlink-strength

Handover Type	Cause Value
Fast-moving micro cell handover	better-cell
Inter-layer handover	better-cell
PBGT handover	better-cell
Capacity- and quality-based TCHH-TCHF handover	uplink-quality
Capacity- and quality-based TCHF-TCHH handover	traffic
SDCCH handover	better-cell, distance, uplink-quality, downlink-quality, traffic, uplink-strength, or downlink-strength

----End

6.4.5 Deactivation

The HUAWEI I Handover feature does not need to be deactivated.

6.5 Deploying HUAWEI II Handover and AMR FR/HR
Dynamic Adjustment

6.5.1 Deployment Requirements

Table 6-4 Deployment requirements

Aspect	Requirement
Related features	See 4 Related Features .
BSC	None
BTS	None
GSM networking	None
MS	None
MSC	None

Aspect	Requirement
License	The license controlling this feature has been activated. For details on how to activate the license, see <i>License Management Feature Parameter Description</i> . For details about license items, see <i>License Control Item Description</i> . NOTE HUAWEI II Handover is a basic feature and is not controlled by a license.
Others	None

6.5.2 Precautions

- After a quick handover, BQ handover, or TA handover is complete, a handover penalty is imposed on the original cell during the specified penalty period to prevent an immediate handover back to the original cell.
- In the Huawei II handover algorithm, do not select the serving cell as the target cell for a BQ handover.
- If **TA Threshold** for a co-site neighboring cell is lower than or equal to that for the serving cell, a TA handover to the neighboring cell is not allowed.
- The interference handover can be an intra-cell handover. However, if intra-cell handovers have occurred several consecutive times, an intra-cell handover is prohibited within a specified period. In the Huawei II handover algorithm, always select the serving cell as the target cell for an interference handover.
- On an enhanced dual-band network, an MS cannot be handed over to a cell in the same group when overlaid/underlaid load handovers, specified by **Load HO From UL Subcell to OL Subcell Allowed** and **Load HO of OL Subcell to UL Subcell Enabled**, respectively, are allowed. This prevents conflicts between a load handover on a common network and an overlaid/underlaid handover on an enhanced dual-band network.
- A PBGT handover to a cell in the same group is not allowed if it is triggered between two cells on an enhanced dual-band network.
- After the fast-moving micro cell handover is successful, the penalty is performed on all the neighboring micro cells.
- The decision on a load handover is not made independently. Instead, it is made during network adjustment. However, the decision on a load handover is made when the decision on a better-cell handover is made. That is, a load handover is triggered only when the conditions for triggering both a load handover and a better-cell handover are met.
- Better cell handover is an optimization of inter-layer handover and PBGT handover in Huawei I handover algorithm. It is specific to Huawei II handover algorithm.
- When **BCCHIntraFreqHOULAssMeasw (BSC6900, BSC6910)** is set to **ON(On)**, the intra-frame bit offset must be set to **0**. Specifically,
 - a. When Soft-Synchronization Network is enabled, **SYMOFFSET (BSC6900, BSC6910)** must be set to **0**.

- b. When BTS GPS Synchronization or Clock over IP support 1588v2 is enabled, **GSM_FBOFFSET** must be set to **0**.

Otherwise, the uplink auxiliary measurement result becomes less accurate, and the handover success rate decreases and call drop rate increases.

- It is recommended that the value of **BQMARGIN (BSC6900, BSC6910)** be differentiated for co-BCCH intra-frequency neighboring cells. Specifically, the value of this parameter for co-BCCH intra-frequency neighboring cells must be 1 dB to 5 dB greater than that for inter-frequency neighboring cells. A larger difference indicates a higher probability that MSs are handed over to co-BCCH intra-frequency neighboring cells.

6.5.3 Activation

- Step 1** Run the **SET GCELLHOBASIC** command with **Current HO Control Algorithm** set to **HOALGORITHM2(Handover algorithm II)**.
- Step 2** Run the **SET GCELLHOEMG** command with **Current HO Control Algorithm** set to **HOALGORITHM2(Handover algorithm II)**.
- Step 3** Run the **SET GCELLHOFITPEN** command with **Current HO Control Algorithm** set to **HOALGORITHM2(Handover algorithm II)**.
- Step 4** Run the **SET GCELLHOIUO** command with **Current HO Control Algorithm** set to **HOALGORITHM2(Handover algorithm II)**.
- Step 5** Run the **SET GCELLHOAD** command with **Max. Num. of HO Attempts** set according to onsite requirements.
- Step 6** (Optional) To improve the handover and power control performance when the BCCH TRX participates in baseband frequency hopping, run the **SET GCELLCCCH** command with **PWRC** set to **YES(Yes)** and run the **SET GCELLTHEXT** command with **Power Control and Handover CMP CON Sw** set to **YES(Yes)**.
- Step 7** Activate the quick handover algorithm by performing the following steps:
 - Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
 - Run the **MOD G2GNCELL** command with its parameters set as follows:
 - Set **Chain Neighbor Cell** to **YES(Yes)**.
 - Set **Chain Neighbour Cell Type** and **Quick Handover Offset for Neighbor Cell** according to onsite requirements.
 - Run the **SET GCELLHOBASIC** command with **Quick Handover Enable** set to **YES(Yes)**. If a fast move handover is required, set **Quick PBGT Handover Enable** to **YES(Yes)**.

NOTE

Quick handovers consist of fast move handovers and frequency offset handovers. **Quick Handover Enable** controls whether to allow fast move handovers and frequency offset handovers. **Quick PBGT Handover Enable** further controls whether to allow fast move handovers.

To enable the BSC to accurately make handover decisions, you are advised to set **Quick PBGT Handover Enabled** to **YES(Yes)** in high-speed railway scenarios or chain cells along the highway. In other scenarios, set **Quick PBGT Handover Enabled** to **NO(No)**.

- Run the **SET GCELLHOFAST** command with the following parameters set according to onsite requirements: **Quick Move Speed Threshold**, **Quick Handover Up Trigger Level**, **Quick Handover Down Trigger Level**, **Quick Handover Offset**, **Quick Handover Punish Time**, **Quick Handover Punish Value**, **Serving Cell Filter Length MR Number**, **Neighbor Cell Filter Length MR Number**, **Ignore Measurement Report Number**, **EN Quick PBGT HO ALG When MS Leaves BTS**, **Handover Direction Forecast Enable**, **Handover Direction Forecast Statistic Times**, **Frequency Shift Handover Statistic Time**, **Frequency Shift Handover Duration**, and **Fast Handover Punish Control Switch**.

 NOTE

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with **Quick Handover Offset**, **Quick Handover Punish Time**, and **Quick Handover Punish Value** set to appropriate values.

- Run the **SET GCELLOTHEXT** command to set extended parameters of a cell. In this step, set **UL Frequency Adjust Switch**, **DL Frequency Adjust Switch**, and **UL Frequency Adjust Value** based on the frequency offset reported by the BTS.

Step 8 Activate the TA handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **TA HO Allowed** set to **YES(Yes)**.
- Run the **SET GCELLHOEMG** command with **TA Threshold** set according to onsite requirements.
- Run the **SET GCELLHOFITPEN** command with **Penalty Level after TA HO** and **Penalty Time after TA HO** set according to onsite requirements.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **TA HO Watch Time** and **TA HO Valid Time** set according to onsite requirements.

Step 9 Activate the BQ handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **BQ HO Allowed** set to **YES(Yes)**.
- (Optional) For a non-AMR call, run the **SET GCELLHOEMG** command with **DL Qual. Threshold** and **UL Qual. Threshold** set according to onsite requirements.
- (Optional) For an AMR FR or AMR HR call, run the **SET GCELLAMRQUL** command with **UL Qual. Limit for AMR FR**, **DL Qual. Limit for AMR FR**, **UL Qual. Limit for AMR HR**, and **DL Qual. Limit for AMR HR** set according to onsite requirements.
- Run the **SET GCELLHOFITPEN** command with **Penalty Level after BQ HO** and **Penalty Time after BQ HO** set according to onsite requirements.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **BQ HO Watch Time** and **BQ HO Valid Time** set according to onsite requirements.

Step 10 Activate the interference handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with its parameters set as follows:
 - Set both **Interference HO Allowed** and **Intracell HO Allowed** to **YES(Yes)**.
 - Set **Inter-layer HO Threshold**, **Interfere HO Static Time**, and **Interfere HO Valid Time** according to onsite requirements.

 NOTE

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with **Inter-layer HO Threshold** set according to onsite requirements.

- Run the **SET GCELLHOAD** command with **Forbidden time after MAX Times**, **MAX Consecutive HO Times**, and **Interval for Consecutive HO Jud.** set according to onsite requirements.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **Adjacent Cell Inter-layer HO Hysteresis** set according to onsite requirements.
- Run the **SET GCELLAMRQUL** command with the following parameters set according to onsite requirements: **Interfere HO Qual. Thresh 1 for Non-AMR FR**, **Interfere HO Qual. Thresh 2 for Non-AMR FR**, **Interfere HO Qual. Thresh 3 for Non-AMR FR**, **Interfere HO Qual. Thresh 4 for Non-AMR FR**, **Interfere HO Qual. Thresh 5 for Non-AMR FR**, **Interfere HO Qual. Thresh 6 for Non-AMR FR**, **Interfere HO Qual. Thresh 7 for Non-AMR FR**, **Interfere HO Qual. Thresh 8 for Non-AMR FR**, **Interfere HO Qual. Thresh 9 for Non-AMR FR**, **Interfere HO Qual. Thresh 10 for Non-AMR FR**, **Interfere HO Qual. Thresh 11 for Non-AMR FR**, **Interfere HO Qual. Thresh 12 for Non-AMR FR**, and **Interfere HO Qual. Thresh Offset for AMR FR**.

Step 11 Activate the no-DL-MR handover algorithm by performing the following steps:

- Run the **SET GCELLHOEMG** command with its parameters set as follows:
 - Set **No DL Mr.HO Allowed** to **YES(Yes)**.
 - Set **UL Qual. Threshold**, **Cons.No DL Mr.HO Allowed Limit**, **No DL Mr. HO Statistic Time**, and **No DL Mr. HO Last Time** according to onsite requirements.
- Run the **SET GCELLAMRQUL** command with **UL Qual. Limit for AMR FR** and **UL Qual. Limit for AMR HR** set according to onsite requirements.
- Run the **SET GCELLHOFITPEN** command with **Filter Length for TCH Qual** and **Filter Length for SDCCH Qual** set according to onsite requirements.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **UL BQ HO Static Time** and **UL BQ HO Last Time** set according to onsite requirements.

Step 12 Activate the enhanced dual-band network handover algorithm by performing the following steps:

- Run the **MOD GCELL** command with its parameters set as follows:

- Set **Cell IUO Type** to **EDB_cell(Enhanced Double Frequency Cell)**.
- Set **Cell Inner/Extra Property** and **Same Group Cell Index Type** according to onsite requirements.
- Set **Same Group Cell Index** or **Same Group Cell Name** to an appropriate value.
- Run the **SET GCELLHOEDBPARA** command with the following parameters set according to onsite requirements: **Load HO Allowed**, **UL Subcell General Overload Threshold**, **Inner Cell Serious OverLoad Thred**, **Subcell HO Allowed Flow Control Level**, **Incoming OL Subcell HO Level TH**, **UL Subcell Serious Overload Threshold**, **UL Subcell Load Hierarchical HO Periods**, **MOD Step LEN of UL Load HO Period**, **Step Length of UL Subcell Load HO**, **UL Subcell Lower Load Threshold**, **Subcell HO Allowed Flow Control Level**, **Outgoing OL Subcell HO Level TH**, **OL Subcell Load Diversity HO Period**, **Step Length of OL Subcell Load HO**, **Distance Between Boundaries of Subcells**, and **Distance Hysteresis Between Boundaries**.

Step 13 Activate the edge handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with its parameters set as follows:
 - Set **Edge HO Allowed** to **YES(Yes)**.
 - Set **Edge HO UL RX_LEV Threshold**, **Edge HO DL RX_LEV Threshold**, **Handover Algorithm II Edge HO Watch Time**, and **Handover Algorithm II Edge HO Valid Time** according to onsite requirements.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with the following parameters set according to onsite requirements: **Inter-cell HO Hysteresis**, **Edge HO AdjCell Watch Time**, and **Edge HO AdjCell Valid Time**.
- (Optional) To set different edge handover thresholds for different neighboring cells, perform the following steps:
- Run the **SET GCELLHOBASIC** command with **Enhanced Outgoing Cell Handover Allowed** set to **YES(Yes)**.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **Enhanced Outgoing Cell Handover Offset** set according to onsite requirements.

Step 14 Activate the fast-moving micro cell handover algorithm by performing the following steps:

- Run the **SET GCELLBASICPARA** command with **Layer of The Cell** set according to onsite requirements.

 NOTE

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with **Layer of the cell** set according to actual conditions.

- Run the **SET GCELLHOBASIC** command with **MS Fast Moving HO Allowed** set to **YES(Yes)** and **Inter-layer HO Threshold** set according to onsite requirements.

 NOTE

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with **Inter-layer HO Threshold** set according to onsite requirements.

- Run the **SET GCELLHOAD** command with the following parameters set according to onsite requirements: **MS Fast-moving Watch Cells**, **MS Fast-moving Valid Cells**, **MS Fast-moving Time Threshold**, **Penalty on MS Fast Moving HO**, and **Penalty Time on Fast Moving HO**.

 NOTE

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with **Penalty Time on Fast Moving HO** and **Penalty on Fast Moving HO** set to appropriate values.

- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with the following parameters set according to onsite requirements: **Adjacent Cell Inter-layer HO Hysteresis**, **HCS HO Watch Time**, and **HCS HO Valid Time**.

Step 15 Activate the better-cell handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **Better Cell Handover Enable** set to **YES(Yes)** and **Pathloss Ho. Enable** set according to onsite requirements.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with **Better Cell HO Watch Time** and **Better Cell HO Valid Time** set according to onsite requirements.
- (Optional) To set different better-cell handover conditions for different neighboring GSM cells, perform the following steps:
- Run the **SET GCELLHOBASIC** command with **Enhanced Outgoing Cell Handover Allowed** set to **YES(Yes)**.
- Run the **MOD G2GNCELL** command with **Index Type**, **Source Cell Index**, **Neighbor 2G Cell Index**, and **Neighboring Cell Type** set according to actual conditions. If a BSC6910 is used, set **Is External Cell** to an appropriate value.
- Run the **MOD G2GNCELL** command with its parameters set as follows for different neighboring GSM cells:
 - Set **Neighboring Cell Penalty Switch** to **YES(Yes)**.
 - Set **Enhanced Outgoing Cell Handover Offset**, **Penalty Stop Level Threshold**, **Penalty Timer Length**, and **Level Penalty Value on Neighboring Cell** according to onsite requirements.

Step 16 (Optional) Activate the load handover penalty inheritance algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **Load Handover Support** set to **YES(Yes)**.
- Run the **SET GCELLHOFITPEN** command with **Load HO Penalty Optimization** set to **ON(On)**.

Step 17 (Optional) Activate the TCHF-TCHH handover optimization algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **F-H HO Load Band Optimization Allowed** set to **ON(On)**.
- Run the **SET GCELLHOBASIC** command with **TCHF-TCHH HO Load Hysteresis** set according to onsite requirements.
- Run the **SET GCELLHOBASIC** command with the following parameters set according to onsite requirements to ensure that the TCHF-TCHH handover quality threshold ranges from 0 to 70: **AMR F-H HO Quality Thres. Fine Tune, Non-AMR F-H HO Quality Thres. Fine Tune, AMR H-F HO Quality Thres. Fine Tune, Non-AMR H-F HO Quality Thres. Fine Tune, AMR F-H Ho Qual. Threshold, AMR H-F Ho Qual. Threshold, Non-AMR F-H Ho Qual. Threshold, and Non-AMR H-F Ho Qual. Threshold.**
- Run the **SET GCELLHOBASIC** command with the following parameters set according to onsite requirements to ensure that the TCHF-TCHH handover ATCB threshold ranges from -63 to 255: **AMR F-H Ho ATCB Threshold, AMR H-F Ho ATCB Threshold, Non-AMR F-H Ho ATCB Threshold, and Non-AMR H-F Ho ATCB Threshold.**

Step 18 Activate the TCHF-TCHH handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with its parameters set as follows:
 - Set **AMR F-H Ho Allowed, Allow AMR H-F Quality-based HO, Non-AMR F-H Ho Allowed, and Allow Non-AMR H-F Quality-based HO** to **YES(Yes)**.
 - Set the following parameters according to onsite requirements: **AMR F-H Traffic Threshold, F-H Ho Duration, F-H Ho Period, AMR F-H Ho Pathloss Adjust Step, AMR F-H Ho ATCB Adjust Step, AMR F-H Ho Pathloss Threshold, F-H Pathloss Offset Overlay, AMR F-H Ho ATCB Threshold, F-H ATCB Offset Overlay, AMR F-H Ho Qual. Threshold, F-H Ho Stat. Time, F-H Ho Last Time, AMR H-F Traffic Threshold, H-F Ho Duration, AMR H-F Ho ATCB Threshold, H-F ATCB Offset Overlay, AMR H-F Ho Pathloss Threshold, H-F Pathloss Offset Overlay, AMR H-F Ho Qual. Threshold, H-F Ho Stat. Time, H-F Ho Last Time, Non-AMR F-H Traffic Threshold, Non-AMR F-H Ho Pathloss Adjust Step, Non-AMR F-H Ho ATCB Adjust Step, Non-AMR F-H Ho Pathloss Threshold, Non-AMR F-H Ho ATCB Threshold, Non-AMR F-H Ho Qual. Threshold, Non-AMR H-F Traffic Threshold, Non-AMR H-F Ho ATCB Threshold, Non-AMR H-F Ho Pathloss Threshold, and Non-AMR H-F Ho Qual. Threshold.**
- Run the **SET GTRXDEV** command with **TCH Rate Adjust Allow** set to **YES(Yes)**.
- Run the **SET GCELLCCACCESS** command with at least **FULL_RATE_VER1(Full-rate VER 1), FULL_RATE_VER3(Full-rate VER 3), and HALF_RATE_VER3(Half-rate VER 3)** selected under **Speech Version**.
- Run the **SET BSCBASIC** command with **A Interface Tag, Um Interface Tag, and Abis Interface Tag** set to **GSM_PHASE_2Plus**.

Step 19 Activate the SDCCH handover algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **SDCCH HO Allowed** set to **YES(Yes)**.

- Run the **SET GCELLHOCTRL** command with **Inter-BSC SDCCH HO ALLOWed** set according to onsite requirements.

----End

MML Command Examples

//Activating HUAWEI II Handover

```
SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2;  
SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2;  
SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2;  
SET GCELLHOIUO: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2;  
SET GCELLHOAD: IDTYPE=BYID, CELLID=0, HOTRYCNT=3;
```

//Activating the quick handover algorithm for the BSC6900

```
MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,  
NCELLTYPE=HANDOVERNCELL, ISCHAINNCELL=YES,  
CHAINNCELLTYPE=QUICK_HO_NCELL_TYPE_A, ADJHOOFFSET=1;  
  
SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,  
QUICKHOEN=YES, QUICKPBGTHOEN=YES;  
  
SET GCELLHOFAST: IDTYPE=BYID, CELLID=0, HOUPTRIGE=50, HODOWNTRIGE=50,  
MOVESPEEDTHRES=35, SCELLFILTER=4, NCELLFILTER=4, IGNOREMRNUM=1,  
TIMEPUNISH=10, HOPUNISHVALUE=63, HOOFFSET=68, MSLEVSTRQPBGT=NO,  
HODIRFORECASTEN=YES, HODIRSTATIME=3, HODIRLASTTIME=2,  
AFCHOSTATIME=4, AFCHOLASTTIME=3;  
  
SET GCELLLOTHEXT: IDTYPE=BYID, CELLID=0, FREQADJ=YES, DLFREQADJ=YES,  
FREQADJVALUE=36671;
```

//Activating the quick handover algorithm for the BSC6910

```
MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,  
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, ISCHAINNCELL=YES,  
CHAINNCELLTYPE=QUICK_HO_NCELL_TYPE_A, ADJHOOFFSET=1;  
  
SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,  
QUICKHOEN=YES, QUICKPBGTHOEN=YES;  
  
SET GCELLHOFAST: IDTYPE=BYID, CELLID=0, HOUPTRIGE=50, HODOWNTRIGE=50,  
MOVESPEEDTHRES=35, SCELLFILTER=4, NCELLFILTER=4, IGNOREMRNUM=1,  
TIMEPUNISH=10, HOPUNISHVALUE=63, HOOFFSET=68, MSLEVSTRQPBGT=NO,  
HODIRFORECASTEN=YES, HODIRSTATIME=3, HODIRLASTTIME=2,  
AFCHOSTATIME=4, AFCHOLASTTIME=3;  
  
SET GCELLLOTHEXT: IDTYPE=BYID, CELLID=0, FREQADJ=YES, DLFREQADJ=YES,  
FREQADJVALUE=36671;
```

//Activating the TA handover algorithm for the BSC6900

```
SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,  
TAHOEN=YES;
```

SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, TALIMIT=125;

SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, SSTAPUNISH=40, TIMETAPUNISH=10;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, NCELLTYPE=HANDOVERNCELL, TASTATTIME=1, TALASTTIME=1;

//Activating the TA handover algorithm for the BSC6910

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, TAHOEN=YES;

SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, TALIMIT=125;

SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, SSTAPUNISH=40, TIMETAPUNISH=10;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, TASTATTIME=1, TALASTTIME=1;

//Activating the BQ handover algorithm for the BSC6900

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, BQHOEN=YES;

SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, DLQUALIMIT=55, ULQUALIMIT=60;

SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, SSBQPUNISH=63, TIMEBQPUNISH=10;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, NCELLTYPE=HANDOVERNCELL, BQSTATTIME=1, BQLASTTIME=1;

//Activating the BQ handover algorithm for the BSC6910

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, BQHOEN=YES;

SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, DLQUALIMIT=55, ULQUALIMIT=60;

SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, SSBQPUNISH=63, TIMEBQPUNISH=10;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, BQSTATTIME=1, BQLASTTIME=1;

//Activating the interference handover algorithm for the BSC6900

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, INTRACELLHOEN=YES, INTERFHOEN=YES, HOTHRES=25, INTERFERESTATTIME=3, INTERFERELASTTIME=2;

```
SET GCELLHOAD: IDTYPE=BYID, CELLID=0, BANTIME=20;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

SET GCELLAMRQUL: IDTYPE=BYID, CELLID=0, RXQUAL1=70, RXQUAL2=60,
RXQUAL3=59, RXQUAL4=58, RXQUAL5=57, RXQUAL6=56, RXQUAL7=55,
RXQUAL8=54, RXQUAL9=53, RXQUAL10=52, RXQUAL11=51, RXQUAL12=50;

//Activating the interference handover algorithm for the BSC6910

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
INTRACELLHOEN=YES, INTERFHOEN=YES, HOTHRES=25, INTERFERESTATTIME=3,
INTERFERELASTTIME=2;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, BANTIME=20;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

SET GCELLAMRQUL: IDTYPE=BYID, CELLID=0, RXQUAL1=70, RXQUAL2=60,
RXQUAL3=59, RXQUAL4=58, RXQUAL5=57, RXQUAL6=56, RXQUAL7=55,
RXQUAL8=54, RXQUAL9=53, RXQUAL10=52, RXQUAL11=51, RXQUAL12=50;

//Activating the no-DL-MR handover algorithm for the BSC6900

SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
NODLMRHOEN=YES, NODLMRHOQUALLIMIT=50, NODLMRHOALLOWLIMIT=8,
NODLMRHOSTATTIME=1, NODLMRHOLASTTIME=1;

SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
DATAQUAFLTLEN=4, SIGQUAFLTLEN=2;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
NCELLTYPE=HANDOVERNCELL, ULBQSTATTIME=1, ULBQLASTTIME=1;

//Activating the no-DL-MR handover algorithm for the BSC6910

SET GCELLHOEMG: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
NODLMRHOEN=YES, NODLMRHOQUALLIMIT=50, NODLMRHOALLOWLIMIT=8,
NODLMRHOSTATTIME=1, NODLMRHOLASTTIME=1;

SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
DATAQUAFLTLEN=4, SIGQUAFLTLEN=2;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, ULBQSTATTIME=1,
ULBQLASTTIME=1;

//Activating the enhanced dual-band network handover algorithm

MOD GCELL: IDTYPE=BYID, CELLID=0, IUOTP=EDB_cell, CELLINEXTP=Inner,
DBLFREQADJIDTYPE=BYID, DBLFREQADJCID=1;

SET GCELLHOEDBPARA: IDTYPE=BYID, CELLID=0, OUTASSOPTEN=YES;

//Activating the edge handover algorithm for the BSC6900
```

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, FRINGEHOEN=YES, ULEDGETHRES=10, DLEDGETHRES=20, EDGESTAT1=6, EDGELAST1=4;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

//Activating the edge handover algorithm for the BSC6910

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, FRINGEHOEN=YES, ULEDGETHRES=10, DLEDGETHRES=20, EDGESTAT1=6, EDGELAST1=4;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, INTELEVHOHYST=67;

//Activating the fast-moving micro cell handover algorithm for the BSC6900

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, LAYER=3;

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, QCKMVHOEN=YES, HOTHRES=25;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, QCKSTATCNT=3, QCKTRUECNT=2, QCKTIMETH=15, SPEEDPUNISH=30, SPEEDPUNISHT=4;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, NCELLTYPE=HANDOVERNCELL, SRCHOCTRLSWITCH=HOALGORITHM2, INTELEVHOHYST=67, HCSSTATTIME=3, HCSLASTTIME=2;

//Activating the fast-moving micro cell handover algorithm for the BSC6910

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, LAYER=3;

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, QCKMVHOEN=YES, HOTHRES=25;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, QCKSTATCNT=3, QCKTRUECNT=2, QCKTIMETH=15, SPEEDPUNISH=30, SPEEDPUNISHT=4;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, SRCHOCTRLSWITCH=HOALGORITHM2, INTELEVHOHYST=67, HCSSTATTIME=3, HCSLASTTIME=2;

//Activating the better-cell handover algorithm for the BSC6900

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, BETTERCELLHOEN=YES, PATHLOSSHOEN=YES;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1, NCELLTYPE=HANDOVERNCELL, SRCHOCTRLSWITCH=HOALGORITHM2, BETTERCELLSTATTIME=6, BETTERCELLLASTTIME=4;

//Activating the better-cell handover algorithm for the BSC6910

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2, BETTERCELLHOEN=YES, PATHLOSSHOEN=YES;

MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL,
SRCHOCTRLSWITCH=HOALGORITHM2, BETTERCELLSTATTIME=6,
BETTERCELLLASTTIME=4;

//Activating the TCHF-TCHH handover algorithm

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
AMRFULLTOHALFHOTALLOW=YES, AMRFULLTOHALFHOTHRESH=75,
FULLTOHALFHODURATION=5, FULLTOHALFHOPERIOD=1,
AMRFULLTOHALFHOPATHADJUSTEP=1, AMRFULLTOHALFHOOATCBADJUSTEP=2,
AMRFULLTOHALFHOPATHTHRESH=100, FULLTOHALFHOPATHOFFSET=10,
AMRFULLTOHALFHOOATCBTHRESH=20, FULLTOHALFHOOATCBOFFSET=20,
AMRFULLTOHALFHOOQUALTHRESH=0, FULLTOHALFHOOSTATTIME=6,
FULLTOHALFHOOLASTTIME=4, AMRHALFTOFULLHOTHRESH=45,
AMRHALFTOFULLHOOQUALALLOW=YES, HALFTOFULLHODURATION=5,
AMRHALFTOFULLHOOATCBTHRESH=4, HALFTOFULLHOOATCBOFFSET=20,
AMRHALFTOFULLHOPATHTHRESH=108, HALFTOFULLHOPATHOFFSET=10,
AMRHALFTOFULLHOOQUALTHRESH=3, HALFTOFULLHOOSTATTIME=6,
HALFTOFULLHOOLASTTIME=4, NOAMRFULLTOHALFHOTALLOW=YES,
NOAMRFULLTOHALFHOTHRESH=85, NOAMRFULLTOHALFHOPATHADJUSTEP=1,
NOAMRFULLTOHALFHOOATCBADJUSTEP=2,
NOAMRFULLTOHALFHOPATHTHRESH=95,
NOAMRFULLTOHALFHOOATCBTHRESH=30,
NOAMRFULLTOHALFHOOQUALTHRESH=0, NOAMRHALFTOFULLHOTHRESH=25,
NOAMRHALFTOFULLHOOQUALALLOW=YES,
NOAMRHALFTOFULLHOOATCBTHRESH=14,
NOAMRHALFTOFULLHOPATHTHRESH=103,
NOAMRHALFTOFULLHOOQUALTHRESH=2;

SET GTRXDEV: TRXID=0, TCHAJFLAG=YES;

SET GCELLCCACCESS: IDTYPE=BYID, CELLID=0,
VOICEVER=FULL_RATE_VER1-1&FULL_RATE_VER2-0&FULL_RATE_VER3-1&HALF_R
ATE_VER1-1&HALF_RATE_VER2-0&HALF_RATE_VER3-1&FULL_RATE_VER5-0;

SET BSCBASIC: AVer=GSM_PHASE_2Plus, UmVer=GSM_PHASE_2Plus,
AbisVer=GSM_PHASE_2Plus;

//Activating the load handover penalty inheritance algorithm

SET GCELLHOFITPEN: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
LoadHoPunishInheritSwitch=ON;

//Activating the TCHF-TCHH handover optimization algorithm

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1,
F2HHOLOADSTFSWITCH=ON;

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
AMRFULLTOHALFHOOATCBTHRESH=-63, AMRFULLTOHALFHOOQUALTHRESH=6,
AMRF2HHOQUALTHFINE=8;

//Activating the SDCCH handover algorithm

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM2,
SIGCHANHOEN=YES;

SET GCELLHOCTRL: IDTYPE=BYID, CELLID=0, INRBSCSDHOEN=YES;

Using the MAE-Deployment

NOTE

When configuring the feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. Configure the parameters of a single object before a batch modification. Perform a batch modification before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in Table 6-5.

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-5 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLHOBASIC	BSC6900/BSC6910	Current HO Control Algorithm	HOCTRLSWITCH	Yes
2	GCELLHOEMG	BSC6900/BSC6910	Current HO Control Algorithm	HOCTRLSWITCH	Yes
3	GCELLHOFITPEN	BSC6900/BSC6910	Current HO Control Algorithm	HOCTRLSWITCH	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
4	GCELLHOIUO	BSC6900/ BSC6910	Current HO Control Algorithm	HOCTRLSWITCH	Yes
5	GCELLHOAD	BSC6900/ BSC6910	Max. Num. of HO Attempts	HOTRYCNT	Yes
6	GCELLCCH Configure this MO if you need to improve the handover and power control performance when the BCCH TRX participates in baseband frequency hopping.	BSC6900/ BSC6910	PWRC	CANPC	Yes
7	GCELLTCH Configure this MO if you need to improve the handover and power control performance when the BCCH TRX participates in baseband frequency hopping.	BSC6900/ BSC6910	Power Control and Handover CMP CON Sw	PCHOCMPCON	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
8	G2GNCELL Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
9	G2GNCELL Configure this MO to activate the quick handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
10	G2GNCELL Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	Chain Neighbor Cell	ISCHAINNCELL	No
			Chain Neighbour Cell Type	CHAINNCELLTYPE	
			Quick Handover Offset for Neighbor Cell	ADJHOOFFSET	
11	GCELLHOBASIC Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	Quick Handover Enable	QUICKHOPEN	Yes
12	GCELLHOBASIC	BSC6900/ BSC6910	Quick PBGT Handover Enable	QUICKPBGTHOEN	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
13	GCELLHOF AST Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	Handover Direction Forecast Enable	HODIRFORECASTEN	Yes
			Quick Handover Up Trigger Level	HOUPTRIGGER	
			Quick Handover Down Trigger Level	HODOWNTRIGGER	
			Quick Move Speed Threshold	MOVESPEEDTHRES	
			Serving Cell Filter Length MR Number	SCELLFILTER	
			Neighbor Cell Filter Length MR Number	NCELLFILTER	
			Ignore Measurement Report Number	IGNOREMRNUM	
			Quick Handover Punish Time	TIMEPUNISH	
			Quick Handover Punish Value	HOPUNISHVALUE	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Quick Handover Offset	HOOFFSET	
			Handover Direction Forecast Statistic Times	HODIRSTA TIME	
			Handover Direction Forecast Last Times	HODIRLAST TIME	
			Frequency Shift Handover Statistic Time	AFCHOSTA TIME	
			Frequency Shift Handover Duration	AFCHOLAST TIME	
			EN Quick PBGT HO ALG When MS Leaves BTS	MSLEVSTR QPBG	
14	GCELLOTH EXT Configure this MO to activate the quick handover algorithm.	BSC6900/ BSC6910	UL Frequency Adjust Switch	FREQADJ	Yes
			DL Frequency Adjust Switch	DLFREQADJ	
			UL Frequency Adjust Value	FREQADJVALUE	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
15	GCELLHOBASIC Configure this MO to activate the TA handover algorithm.	BSC6900/ BSC6910	TA HO Allowed	TAHOEN	Yes
16	GCELLHOEMG Configure this MO to activate the TA handover algorithm.	BSC6900/ BSC6910	TA Threshold	TALIMIT	Yes
17	GCELLHOFITPEN Configure this MO to activate the TA handover algorithm.	BSC6900/ BSC6910	Penalty Level after TA HO	SSTAPUNISH	Yes
			Penalty Time after TA HO	TIMETAPUNISH	
18	G2GNCELL Configure this MO to activate the TA handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
19	G2GNCELL Configure this MO to activate the TA handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
20	G2GNCELL Configure this MO to activate the TA handover algorithm.	BSC6900/ BSC6910	TA HO Watch Time	TASTATTIME	No
			TA HO Valid Time	TALASTTIME	
21	GCELLHOBASIC Configure this MO to activate the BQ handover algorithm.	BSC6900/ BSC6910	BQ HO Allowed	BQHOEN	Yes
22	GCELLHOEMG Configure this MO to activate the BQ handover algorithm.	BSC6900/ BSC6910	DL Qual. Threshold	DLQUALIMIT	Yes
			UL Qual. Threshold	ULQUALIMIT	
23	GCELLAMRQUL Configure this MO to activate the BQ handover algorithm.	BSC6900/ BSC6910	UL Qual. Limit for AMR FR	ULQUALIMITAMRFR	Yes
			DL Qual. Limit for AMR FR	DLQUALIMITAMRFR	
			UL Qual. Limit for AMR HR	ULQUALIMITAMRHR	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			DL Qual. Limit for AMR HR	DLQUALIMITAMRHR	
24	GCELLHOFTPEN Configure this MO to activate the BQ handover algorithm.	BSC6900/ BSC6910	Penalty Level after BQ HO	SSBQPUNISH	Yes
			Penalty Time after BQ HO	TIMEBQPUNISH	
25	G2GNCELL Configure this MO to activate the BQ handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
26	G2GNCELL Configure this MO to activate the BQ handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
27	G2GNCELL Configure this MO to activate the BQ handover algorithm.	BSC6900/ BSC6910	BQ HO Watch Time	BQSTATTIME	No
			BQ HO Valid Time	BQLASTTIME	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
28	GCELLHOBASIC Configure this MO to activate the interference handover algorithm.	BSC6900/ BSC6910	Interference HO Allowed	INTERFHOPEN	Yes
			Intracell HO Allowed	INTRACELHOPEN	
			Inter-layer HO Threshold	HOTHRES	
			Interference HO Static Time	INTERFERESTATTIME	
			Interference HO Valid Time	INTERFERELASTTIME	
29	GCELLHOAD Configure this MO to activate the interference handover algorithm.	BSC6900/ BSC6910	Forbidden time after MAX Times	BANTIME	Yes
			MAX Consecutive HO Times	MAXCNTNUM	
			Interval for Consecutive HO Jud.	CONTINTV	
30	G2GNCELL Configure this MO to activate the interference handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
31	G2GNCELL Configure this MO to activate the interference handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
32	G2GNCELL Configure this MO to activate the interference handover algorithm.	BSC6900/ BSC6910	Adjacent Cell Inter-layer HO Hysteresis	INTELEVH OHYST	No
33	GCELLAMR QUL Configure this MO to activate the interference handover algorithm.	BSC6900/ BSC6910	Interfere HO Qual. Thresh 1 for Non-AMR FR	RXQUAL1	Yes
			Interfere HO Qual. Thresh 2 for Non-AMR FR	RXQUAL2	
			Interfere HO Qual. Thresh 3 for Non-AMR FR	RXQUAL3	
			Interfere HO Qual. Thresh 4 for Non-AMR FR	RXQUAL4	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Interfere HO Qual. Thresh 5 for Non-AMR FR	RXQUAL5	
			Interfere HO Qual. Thresh 6 for Non-AMR FR	RXQUAL6	
			Interfere HO Qual. Thresh 7 for Non-AMR FR	RXQUAL7	
			Interfere HO Qual. Thresh 8 for Non-AMR FR	RXQUAL8	
			Interfere HO Qual. Thresh 9 for Non-AMR FR	RXQUAL9	
			Interfere HO Qual. Thresh 10 for Non-AMR FR	RXQUAL10	
			Interfere HO Qual. Thresh 11 for Non-AMR FR	RXQUAL11	
			Interfere HO Qual. Thresh 12 for Non-AMR FR	RXQUAL12	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Interfere HO Qual. Thresh Offset for AMR FR	RXLEVOFF	
34	GCELLHOE MG Configure this MO to activate the no-DL-MR handover algorithm.	BSC6900/ BSC6910	No DL Mr.HO Allowed	NODLMRH OEN	Yes
			UL Qual. Threshold	ULQUALIMIT	
			Cons.No DL Mr.HO Allowed Limit	NODLMRH OALLOWLIMIT	
			No DL Mr. HO Statistic Time	NODLMRH OSTATTIME	
			No DL Mr. HO Last Time	NODLMRH OLASTTIME	
35	GCELLAMR QUL Configure this MO to activate the no-DL-MR handover algorithm.	BSC6900/ BSC6910	UL Qual. Limit for AMR FR	ULQUALIMITAMRFR	Yes
			UL Qual. Limit for AMR HR	ULQUALIMITAMRHR	
36	GCELLHOFI TPEN Configure this MO to activate the no-DL-MR handover algorithm.	BSC6900/ BSC6910	Filter Length for TCH Qual	DATAQUAF LTLEN	Yes
			Filter Length for SDCCCH Qual	SIGQUAFLT LEN	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
37	G2GNCELL Configure this MO to activate the no-DL-MR handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
38	G2GNCELL Configure this MO to activate the no-DL-MR handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
39	G2GNCELL Configure this MO to activate the no-DL-MR handover algorithm.	BSC6900/ BSC6910	UL BQ HO Static Time	ULBQSTAT TIME	No
			UL BQ HO Last Time	ULBQLAST TIME	
40	GCELL Configure this MO to activate the enhanced dual-band network handover algorithm.	BSC6900/ BSC6910	Cell IUO Type	IUOTP	Yes
			Cell Inner/Extra Property	CELLINEXT P	
			Same Group Cell Index Type	DBLFREQA DJIDTYPE	
			Same Group Cell Index Type	DBLFREQA DJIDTYPE	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Same Group Cell Index	DBLFREQA DJCID	
			Same Group Cell Name	DBLFREQA DJCNAME	
41	GCELLHOE DBPARA Configure this MO to activate the enhanced dual-band network handover algorithm.	BSC6900/ BSC6910	Load HO Allowed	OUTLOAD HOENABLE	Yes
			UL Subcell General Overload Threshold	OUTGENOVERLDTHREED	
			Inner Cell Serious OverLoad Thred	INNSEIOVERLDTHREED	
			Subcell HO Allowed Flow Control Level	EDBSYSFLOWLEV	
			Incoming OL Subcell HO Level TH	INTOINNREXLEVTHREED	
			UL Subcell Serious Overload Threshold	OUTSEIOVERLDTHREED	
			UL Subcell Load Hierarchical HO Periods	OUTLOADHOPERIOD	
			MOD Step LEN of UL Load HO Period	OUTLOADHOMODPERI	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Step Length of UL Subcell Load HO	OUTLOADHOSTEP	
			UL Subcell Lower Load Threshold	OUTLOWLOADTHREDD	
			Subcell HO Allowed Flow Control Level	EDBSYSFLOWLEV	
			Outgoing OL Subcell HO Level TH	OUTINNREXLEVTHREDD	
			OL Subcell Load Diversity HO Period	INNLOADHOPERI	
			Step Length of OL Subcell Load HO	INNLOADHOSTEP	
			Distance Between Boundaries of Subcells	ATCBTHREDD	
			Distance Hysteresis Between Boundaries	ATCBHYST	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
42	GCELLHOBASIC Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Edge HO Allowed	FRINGEHOEN	Yes
			Edge HO UL RX_LEV Threshold	ULEDGETHRES	
			Edge HO DL RX_LEV Threshold	DLEDGETHRES	
			Handover Algorithm II Edge HO Watch Time	EDGESTAT	No
			Handover Algorithm II Edge HO Valid Time	EDGELAST	
43	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
44	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6910	Is External Cell	ISNCELL	No

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification on Center
45	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Adjacent Cell Inter-layer HO Hysteresis	INTELEVH OHYST	No
			Edge HO AdjCell Watch Time	EDGEADJS TATTIME	
			Edge HO AdjCell Valid Time	EDGEADJL ASTTIME	
46	GCELLHOB ASIC Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Enhanced Outgoing Cell Handover Allowed	EDOUTH OADEN	Yes
47	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCE LLID	
			Neighbor 2G Cell Index	NBR2GNCE LLID	
			Neighboring Cell Type	QUERYTYPE	
48	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6910	Is External Cell	ISNCELL	No

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
49	G2GNCELL Configure this MO to activate the edge handover algorithm.	BSC6900/ BSC6910	Enhanced Outgoing Cell Handover Offset	EDOUTHO OFFSET	No
50	GCELLBASIC CPARA Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	Layer of The Cell	LAYER	Yes
51	GCELLHOB ASIC Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	MS Fast Moving HO Allowed	QCKMVH OEN	Yes
			Inter-layer HO Threshold	HOTHRES	
52	GCELLHOA D Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	MS Fast-moving Watch Cells	QCKSTATC NT	Yes
			MS Fast-moving Valid Cells	QCKTRUEC NT	
			MS Fast-moving Time Threshold	QCKTIMET H	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Penalty on MS Fast Moving HO	SPEEDPUNISH	
			Penalty Time on Fast Moving HO	SPEEDPUNISHT	
53	G2GNCELL Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighborin g Cell Type	QUERYTYPE	
54	G2GNCELL Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
55	G2GNCELL Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	Adjacent Cell Inter-layer HO Hysteresis	INTELEVH OHYST	No
			HCS HO Watch Time	HCSSTATTIME	
			HCS HO Valid Time	HCSLASTTIME	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
56	GCELLHOBASIC Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	Better Cell Handover Enable	BETTERCELLHOEN	Yes
			Pathloss Ho. Enable	PATHLOSSHOEN	
57	G2GNCELL Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
58	G2GNCELL Configure this MO to activate the fast-moving micro cell handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
59	G2GNCELL Configure this MO to activate the better-cell handover algorithm.	BSC6900/ BSC6910	Better Cell HO Watch Time	BETTERCELLSTATTIME	No
			Better Cell HO Valid Time	BETTERCELLLASTTIME	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification on Center
60	GCELLHOBASIC Configure this MO to activate the better-cell handover algorithm.	BSC6900/ BSC6910	Enhanced Outgoing Cell Handover Allowed	EDOUTHODEN	Yes
61	G2GNCELL Configure this MO to activate the better-cell handover algorithm.	BSC6900/ BSC6910	Index Type	IDTYPE	No
			Source Cell Index	SRC2GNCELLID	
			Neighbor 2G Cell Index	NBR2GNCELLID	
			Neighboring Cell Type	QUERYTYPE	
62	G2GNCELL Configure this MO to activate the better-cell handover algorithm.	BSC6910	Is External Cell	ISNCELL	No
63	G2GNCELL Configure this MO to activate the better-cell handover algorithm.	BSC6900/ BSC6910	Neighboring Cell Penalty Switch	NCELLPUNEN	No
			Enhanced Outgoing Cell Handover Offset	EDOUTHODOFFSET	
			Penalty Stop Level Threshold	NCELLPUNSTPTH	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification on Center	
			Penalty Timer Length	NCELLPUNTM		
			Level Penalty Value on Neighboring Cell	NCELLPUNLEV		
64	GCELLHOBASIC Configure this MO to activate the load handover penalty inheritance algorithm.	BSC6900/BSC6910	Load Handover Support	LoadHoEn	Yes	
65	GCELLHOFITPEN Configure this MO to activate the load handover penalty inheritance algorithm.	BSC6900/BSC6910	Load HO Penalty Optimization	LoadHoPunishInheritSwitch	Yes	
	GCELLHOBASIC Configure this MO to activate the TCHF-TCHH handover optimization algorithm.		F-H HO Load Band Optimization Allowed	F2HHOLOADSTFSWITCH		
			TCHF-TCHH HO Load Hysteresis	FullToHalfHoLoadStf		
			AMR F-H HO Quality Thres. Fine Tune	AMRF2HHOQUALTHFINE		

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Non-AMR F-H HO Quality Thres. Fine Tune	NOAMRF2HHOQUALTHFINE	
			AMR H-F HO Quality Thres. Fine Tune	AMRH2FHOQUALFINE	
			Non-AMR H-F HO Quality Thres. Fine Tune	NOAMRH2FHOQUALFINE	
			AMR F-H Ho Qual. Threshold	AMRFULLTOHALFHQUALTHRESH	
			AMR H-F Ho Qual. Threshold	AMRHALFTOFULLHQUALTHRESH	
			Non-AMR F-H Ho Qual. Threshold	NOAMRFULLTOHALFHQUALTHRESH	
			Non-AMR H-F Ho Qual. Threshold	NOAMRHALFTOFULLHQUALTHRESH	
			AMR F-H Ho ATCB Threshold	AMRFULLTOHALFHATCBTHRESH	
			AMR H-F Ho ATCB Threshold	AMRHALFTOFULLHOATCBTHRESH	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Non-AMR F-H Ho ATCB Threshold	NOAMRFULLTOHALFHOTHRESH	
			Non-AMR H-F Ho ATCB Threshold	NOAMRHALLFTOFULLHOTHOTHRESH	
66	GCELLHOBASIC Configure this MO to activate the TCHF-TCHH handover algorithm.	BSC6900/ BSC6910	AMR F-H Ho Allowed	AMRFULLTOHALFHOTALLOW	Yes
			Allow AMR H-F Quality-based HO	AMRHALLFTOFULLHOQUALALLOW	
			Non-AMR F-H Ho Allowed	NOAMRFULLTOHALFHOTALLOW	
			Allow Non-AMR H-F Quality-based HO	NOAMRHALLFTOFULLHOQUALALLOW	
			AMR F-H Traffic Threshold	AMRFULLTOHALFHOTHRESH	
			F-H Ho Duration	FULLTOHALFHODURATION	
			F-H Ho Period	FULLTOHALFHOPERIOD	
			AMR F-H Ho Pathloss Adjust Step	AMRFULLTOHALFHOPATHADJUSTEP	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			AMR F-H Ho ATCB Adjust Step	AMRFULLT OHALFHO ATCBADJUST EP	
			AMR F-H Ho Pathloss Threshold	AMRFULLT OHALFHO PATHTHRESH	
			F-H Pathloss Offset Overlay	FULLTOHA LFHOPATH OFFSET	
			AMR F-H Ho ATCB Threshold	AMRFULLT OHALFHO ATCBTHRESH	
			F-H ATCB Offset Overlay	FULLTOHA LFHOATCB OFFSET	
			AMR F-H Ho Qual. Threshold	AMRFULLT OHALFHO QUALTHRESH	
			F-H Ho Stat. Time	FULLTOHA LFHOSTAT TIME	
			F-H Ho Last Time	FULLTOHA LFHOLAST TIME	
			AMR H-F Traffic Threshold	AMRHALFT OFULLHOT HRESH	
			H-F Ho Duration	HALFTOFU LLHODURATION	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			AMR H-F Ho ATCB Threshold	AMRHALFTOFULLHOATCBTHRESH	
			H-F ATCB Offset Overlay	HALFTOFULLATCBOFFSET	
			AMR H-F Ho Pathloss Threshold	AMRHALFTOFULLHOPATHTHRESH	
			H-F Pathloss Offset Overlay	HALFTOFULLHOPATHOFFSET	
			AMR H-F Ho Qual. Threshold	AMRHALFTOFULLHOQUALTHRESH	
			H-F Ho Stat. Time	HALFTOFULLHOSTATTIME	
			H-F Ho Last Time	HALFTOFULLHOLASTTIME	
			Non-AMR F-H Traffic Threshold	NOAMRFULLTOHALFTHRESH	
			Non-AMR F-H Ho Pathloss Adjust Step	NOAMRFULLTOHALFHOPATHADJUSTSTEP	
			Non-AMR F-H Ho ATCB Adjust Step	NOAMRFULLTOHALFHOATCBADJUSTSTEP	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Non-AMR F-H Ho Pathloss Threshold	NOAMRFU LLTOHALF HOPATHT HRESH	
			Non-AMR F-H Ho ATCB Threshold	NOAMRFU LLTOHALF HOATCBTH RESH	
			Non-AMR F-H Ho Qual. Threshold	NOAMRFU LLTOHALF HOQUALT HRESH	
			Non-AMR H-F Traffic Threshold	NOAMRHA LFTOFULLT HRESH	
			Non-AMR H-F Ho ATCB Threshold	NOAMRHA LFTOFULL HOATCBTH RESH	
			Non-AMR H-F Ho Pathloss Threshold	NOAMRHA LFTOFULL HOPATHT HRESH	
			Non-AMR H-F Ho Qual. Threshold	NOAMRHA LFTOFULL HOQUALT HRESH	
67	GTRXDEV Configure this MO to activate the TCHF-TCHH handover algorithm.	BSC6900/ BSC6910	TCH Rate Adjust Allow	TCHAJFLAG	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
68	GCELLCCACCESS Configure this MO to activate the TCHF-TCHH handover algorithm.	BSC6900/ BSC6910	Speech Version	VOICEVER	Yes
69	BSCBASIC Configure this MO to activate the TCHF-TCHH handover algorithm.	BSC6900/ BSC6910	A Interface Tag	Aver	Yes
			Abis Interface Tag	AbisVer	
			Um Interface Tag	UmVer	
70	GCELLHOBASIC Configure this MO to activate the SDCCH handover algorithm.	BSC6900/ BSC6910	SDCCH HO Allowed	SIGCHANH OEN	Yes
71	GCELLHOC TRL Configure this MO to activate the SDCCH handover algorithm.	BSC6900/ BSC6910	Inter-BSC SDCCH HO Allowed	INRBSCSD HOEN	Yes

6.5.4 Activation Observation

Step 1 Run the **LST GCELLHOBASIC** command to query the value of **Current HO Control Algorithm**.

Expected result: The value of **Current HO Control Algorithm** is **Handover algorithm II**.

Step 2 Trace BSSAP messages over the A interface.

Expected result: The BSC sends a Handover Performed message to the MSC.

- In the case of a quick handover, the handover cause value is "better-cell".
- In the case of a TA handover, the handover cause value is "distance".
- In the case of a BQ handover, the handover cause value is "uplink-quality" or "downlink-quality".
- In the case of an interference handover, the handover cause value is "uplink-quality" or "downlink-quality".
- In the case of a no-DL-MR handover, the handover cause value is "uplink-quality".
- In the case of an enhanced dual-band network handover, the handover cause value is "better-cell" or "traffic".
- In the case of a load handover, the handover cause value is "traffic".
- In the case of an edge handover, the handover cause value is "uplink-strength" or "downlink-strength".
- In the case of a fast-moving micro cell handover, the handover cause value is "better-cell".
- In the case of a better-cell handover, the handover cause value is "better-cell".
- In the case of a TCHF-TCHH handover, the handover cause value is "downlink-quality".
- In the case of an SDCCH handover, the handover cause value is "better-cell", "distance", "uplink-quality", "downlink-quality", "traffic", "uplink-strength", or "downlink-strength".

Step 3 (Optional) Verify the load handover penalty inheritance algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with **Load Handover Support** set to **YES(Yes)** for cell A.
- Run the **SET GCELLHOFITPEN** command with **Load HO Penalty Optimization** set to **ON(On)** for cell A. Ensure that a load handover is performed from cell C to cell B, then to cell A.
- Run the **SET GCELLHOAD** command with the following parameters set to appropriate values for cell A to ensure that the conditions for triggering a load handover from cell A to cell C are met: **Load HO Threshold**, **Load handover Load Accept Threshold**, **Load HO Bandwidth**, and **Load HO Step Level**.

Expected result: A load handover from cell A to cell C is not triggered.

Step 4 (Optional) Verify the TCHF-TCHH handover optimization algorithm by performing the following steps:

- Run the **SET GCELLHOBASIC** command with all the following parameters set to **YES(Yes)**: **AMR F-H Ho Allowed**, **Allow AMR H-F Quality-based HO**, **Non-AMR F-H Ho Allowed**, and **Allow Non-AMR H-F Quality-based HO**.
- Run the **SET GCELLHOBASIC** command with **F-H HO Load Band Optimization Allowed** set to **ON(On)**.

- Run the **SET GCELLHOBASIC** command with **AMR F-H Traffic Threshold** and **TCHF-TCHH HO Load Hysteresis** set according to onsite requirements for cell A. Ensure that the following condition is met after some AMR FR calls are converted into AMR HR calls in the cell: **Cell load < AMR F-H Traffic Threshold – TCHF-TCHH HO Load Hysteresis**
Expected result: Only some AMR FR calls support TCHF-to-TCHH handovers.
- Run the **SET GCELLHOBASIC** command with **Non-AMR H-F Ho ATCB Threshold** set to **-10**.
Expected result: Only the AMR HR calls whose ATCB values are less than or equal to -10 support TCHH-to-TCHF handovers.
- Run the **SET GCELLHOBASIC** command with **Non-AMR H-F Ho Qual. Threshold** set to **3** and **Non-AMR H-F HO Quality Thres. Fine Tune** set to **9**.
Expected result: Only the AMR HR calls whose TCHF-TCHH handover quality thresholds exceed 39 support TCHH-to-TCHF handovers

----End

6.5.5 Deactivation

Using MML Commands

Run the **SET GCELLHOBASIC** command with **Current HO Control Algorithm** set to **HOALGORITHM1 (Handover algorithm 1)**.

MML Command Examples

//Deactivating HUAWEI II Handover

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1;

Using the MAE-Deployment

NOTE

When configuring the feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. Configure the parameters of a single object before a batch modification. Perform a batch modification before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-6](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-6 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification on Center
1	GCELLHOB ASIC	BSC6900/BSC6910	Current HO Control Algorithm	HOCTRLSWITCH	Yes

6.6 Deploying Direct Retry

6.6.1 Deployment Requirements

Table 6-7 Deployment requirements

Aspect	Requirement
Related features	See 4 Related Features .
BSC	None
BTS	None
GSM networking	None
MS	None
MSC	None
License	None
Others	None

6.6.2 Activation

Using MML Commands

Step 1 Run the **SET GCELLBASICPARA** command with **Directed Retry** set to **YES(Yes)**.

- Step 2** Run the **SET GCELLCCBASIC** command with **Directed Retry Load Access Threshold** set according to the actual load of candidate neighboring cells.
- Step 3** Run the **SET GCELLHOBASIC** command with **Current HO Control Algorithm** set to an appropriate value.
- If **Current HO Control Algorithm** is set to **HOALGORITHM1(Handover algorithm I)**, run the **SET GCELLHOCTRL** command with **Min Power Level For Directed Retry** set according to the actual receive level of neighboring cells.
 - If **Current HO Control Algorithm** is set to **HOALGORITHM2(Handover algorithm II)**, run the **ADD G2GNCELL** command with cells 0 and 1 set to neighboring GSM cells. Then, set **Inter-cell HO Hysteresis** and **Directed Retry Handover Level Range** for cell 1 according to onsite requirements. If a BSC6910 is used, set **Is External Cell**.
- Step 4 Optional:** Set the number of cells for directed retry and the priorities of these cells.
- Run the **SET GCELLOTHEXT** command with **DRD HO Target Cell Selection Optimized** set to **YES(Yes)**.
 - Run the **SET GCELLHOAD** command with **Max. Num. of HO Attempts** set according to network plan.

----End

MML Command Examples

//Activating Direct Retry for the BSC6900

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, DIRECTRYEN=YES;

SET GCELLOTHEXT: IDTYPE=BYID, CELLID=0, DRTAGCELLSEL=YES;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, HOTRYCNT=3;

SET GCELLCCBASIC: IDTYPE=BYID, CELLID=0, DTLOADTHRED=85;

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1;

SET GCELLHOCTRL: IDTYPE=BYID, CELLID=0, MINPWRLEVDIRTRY=16;

ADD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
NCELLTYPE=HANDOVERNCELL, SRCHOCTRLSWITCH=HOALGORITHM1;

//Activating Direct Retry for the BSC6910

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, DIRECTRYEN=YES;

SET GCELLOTHEXT: IDTYPE=BYID, CELLID=0, DRTAGCELLSEL=YES;

SET GCELLHOAD: IDTYPE=BYID, CELLID=0, HOTRYCNT=3;

SET GCELLCCBASIC: IDTYPE=BYID, CELLID=0, DTLOADTHRED=85;

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=0, HOCTRLSWITCH=HOALGORITHM1;

SET GCELLHOCTRL: IDTYPE=BYID, CELLID=0, MINPWRLEVDIRTRY=16;

ADD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=0, NBR2GNCELLID=1,
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL,
SRCHOCTRLSWITCH=HOALGORITHM1;

Using the MAE-Deployment

 NOTE

When configuring the feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. Configure the parameters of a single object before a batch modification. Perform a batch modification before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-8](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-8 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLBASICPARA	BSC6900/BSC6910	Directed Retry	DIRECTRYEN	Yes
2	GCELLCCBASIC	BSC6900/BSC6910	Directed Retry Load Access Threshold	DTLOADTHRED	Yes
		BSC6900/BSC6910	Current HO Control Algorithm	HOCTRLSWITCH	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification on Center
3	GCELLHOC TRL NOTE Configure this MO if Current HO Control Algorithm is set to HOALGORITHM1 (Handover algorithm I)	BSC6900/ BSC6910	Min Power Level For Directed Retry	MINPWRLE VDIRTRY	Yes
4	G2GNCELL NOTE Configure this MO if Current HO Control Algorithm is set to HOALGORITHM2 (Handover algorithm II).	BSC6900/ BSC6910	Inter-cell HO Hysteresis	INTERCELL HYST	No
			Directed Retry Handover Level Range	DRHOLEVR ANGE	
		BSC6910	Is External Cell	ISNCELL	
5	GCELLOTH EXT NOTE Configure this MO to set the number of cells for directed retry and the priorities of these cells.	BSC6900/ BSC6910	DRD HO Target Cell Selection Optimized	DRTAGCEL LSEL	Yes

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
6	GCELLHOAD NOTE Configure this MO to set the number of cells for directed retry and the priorities of these cells.	BSC6900/ BSC6910	Max. Num. of HO Attempts	HOTRYCNT	Yes

6.6.3 Activation Observation

Step 1 Block all TCHs in cell 0. Meanwhile, ensure that the SDCCH in cell 0 is available.

Step 2 Trace BSSAP messages over the A interface and RSL messages over the Abis interface in cells 0 and 1.

Step 3 Use an MS to initiate a call in cell 0.

Step 4 View the BSSAP messages traced over the A interface and RSL messages traced over the Abis interface in cells 0 and 1.

Expected result: The call has been set up and the voice quality is good. A Handover Detection message is traced over the Abis interface and an Assignment Complete message is traced over the A interface in cell 1. For an intra-BSC directed retry, after you enable CDT during a single user trace task, the value of ucHoCause in the crdlc-msip-ho-trig-ind message is "directed-retry". For an inter-BSC directed retry, the handover cause value in the Handover Required message traced on the A interface in cell 0 is "directed-retry".

----End

6.6.4 Deactivation

Using MML Commands

Run the **SET GCELLBASICPARA** command with **Directed Retry** set to **NO(No)**.

MML Command Examples

```
//Deactivating Direct Retry
```

SET GCELLBASICPARA: IDTYPE=BYID, CELLID=0, DIRECTRYEN=NO;

Using the MAE-Deployment

NOTE

When configuring the feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. Configure the parameters of a single object before a batch modification. Perform a batch modification before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-9](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-9 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLBASICPARA	BSC6900/BSC6910	Directed Retry	DIRECTRYEN	Yes

6.7 Deploying Pre-processing of Measurement Report

6.7.1 Deployment Requirements

Table 6-10 Deployment requirements

Aspect	Requirement
Related features	See 4 Related Features .
BSC	None
BTS	The BTS supports this feature.
GSM networking	None
MS	None
MSC	None
License	None
Others	None

6.7.2 Activation

Using MML Commands

On the BSC LMT, run the **SET GCELLHOCTRL** command with its parameters set as follows:

- Set **MR.Preprocessing** to **BTS_Preprocessing(BTS preprocessing)**.
- Set **Transfer Original MR**, **Transfer BTS/MS Power Class**, and **Sent Freq.of preprocessed MR** according to network plan.

MML Command Examples

//Activating Pre-processing of Measurement Report

```
SET GCELLHOCTRL: IDTYPE=BYID, CELLID=0,  
BTSMESRPTPREPROC=BTS_Preprocessing, PRIMMESPT=NO, BSMSPWRLEV=YES,  
MRPREPROCFREQ=Once_ps;
```

Using the MAE-Deployment

NOTE

When configuring the feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. Configure the parameters of a single object before a batch modification. Perform a batch modification before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-11](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

Table 6-11 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLHOC TRL	BSC6900/ BSC6910	MR.Preprocessing	BTSMESRPTPREPROC	Yes
			Transfer Original MR	PRIMMESPT	
			Transfer BTS/MS Power Class	BSMSPWRL EV	
			Sent Freq.of preprocessed MR	MRPREPROCFREQ	

----End

6.7.3 Activation Observation

- Step 1** Trace RSL messages over the Abis interface. For details, see section "Tracing CS Domain RSL Messages on the Abis Interface" in *BSC6910 LMT User Guide*.
- Step 2** Use an MS to call a fixed-line phone in the test cell.
- Step 3** Check whether the Preprocessed Measurement Result message is displayed in the trace window. If displayed, this feature has been activated.

----End

6.7.4 Deactivation

Using MML Commands

Run the **SET GCELLHOCTRL** command with **MR.Preprocessing** set to **BSC_Preprocessing (BSC Preprocessing)**.

MML Command Examples

//Deactivating Pre-processing of Measurement Report

```
SET GCELLHOCTRL: IDTYPE=BYID, CELLID=0,
BTSMESSRPTPREPROC=BSC_Preprocessing;
```

Using the MAE-Deployment

NOTE

When configuring the feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. Configure the parameters of a single object before a batch modification. Perform a batch modification before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-12](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-12 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLHOCTRL	BSC6900/ BSC6910	MR.Preprocessing	BTSMESSRPTPREPROC	Yes

6.8 Deploying Fast Move Handover

6.8.1 Deployment Requirements

Table 6-13 Deployment requirements

Aspect	Requirement
Related features	See 4 Related Features .
BSC	None
BTS	BTSs other than the BTS3900B and BTS3900E are used.
GSM networking	None
MS	None
MSC	None
License	The license controlling this feature has been activated. For details on how to activate the license, see <i>License Management Feature Parameter Description</i> . For details about license items, see <i>License Control Item Description</i> .
Others	None

6.8.2 Activation

Using MML Commands

- Step 1** Run the **SET GCELLHOBASIC** command with both **Quick Handover Enable** and **Quick PBGT Handover Enable** set to **YES(Yes)**.
- Step 2** Run the **SET GCELLHOFAST** command with the following parameters set according to onsite requirements: **Quick Move Speed Threshold**, **Quick Handover Up Trigger Level**, **Quick Handover Down Trigger Level**, **Quick Handover Offset**, **Quick Handover Punish Time**, **Quick Handover Punish Value**, **Serving Cell Filter Length MR Number**, **Neighbor Cell Filter Length MR Number**, **Ignore Measurement Report Number**, **EN Quick PBGT HO ALG When MS Leaves BTS**, **Handover Direction Forecast Enable**, **Handover Direction Forecast Statistic Times**, **Frequency Shift Handover Statistic Time**, **Frequency Shift Handover Duration**, and **Fast Handover Punish Control Switch**.

 NOTE

Run the **LST G2GNCELL** command to query information about neighboring GSM cells.

If a cell is configured with external neighboring GSM cells, run the **MOD GEXT2GCELL** command with **Quick Handover Offset**, **Quick Handover Punish Time**, and **Quick Handover Punish Value** set according to onsite requirements.

Step 3 Run the **SET GCELLOTHEXT** command with **UL Frequency Adjust Switch** and **UL Frequency Adjust Value** set according to onsite requirements.

----End

MML Command Examples

//Activating Fast Move Handover

SET GCELLHOBASIC: IDTYPE=BYID, CELLID=1024, QUICKHOEN=YES,
QUICKPBGTHOEN=YES;

SET GCELLHOFAST: IDTYPE=BYID, CELLID=1024, HOUPTRIGE=50,
HODOWNTRIGE=50, MOVESPEEDTHRES=35, MSLEVSTRQPBGT=YES,
QuickHoPunishSw=ON;

SET GCELLOTHEXT: IDTYPE=BYID, CELLID=1024, FREQADJ=YES,
FREQADJVALUE=255;

Using the MAE-Deployment

 NOTE

When configuring the Fast Move Handover feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. You must perform a single configuration for a parameter before batch modifications of the parameter. You are advised to perform batch modifications before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-14](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-14 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLHOBASIC	BSC6900/BSC6910	Quick Handover Enable	QUICKHOEN	Yes
			Quick PBGT Handover Enable	QUICKPBGTHOEN	
2	GCELLHOFAST	BSC6900/BSC6910	Quick Handover Up Trigger Level	HOUPTRIG E	Yes
			Quick Handover Down Trigger Level	HODOWNTRIGE	
			Quick Handover Offset	HOOFFSET	
			Fast Handover Punish Control Switch	QuickHoPunishSw	
			Quick Handover Punish Time	TIMEPUNISH	
			Quick Handover Punish Value	HOPUNISHVALUE	
			Serving Cell Filter Length MR Number	SCELLFILTER	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
			Neighbor Cell Filter Length MR Number	NCELLFILTER	
			Ignore Measurement Report Number	IGNOREMRNUM	
			EN Quick PBGT HO ALG When MS Leaves BTS	MSLEVSTRQPBGT	
			Handover Direction Forecast Enable	HODIRFORECASTEN	
			Handover Direction Forecast Statistic Times	HODIRSTATIME	
			Frequency Shift Handover Statistic Time	AFCHOSTATIME	
			Frequency Shift Handover Duration	AFCHOLASTIME	
			Fast Handover Punish Control Switch	QuickHoPunishSw	

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification on Center
3	GCELLOTH EXT	BSC6900/ BSC6910	UL Frequency Adjust Switch	FREQADJ	Yes
			UL Frequency Adjust Value	FREQADJVALUE	

6.8.3 Activation Observation

Step 1 Move an MS away from the serving cell at a speed greater than **Quick Move Speed Threshold** until the frequency offset in the measurement report, CH RQD, or HO DETECT message is a negative value. If the frequency offset information fails to be reported or the frequency offset information is invalid, the uplink level in the serving cell is lower than **Quick Handover Up Trigger Level**, the downlink level in the serving cell is lower than **Quick Handover Down Trigger Level**, and the path loss of chain neighboring cells is lower than the path loss of the serving cell minus the sum of **Quick Handover Offset** and **Quick Handover Offset for Neighbor Cell**.

Step 2 Trace BSSAP messages over the A interface.

Expected result: The BSC sends a Handover Performed message to the MSC, with the cause value "better-cell".

----End

6.8.4 Deactivation

Using MML Commands

Run the **SET GCELLHOBASIC** command with **Quick PBGT Handover Enable** set to **NO(No)**.

NOTE

Quick handovers consist of fast move handovers and frequency offset handovers. **Quick Handover Enable** controls whether to allow fast move handovers and frequency offset handovers. In this step, you can also deactivate the fast move handover by setting **Quick Handover Enable** to **NO(No)**, but the frequency offset handover is deactivated at the same time.

MML Command Examples

```
//Deactivating Fast Move Handover
```

```
SET GCELLHOBASIC: IDTYPE=BYID, CELLID=1024, QUICKPBGTHOEN=NO;
```

Using the MAE-Deployment

NOTE

When configuring the Fast Move Handover feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. You must perform a single configuration for a parameter before batch modifications of the parameter. You are advised to perform batch modifications before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-15](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-15 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLHOBASIC	BSC6900/BSC6910	Quick PBGT Handover Enable	QUICKPBGTHOEN	Yes

6.9 Deploying Chain Cell Handover

6.9.1 Deployment Requirements

Table 6-16 Deployment requirements

Aspect	Requirement
Related features	See 4 Related Features .
BSC	None
BTS	None
GSM networking	None
MS	None
MSC	None
License	The license controlling this feature has been activated. For details on how to activate the license, see <i>License Management Feature Parameter Description</i> . For details about license items, see <i>License Control Item Description</i> .
Others	None

6.9.2 Activation

Using MML Commands

Run the **ADD G2GNCELL** command with **Chain Neighbor Cell** set to **YES(Yes)**. If a cell has been added as a neighboring cell, run the **MOD G2GNCELL** command with **Neighboring Cell Type** to **HANDOVERNCELL(Handover Neighboring cell)** and **Chain Neighbor Cell** to **YES(Yes)**.

MML Command Examples

```
//Activating Chain Cell Handover
```

```
ADD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=1, NBR2GNCELLID=2,  
NCELLTYPE=HANDOVERNCELL, SRCHOCTRLSWITCH=HOALGORITHM1,  
ISCHAINNCELL=YES;
```

Using the MAE-Deployment

NOTE

When configuring the Chain Cell Handover feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. You must perform a single configuration for a parameter before batch modifications of the parameter. You are advised to perform batch modifications before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-17](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-17 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	G2GNCELL	BSC6900/ BSC6910	Chain Neighbor Cell	ISCHAINNCELL	No
			Neighboring Cell Type	NCELLTYPE	
			Chain Neighbor Cell	ISCHAINNCELL	

6.9.3 Activation Observation

Run the **LST G2GNCELL** command to query the value of **Chain Neighbor Cell**.

Expected result: The value of **Chain Neighbor Cell** is **Yes**.

6.9.4 Deactivation

Using MML Commands

Run the **MOD G2GNCELL** command with **Chain Neighbor Cell** set to **NO(No)**.

MML Command Examples

```
//Deactivating Chain Cell Handover for the BSC6900
```

```
MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=1024, NBR2GNCELLID=2542,  
NCELLTYPE=HANDOVERNCELL, ISCHAINNCELL=NO;
```

```
//Deactivating Chain Cell Handover for the BSC6910
```

```
MOD G2GNCELL: IDTYPE=BYID, SRC2GNCELLID=1024, NBR2GNCELLID=2542,  
ISNCELL=INNCELL, NCELLTYPE=HANDOVERNCELL, ISCHAINNCELL=NO;
```

Using the MAE-Deployment

NOTE

When configuring the Chain Cell Handover feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. You must perform a single configuration for a parameter before batch modifications of the parameter. You are advised to perform batch modifications before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-18](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-18 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	G2GNCELL	BSC6900/ BSC6910	Chain Neighbor Cell	ISCHAINNCELL	No

6.10 Deploying BTS power lift for handover

6.10.1 Deployment Requirements

Table 6-19 Deployment requirements

Aspect	Requirement
Related features	For details, see 4 Related Features .
BSC	None
BTS	None
GSM networking	None
MS	None
MSC	None
License	The license controlling this feature has been activated. For details on how to activate the license, see <i>License Management Feature Parameter Description</i> . For details about license items, see <i>License Control Item Description</i> .
Others	None

6.10.2 Activation

Using MML Commands

Run the **SET GCELLOTHBASIC** command with **Power Boost before HO Enabled or Not** set to **YES(StartUp)**.

MML Command Examples

```
//Activating BTS power lift for handover
```

```
SET GCELLOTHBASIC: IDTYPE=BYID, CELLID=0, HOPOWERBOOST=YES;
```

Using the MAE-Deployment

NOTE

When configuring the feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. Configure the parameters of a single object before a batch modification. Perform a batch modification before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-20](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-20 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLOTHBASIC	BSC6900/BSC6910	Power Boost before HO Enabled or Not	HOPOWER BOOST	Yes

6.10.3 Activation Observation

Run the **LST GCELLOTHBASIC** command to query the value of **Power Boost before HO Enabled or Not**.

Expected result: The value of **Power Boost before HO Enabled or Not** is **StartUp**.

6.10.4 Deactivation

Using MML Commands

Run the **SET GCELLOTHBASIC** command with **Power Boost before HO Enabled or Not** set to **NO(Not StartUp)**.

MML Command Examples

//Deactivating BTS power lift for handover

SET GCELLOTHBASIC: IDTYPE=BYID, CELLID=0, HOPOWERBOOST=NO;

Using the MAE-Deployment

 NOTE

When configuring the feature on the MAE-Deployment, you must perform a single configuration first, and then perform batch modifications if required. Configure the parameters of a single object before a batch modification. Perform a batch modification before logging out of the parameter setting interface.

Step 1 Configure a single object on the MAE-Deployment.

Set parameters on the MAE-Deployment configuration interface according to the operation sequence described in [Table 6-21](#).

For detailed operations on the MAE-Deployment, see Feature Configuration Using the MAE-Deployment.

Step 2 (Optional) Modify objects in batches on the MAE-Deployment.

To modify objects, such as BSCs, BTSs, cells, and TRXs, in batches, click the  icon on the MAE-Deployment configuration interface to start the batch modification wizard.

For instructions on how to perform a batch modification, click  on the wizard interface to obtain online help.

----End

Table 6-21 Configuring parameters on the MAE-Deployment

SN	MO	NE	Parameter Name	Parameter ID	Configurable in MAE-Deployment Batch Modification Center
1	GCELLOTHBASIC	BSC6900/BSC6910	Power Boost before HO Enabled or Not	HOPOWERBOOST	Yes

7 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

Base station controller parameter reference

- BSC6900 GSM Parameter Reference: contains the parameters related to the BSC6900 equipment, transport, and radio access functions.
- BSC6910 GSM Parameter Reference: contains the parameters related to the BSC6910 equipment, transport, and radio access functions.

NOTE

You can find the EXCEL files of parameter reference for the software version used on the live network from the "Operation and Maintenance > Configuration Management" node in the *BSC69xx Product Documentation* delivered with that version.

eGBTS Parameter reference

- Node Parameter Reference: contains base station equipment and transport parameters.
- GBTSFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.

NOTE

You can find the EXCEL files of parameter reference for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

1. Open the EXCEL file of parameter reference.
2. On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, GBFD-200201.
3. Click **OK**. All parameters related to the feature are displayed.

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

1. Open the EXCEL file of the used reserved parameter list.
2. On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

Base station controller performance counter reference

- BSC6900 GSM Performance Counter Reference: contains the counters related to the BSC6900 equipment, transport, and radio access functions.
- BSC6910 GSM Performance Counter Reference: contains the counters related to the BSC6910 equipment, transport, and radio access functions.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the "Operation and Maintenance > Performance Management" node in the *BSC69xx GSM Product Documentation* delivered with that version.

eGBTS performance counter reference

- Node Performance Counter Summary: contains base station equipment and transport counters.
- GBTSFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

1. Open the EXCEL file of performance counter reference.
2. On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, GBFD-200201.
3. Click **OK**. All counters related to the feature are displayed.

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

The reference documents are as follows:

1. 3GPP TS 08.58 Base Station Controller - Base Transceiver Station (BSC-BTS) Interface Layer 3 Specification
2. 3GPP TS 04.08 Mobile radio interface layer 3 specification
3. BSC6900 Feature List
4. BSC6900 Basic Feature Description
5. BSC6900 Optional Feature Description
6. BSC6900 GSM Parameter Reference
7. BSC6900 GSM MML Command Reference
8. BSC6900 GSM Performance Counter Reference